

Reviewed Preprint

v2 • November 15, 2024

Revised by authors

Reviewed Preprint

v1 • June 14, 2024

Reversing protonation of weakly basic drugs greatly enhances intracellular diffusion and decreases lysosomal sequestration

Debabrata Dey, Shir Marciano, Anna Poryvai, Ondřej Groborz, Lucie Wohlrábová, Tomáš Slanina, Gideon Schreiber 

Department of Biomolecular Sciences, Weizmann Institute of Science, Rehovot, Israel • Institute of Organic Chemistry and Biochemistry of the Czech Academy of Sciences, Prague, Czech Republic

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

eLife Assessment

This is a **valuable** study on the diffusion rates of drug molecules in human-derived cells, presenting **convincing** data indicating that their diffusion behavior depends on their charged state. It proposes that blocking drug protonation enhances diffusion and fractional recovery, suggesting improved intracellular availability of weakly basic drugs. The findings are significant for drug design and understanding the biophysical behavior of small molecules in cells.

<https://doi.org/10.7554/eLife.97255.2.sa3>

Abstract

For drugs to be active they have to reach their targets. Within cells this requires crossing the cell membrane, and then free diffusion, distribution, and availability. Here, we explored the in-cell diffusion rates and distribution of a series of small molecular fluorescent drugs, in comparison to proteins, by microscopy and fluorescence recovery after photobleaching (FRAP). While all proteins diffused freely, we found a strong correlation between pK_a and the intracellular diffusion and distribution of small molecule drugs. Weakly basic, small-molecule drugs displayed lower fractional recovery after photobleaching and 10-to-20-fold slower diffusion rates in cells than in aqueous solutions. As, more than half of pharmaceutical drugs are weakly basic, they, are protonated in the cell cytoplasm. Protonation, facilitates the formation of membrane impermeable ionic form of the weak base small molecules. This results in ion trapping, further reducing diffusion rates of weakly basic small molecule drugs under macromolecular crowding conditions where other nonspecific interactions become more relevant and dominant. Our imaging studies showed that acidic organelles, particularly the lysosome, captured these molecules. Surprisingly, blocking lysosomal import only slightly increased diffusion rates and fractional recovery. Conversely, blocking protonation by *N*-acetylated analogues, greatly enhanced their diffusion and fractional recovery after FRAP.

Based on these results, *N*-acetylation of small molecule drugs may improve the intracellular availability and distribution of weakly basic, small molecule drugs within cells.

Introduction

Drug molecules alter many biochemical reaction pathways inside the cell by interacting with proteins, DNAs, RNAs, or others. By this, they influence one or more specific cellular activities (Ong et al., 2009 [↗](#); Schreiber, 2005 [↗](#); Stockwell, 2004 [↗](#)). Due to their bioactivity, stability, rather simple chemical synthesis allowing for industrial bulk scale production and the easiness of administration, most drugs are small molecules (< 1 kDa) (Ngo and Garneau-Tsodikova, 2018 [↗](#); Veber et al., 2002 [↗](#)). Over half of these drugs have a significant degree of hydrophobicity and are weakly basic at intracellular pH of 7.4 (Charifson and Walters, 2014 [↗](#); Lipinski et al., 1997 [↗](#)). Hydrophobicity is required to cross the lipid bilayer of the cell membranes, and polar functional groups solubilize the drug in the cellular medium (Lipinski et al., 1997 [↗](#)). The balance between pK_a and logP (partition coefficient of small molecules as measured from their partition between octanol/water) is critical for successful drug design (Manallack, 2007 [↗](#); Meanwell, 2011 [↗](#)). Lipinski's "Rules of 5" suggests that logP should be between 1 and 6 to be a candidate for successful oral administration. Small molecule drugs can be acidic or basic depending on their pK_a values (Lipinski et al., 1997 [↗](#)). Drugs with $pK_a \sim 0-7$ are considered acidic, and $pK_a \sim 8-14$ are regarded as basic or alkaline. Basic drugs with $pK_a > 8$ are more common than their acidic counterparts ($pK_a < 7$, $\sim 60:40$ ratios). In some cases, lowering their pK_a to the range between 6 and 7, has proven beneficial (Charifson and Walters, 2014 [↗](#)). For a long time, mechanisms such as passive diffusion (Di et al., 2012 [↗](#)) and carrier-mediated drug delivery (Dobson and Kell, 2008 [↗](#)) or a combination of both (Sugano et al., 2010 [↗](#)) have been used to explain intracellular drug delivery. However, transport of a drug across the cellular membrane is only one part of a successful delivery. Intracellular diffusion and bio-distribution needs to be taken in account when assessing the drug mode of action.

In previous work, we studied the diffusion of fluorescent drug molecules in crowded environments *in vitro* (Dey et al., 2022 [↗](#)). However, the cellular cytoplasm of eukaryotic cells is by far more complex (Model et al., 2021 [↗](#)). Intracellular membranes, multiple organelles (Ellis, 2001 [↗](#); Fulton, 1982 [↗](#)) and a large diversity of macromolecules, in terms of size, charge and hydrophobicity, render the cellular interior extremely complex and heterogeneous (Milo and Phillips, 2015 [↗](#); Neurohr and Amon, 2020 [↗](#)). Moreover, the pH inside cellular organelles can vastly differ from the cytosolic pH (Madhus, 1988 [↗](#)). Organelles like lysosomes, late endosomes, Golgi, and secretory vesicles are acidic and have been known for sequestering weakly basic drugs (Asokan and Cho, 2002 [↗](#); Proksch, 2018 [↗](#)). Therefore, predicting diffusion of a particular drug molecule in the dense cytoplasm is a complex task and requires sub-organelle resolution. For example, positively charged (basic) drugs designed to target nuclear DNA may bind cytoplasmic matter through nonspecific interactions (Long et al., 2022 [↗](#); Tiwari et al., 2018 [↗](#)), they could be sequestered within the lysosome, or diffuse slower than expected due to crowding and nonspecific interactions (Gotink et al., 2011 [↗](#); Jansen et al., 1999 [↗](#); Zhitomirsky and Assaraf, 2015 [↗](#)).

The mobility of a drug in the cellular interior has been shown to be a good proxy for intracellular drug delivery and other processes, such as nonspecific binding or aggregation (Dey et al., 2022 [↗](#)). The primary tool for measuring small molecule diffusion inside the cell is fluorescence recovery after photobleaching (FRAP), which inherently requires fluorescent drug molecules. FRAP with a confocal microscope is a sufficiently sensitive, versatile, and easy-to-use method suitable for measuring the diffusion of weakly fluorescent molecules. Unfortunately, single-molecule fluorescence correlation spectroscopy (FCS), a gold-standard method for measuring diffusion of fluorescent particles, cannot be applied for most small molecule drugs due to their weak fluorescence (Dey et al., 2022 [↗](#), 2021 [↗](#); Zotter et al., 2017).

In this work, we determined the intracellular diffusion coefficients of a series of fluorescent dyes and drugs compared to protein diffusion by FRAP (**Figure 1** and **Table 1**). Specifically, we used Line-FRAP, which has a much higher time resolution and therefore can measure faster diffusion rates in comparison to standard FRAP (800 Hz versus 50 Hz, for detailed explanations, see (Dey et al., 2022) (Dey et al., 2021)). A pictorial representation of Line FRAP performed in a live HeLa cell treated with a fluorescent drug is shown in **Figure 2A**. The small red circle denotes the bleached area which is done by using a Clip tornado (of 2x2 pixel size). The white line indicates the scanning region which was performed along at a rate of 800 Hz in order to record the FRAP as a function of time. The Line-FRAP cross section profile as a function of time is shown in **Figure 2B**. The vertical direction of the line denotes the time progress. The vertically marked white rectangular Region of Interest (ROI), in the left panel (**Figure 2B**) shows the fluorescence intensity recovery over time, which is plotted in **Figure 2C**. The horizontal rectangular region of interest (ROI) outlined in white (in the right panel of **Figure 2B**) denotes the time frame immediately after the photo-bleach pulse (the highly intense green is the bleach pulse). The time interval between each line scanning measurement is ~1.25 milliseconds. This horizontal white ROI generates the fluorescence intensity over the scanning line distance which is plotted in **Figure 2D**. This profile gives the effective bleach radius when fitted with a gaussian function (Dey et al., 2022, 2021). The apparent diffusion coefficients derived from FRAP measurements are termed D_{confocal} are calculated using equation 1:

$$D_{\text{confocal}} = \frac{r_e^2 + r_n^2}{8\tau_{1/2}} \quad \text{eq. 1}$$

where $\tau_{1/2}$ is the half-time of recovery and r_e and r_n are the effective and nominal bleach radii values (for details see (Dey et al., 2021)). D_{confocal} values for proteins determined by Line-FRAP agreed with the diffusion rates measured by FCS (Dey et al., 2021). For small molecules, Line-FRAP only provides relative 3D diffusion coefficients (Dey et al., 2022). We studied the following fluorescent drugs: the antimalarial drugs quinacrine dihydrochloride (Ehsanian et al., 2011), primaquine phosphate (Camarda et al., 2019) and amidoquine (Olliaro and Mussano, 2009). The anti-cancer drug mitoxantrone (Shenkenberg and Von Hoff, 1986), and the glycogen synthase kinase GSK3 inhibitor SB216763 (Wagman et al., 2005). In addition, we monitored the diffusion of the diagnostic staining agent fluorescein disodium salt (Jampol and Cunha-Vaz, 1984), the fluorogenic substrate CCF2 (Zuverink and Barbieri, 2015) and boron dipyrromethene (BODIPY) analogues, which are used as fluorescent markers for lipids, membranes, and other lipophilic compounds (Poryvai et al., 2022).

The diffusion coefficients and their cellular distributions showed very slow diffusion of the basic compounds, i.e., low FRAP recovery, combined with an accumulation of the molecules in lysosomes. Surprisingly, while inhibiting the V-ATPase H^+ pump using Bafilomycin A1 (Spugnini et al., 2010; Wang et al., 2021) or sodium azide (Hiruma et al., 2007) inhibited accumulation in lysosomes, cellular diffusion and fractional recovery increased only slightly. Contrary, blocking protonatable amino groups by acetylation greatly enhanced diffusion and FRAP recovery of studied small molecules. As over half the small molecule drugs are basic, our findings should be considered in drug assessment and development, as slow diffusion and low FRAP recovery is directly related to reduced cellular activity. Contrary to small molecules, we show here that diffusion rates within the cytoplasm of HeLa cells of 16 different *E.coli* proteins with pK_a values of 4.5-8, are within the expected range, suggesting that hindered diffusion is mainly a problem of small molecules.

Results

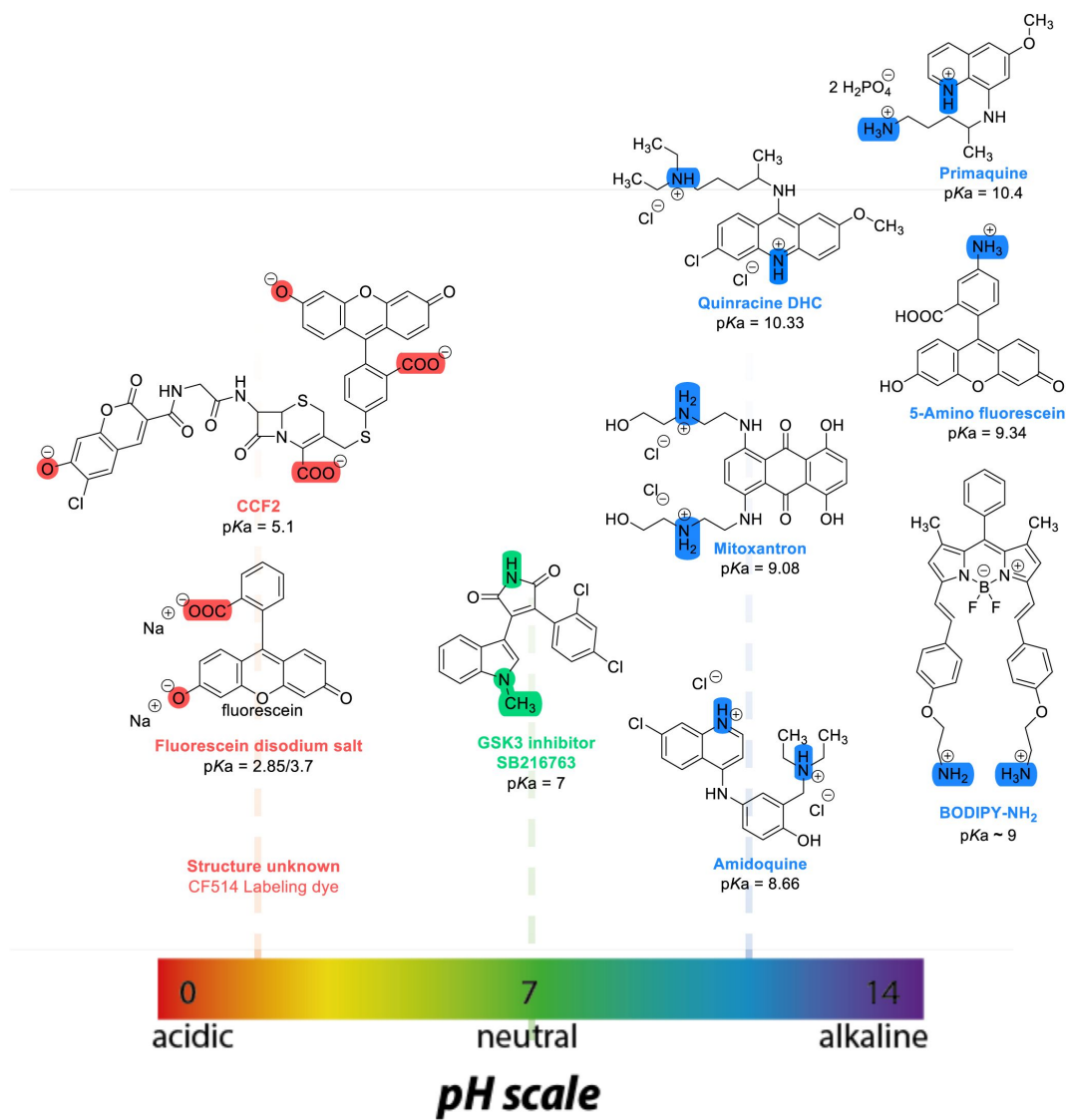


Figure 1.

Chemical structures and pKa values (prediction from ChemAxon software) of the small molecules and drugs used in this study.

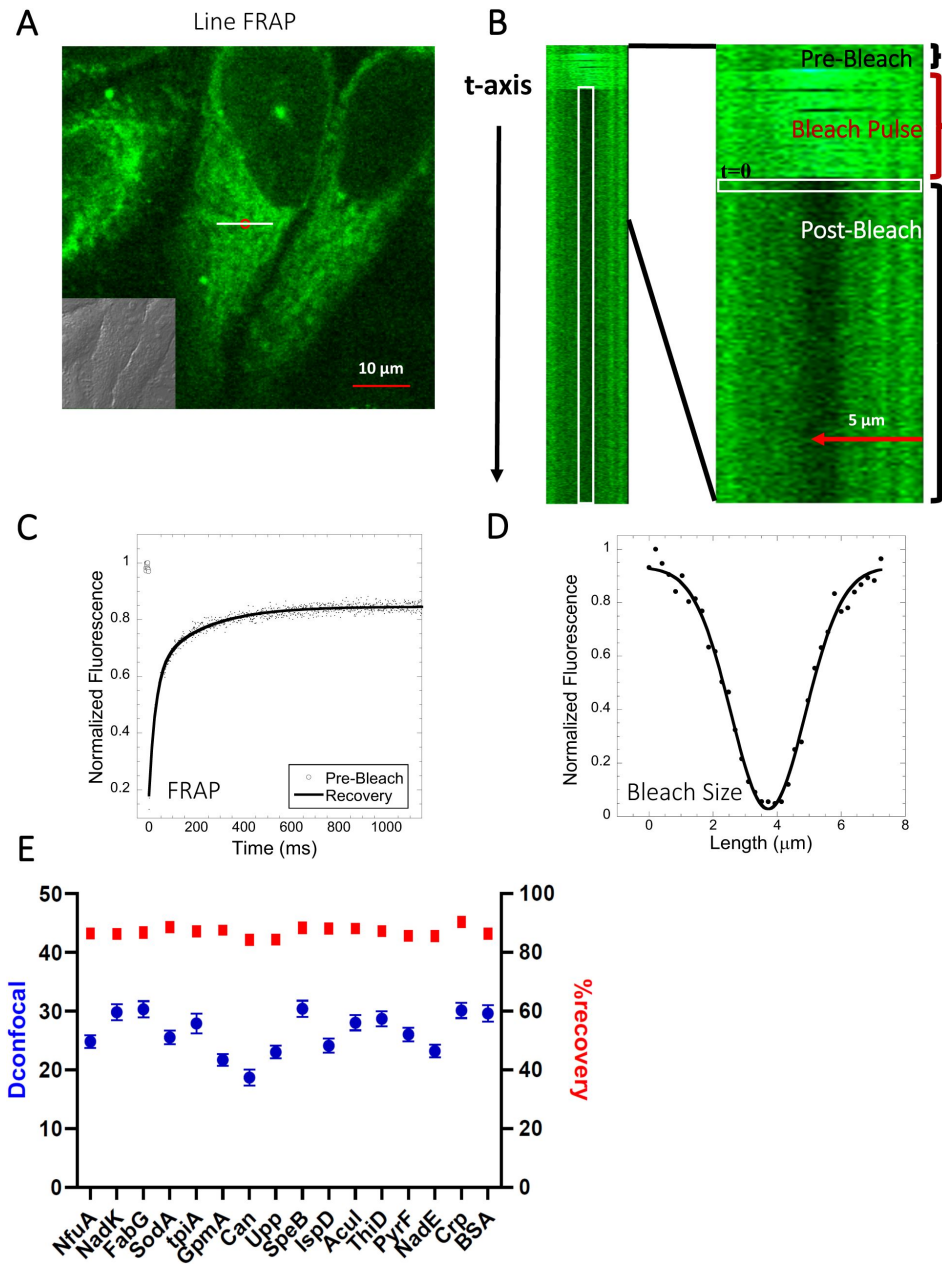


Figure 2.

Schematic representation of Line-FRAP & Diffusion coefficients of proteins in the HeLa cells.

(A) Line-FRAP performed in a fluorescent drug labeled HeLa cell. The white line represents the scanning line; The red circular area denotes the bleaching area. (B): A single scanning line as a function of time (in the vertical direction) including the bleach is shown. The white vertical rectangular region of interest (ROI), which was used for analysis. A close-up view of the Line-FRAP profile, is also shown where the horizontal rectangular area (white colored marks ROI) denotes the fluorescence intensity as a function of length across the scanning line. (C) Average of 30 FRAP recovery curves as a function of time ($N=30$; Correlation coefficient of exponential fit: $R=0.98$) and (D) Averaged bleach size profile with gaussian fits ($N=30$; $R=0.99$) are shown. (E) Comparison of diffusion coefficients (D_{confocal} in blue circles) and percentage of recoveries (in red squares) of bacterial proteins and BSA as measured in HeLa cell cytoplasm are shown. Error bars represent SE calculated from fitting the FRAP progression curves, which are averaged over at least 30 independent measurements.

Therapeutic Indication	Drug/Small Molecules	Calculated LogP	Calculated pKa (Strongest basic)	Lysosomal sequestration
Anti-cancer	mitoxantrone	1.19	9.08	+
Anti-Malarial	quinacrine	5.15	10.33	+
	primaquine-biphosphate	1.64	10.4	+
	amodiaquine	4.83	8.66	+
	5-Amino fluorescein	3.02	9.34	+
Other Small molecules	fluorescein-Sodium Salt	2.65	2.85/3.7	-
	GSK3 inhibitor (SB216763)	4.5	7	+
	BODIPY-NH ₂	—	9.00	—
	BODIPY-NHAc	—	—	—
	CCF2	4.5	5.1	—
	CF514	—	~3.0 (from UV spectrum)	—

Values of pKa were estimated using ChemAxon. Lysosomal sequestration was taken from the in-cell micrographs in this work.

Table 1

LogP, pKa and lysosomal sequestration of drugs used in this study (Predicted by ChemAxon software, and obtained from drug bank data base (Zhitomirsky and Assaraf, 2015 [\[15\]](#)))

Acidic Proteins diffuse freely in the HeLa cytoplasm

We have previously shown that diffusion in the cellular cytoplasm of E-fTs and baeR, two bacterial globular proteins, aligns with the expected values considering the ~2-3-fold higher macroviscosity (Dey et al., 2021 [DOI](#)). Here, we extended this study to include 16 additional proteins (mostly from *E. coli*) whose oligomeric states in solution were previously characterized (Table S1 and (Marciano et al., 2022 [DOI](#))). The aim was to determine how different heterologous proteins (*E. coli* proteins in HeLa cells) diffuse. The proteins showed D_{confocal} values of $18\text{--}30\ \mu\text{m}^2\text{s}^{-1}$ in HeLa cell cytoplasm, which is typical for a cytoplasmic protein diffusion coefficient (Figure 2E [DOI](#)). Moreover, full FRAP recovery was observed, indicating free diffusion. Table S1 provides the predicted isoelectric points and net charges at pH 7.4 of the purified proteins as calculated by ProtPi, showing pKa values of 4.5-8, which are representative of the majority of *E. coli* cytosolic proteins. The results show that acidic and neutral proteins, even if not in their native environment, do not stick to the HeLa cell cytoplasm or membranes.

Negatively charged small molecules rapidly diffuse in live cells

Fluorescein is a negatively charged small organic molecule (pK_a 's = 2.85, 3.7, M.W. = 332 Da, Figure 1 [DOI](#) and Table 1 [DOI](#)). We showed in previous work that fluorescein diffusion is slower in the presence of crowder proteins like BSA, lysozyme and, myoglobin, even at relatively low protein concentrations (Dey et al., 2022 [DOI](#)). Here, the Fluorescein dye was micro-injected in live HeLa cells, followed by FRAP measurements (Figure 3 A-D [DOI](#)). We found that the diffusion coefficient of fluorescein in PBS (D_{confocal} of $56.5 \pm 2.4\ \mu\text{m}^2\text{s}^{-1}$) was reduced to D_{confocal} of $38.5 \pm 2.0\ \mu\text{m}^2\text{s}^{-1}$ in HeLa cytoplasm (Figure 3C [DOI](#)). In the cell nucleus, a value of D_{confocal} of $43.2 \pm 2.2\ \mu\text{m}^2\text{s}^{-1}$ was determined, while in concentrated HeLa cell extract (equivalent protein concentration of ~100 mg/mL), D_{confocal} of $49.6 \pm 1.9\ \mu\text{m}^2\text{s}^{-1}$ was measured. The faster D_{confocal} in cell extract compared to live-cell cytoplasm may result from the higher crowding density of the latter. The lowering of diffusion coefficients of negatively charged fluorescein correlates with the increasing crowding density of the medium moving from simple buffer to cells. For comparison, we also incubated HeLa cells for 24 hrs with fluorescein, which is sufficient time for it to accumulate in the cells by diffusion (Figure S1 A-B). While micrographs did not show dye aggregation inside the cell cytoplasm (Figure S1A-B) and diffusion rates are similar to those determined by microinjection (Figure S1C and S1D), the fractional recovery dropped from 0.95 to 0.65. This suggests that a fraction of fluorescein is immobile in the cell after 24 h incubation, and is not diffusing.

For calculating the D_{confocal} values shown in Figure 3C [DOI](#), 30 individual Line-FRAP curves were merged for signal to noise reduction (Figure 3A [DOI](#)) (Dey et al., 2022 [DOI](#), 2021 [DOI](#)). For comparison, we calculated D_{confocal} values from individual Line FRAP progression curves in HeLa cytoplasm (Figure S2A and S2B), obtaining the same mean value as for the merged curve (Figure S2B and S2D), however, with much larger variation over the mean. To estimate whether the large variation from the mean of the individual curves is due to different fluorescein concentrations in the individual measurements, we plotted the fluorescence signal (which is directly related to the concentration) versus the FRAP recovery rate (Figure S3A) and versus the percent recovery (Figure S3E). The lack of correlation suggests that the variation in FRAP rates of the individual measurements is due to the low signal to noise of the individual curves versus the average. Next, we calculated D_{confocal} also for fluorescein in PBS, nucleus and HeLa cell extract (Figure 3B [DOI](#)). The mean values from individual curves are very similar to those calculated from the merged curves (Figure 3C [DOI](#)), with a normal distribution of the individual calculated D_{confocal} values around the mean. Moreover, also for the values measured in the nucleus, no relation between fluorescein concentration and FRAP rates or percent recovery was observed (Figure S3B and F). Overall, this suggest that merged curves can be used to calculate D_{confocal} values, as they give much better signal to noise, with apparent minimal loss of information.

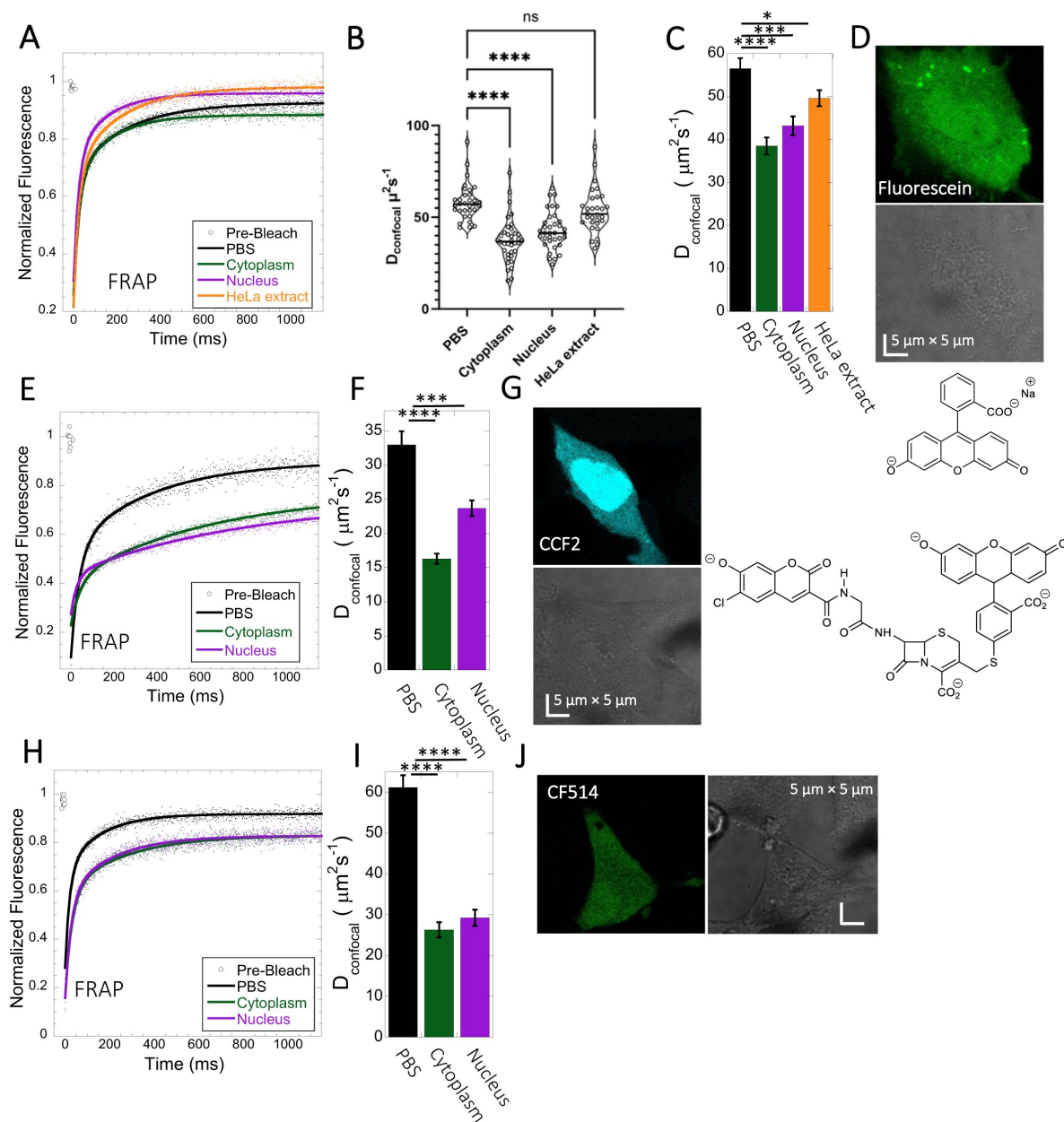


Figure 3.

Fluorescein, CCF2 and CF514 diffusion in PBS and inside HeLa cells.

Comparative merged FRAP profiles ($N = 30$; $R = 0.99$ for each of the fits) in PBS and inside HeLa cells with exponential fits are shown for (A) Fluorescein, (E) CCF2 and (H) CF514. Comparative D_{confocal} values for (C) Fluorescein (F) CCF2 and (I) CF514 are also shown. (B) D_{confocal} values as calculated from individual FRAP curves for Fluorescein (see also Fig. S2). D_{confocal} values as calculated from the merged FRAP curves shown in panel A. (D) HeLa cells after Micro-injection of Fluorescein; (G) CCF2; and (J) for CF514. Error bars represent SE calculated from fitting the FRAP progression curves of 30 merged independent measurements. Statistical significance calculations are detailed in the materials and methods section.

CCF2 is a fluorogenic substrate used for monitoring β -lactamase activity, with a pK_a of ~ 5.1 (Zlokarnik et al., 1998 [\[1\]](#)). It is composed of cephalosporin core linking B7-hydroxycoumarin to fluorescein (**Figure 1** [\[1\]](#)). Its calculated D_{confocal} values in PBS and inside cells are about half as fast as those measured for fluorescein (**Figure 3E-G** [\[1\]](#)), and its fractional recovery is ~ 0.7 . No sign of aggregation is seen in micrographs inside the cell cytoplasm or nucleus (**Figure 3G** [\[1\]](#)). The substantial slower rate of recovery, and lower fractional recovery of CCF2 relative to Fluorescein suggests its diffusion to be obstructed within the cell. This is in line with our prediction, that the observed substrate limited enzymatic degradation of CCF2 in HeLa cells is a consequence of its occlusion within the cellular cytoplasm (Zotter et al., 2017).

A third molecule for which we determined the diffusion coefficient is the labelling dye CF514. It is used to label proteins and has high quantum yield. As its structure has not been published, we estimated the pK_a value from the pH dependent UV-spectrophotometry measurements. The sharp changes of the UV-spectrum at $pH \sim 3$ (Figure S3I-J), as well as the redshift confirms a pK_a value of ~ 3 . The FRAP curves look similar to fluorescein (**Figure 3H-J** [\[1\]](#)), with $\sim 90\%$ recovery after FRAP. D_{confocal} in PBS was $\sim 61.1 \pm 3.0 \mu\text{m}^2\text{s}^{-1}$, and 26.3 ± 1.9 and $29.2 \pm 2.0 \mu\text{m}^2\text{s}^{-1}$ in the cell cytoplasm and nucleus of HeLa cells, respectively (**Figure 3I** [\[1\]](#)). The drop of D_{confocal} is attributed to cellular crowding. The results so far show that acidic small molecules diffuse well inside cells. Next, we focus on basic small molecules with higher pK_a values.

High FRAP recovery but slow diffusion measured for the glycogen synthase kinase (GSK3) inhibitor SB216763

The GSK3 inhibitor SB216763 ($pK_a=7.0$, M.W.=371 Da, **Figure 1** [\[1\]](#)) aggregates when dissolved in PBS buffer solution. However, in our previous study we have seen that aggregation of this particular small molecule drug can be reversed by adding BSA and the extent of de-aggregation depends on the amount of BSA added (Dey et al., 2022 [\[2\]](#)). Here, our aim is to compare diffusion behavior of the GSK3 inhibitor in solution phase to that found in live cell cytoplasm. Since the molecule has some inherent solubility issues in PBS buffer, we compared its diffusion also in BSA protein solution and in cell culture media DMEM (Dulbecco's modified eagle medium). Comparative FRAP profiles and D_{confocal} rates of SB216763 in different mediums are shown in **Figure 4A-C** [\[1\]](#). The FRAP profiles and relative D_{confocal} values show very slow diffusion of SB216763 in HeLa cell cytoplasm of $\sim 0.61 \pm 0.03 \mu\text{m}^2\text{s}^{-1}$ with a fraction recovery of 0.8 after 10s (**Figure 4A-C** [\[1\]](#)). This is a 40-fold reduction in D_{confocal} compared to measurements in 100 mg/mL BSA protein solution, where SB216763 is not aggregating ($D_{\text{confocal}} \sim 24.3 \pm 0.5 \mu\text{m}^2\text{s}^{-1}$ and fractional recovery of ~ 0.9 - **Figure 4A-C** [\[1\]](#)). In cell culture DMEM media (containing 5% serum protein) fractional recovery after FRAP is very low (**Figure 4A** [\[1\]](#)), indicating that the drug is still aggregating (Dey et al., 2022 [\[2\]](#)). Interestingly, the D_{confocal} value in HeLa cell extract was higher $\sim 11.3 \pm 0.7 \mu\text{m}^2\text{s}^{-1}$ as is its fractional recovery (0.9), showing that HeLa cell extract, like BSA is disaggregating SB216763. Co-localization with Lysotracker-633 (**Figure 4D-E** [\[1\]](#)) shows that SB216763 colocalizes and is sequestered in acidic chambers of the cell lysosomes. Moreover, SB216763 accumulation into the lysosome increased over time (**Figure 4D and E** [\[1\]](#), 45-minute and 2-hour treatment). Varying drug dosages resulted in D_{confocal} values of 0.42 - $0.61 \mu\text{m}^2\text{s}^{-1}$, suggesting that the slow diffusion is not related to SB216763 concentration (Figure S4A-B). To validate D_{confocal} values determined from merged progression curves, we calculated them also using individual Line FRAP profiles (**Figure 4C** [\[1\]](#) vs 4B), with both methods showing the same mean values. Individual FRAP progression curves are shown in Figure S2E, which have a much lower signal to noise ratio in HeLa cell cytoplasm, but result in a similar mean D_{confocal} value as determined from the merged curves (**Figure 4A-C** [\[1\]](#) and S2E-H). There may be a slight decrease in FRAP rates and percent recovery at higher SB216763 concentrations as determined from the individual profiles (Figure S3C and G).

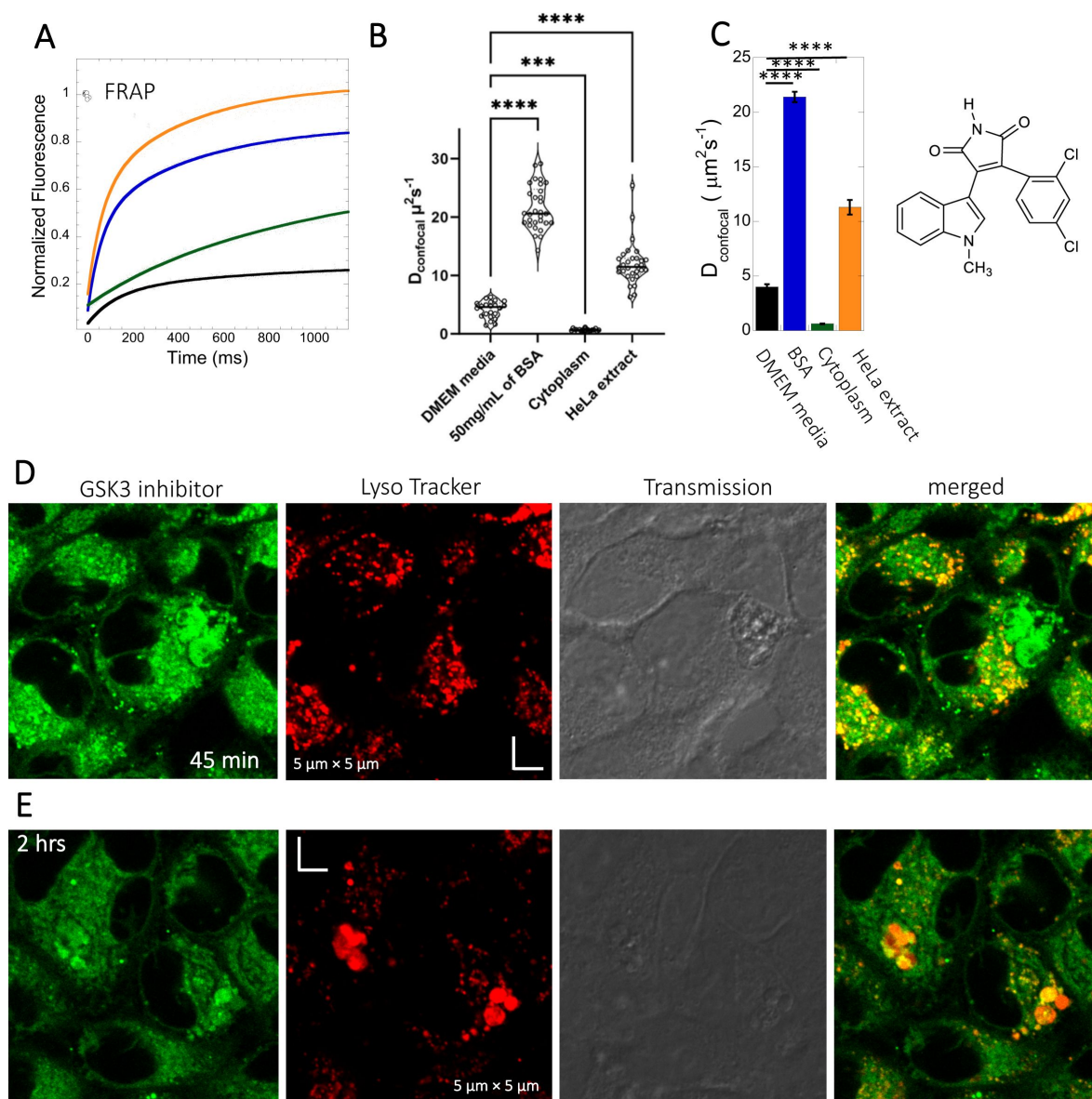


Figure 4.

Diffusion of GSK3 inhibitor in PBS and inside HeLa cells.

Comparative (A) averaged FRAP recovery profiles with exponential fits for GSK3 inhibitor (SB216763) (N= 30; R= 0.99 for each of the fits). Dots show prebleach fluorescence, Black line for DMEM, blue for PBS buffer + 50 mg/mL of BSA, green for HeLa cells and orange for HeLa extract. (B) D_{confocal} values calculated from individual FRAP curves. (C) D_{confocal} values calculated from merger of 30 independent FRAP curves. (D-E) Colocalization of GSK3 inhibitor in lysosomal compartments. HeLa cells were treated with Lyso Tracker 633 for 30 mins. GSK3 inhibitor treatment for (D) 45 mins and (E) 2 hours respectively at 10 μM concentrations are also shown. Error bars represent SE calculated from fitting the FRAP progression curves, which are averaged over at least 30 independent measurements. Statistical significance calculations are detailed in the materials and methods section.

Despite its high solubility in aqueous solution, Quinacrine diffusion in HeLa cells is slow

Quinacrine DHC (M.W.= 472.9 Da, **Figure 1**) is a compound highly soluble in PBS (Dey et al., 2022) with a fast D_{confocal} rate of $41.9 \pm 3.0 \mu\text{m}^2\text{s}^{-1}$ (**Figure 5A-C**). This rate dropped to $D_{\text{confocal}} \sim 2.5 \pm 0.1 \mu\text{m}^2\text{s}^{-1}$ in HeLa cell cytoplasm, with fraction recovery of ~ 0.8 , in comparison to one in buffer. In HeLa cell extract a D_{confocal} rate of $15.1 \pm 0.6 \mu\text{m}^2\text{s}^{-1}$ was determined, which lays between PBS and HeLa cytoplasm. D_{confocal} estimated from individual Line FRAP profiles and the merged Line FRAP profile (**Figure 5B** vs 5C) are also shown. Figure S2I-L shows that the merged FRAP profile yields less noisy data and gives a better fit, yet the mean D_{confocal} values are the same. As with SB216763, lower fraction recovery is observed at high quinacrine concentrations, but the FRAP rates seem to be concentration independent (Figure S3D and H). Micrographs of live HeLa cells show that the quinacrine DHC molecules aggregate in the cytoplasm. Co-localization with LysoTracker-633 (**Figure 5D-E**) show that quinacrine colocalizes and is sequestered in acidic chambers of the cell lysosomes. Moreover, quinacrine aggregation into the lysosome increased over time (**Figure 5D and E**, 45-minute and 2 hours treatment). Therefore, we measured diffusion instantaneously following micro-injection, and compared it to 2 and 24 hours of incubation (Figure S5A-B). Quinacrine accumulation in the lysosome was observed also immediately after micro-injection, with aggregation increasing over time. D_{confocal} of $4.2 \pm 0.2 \mu\text{m}^2\text{s}^{-1}$ was calculated from line-FRAP immediately after micro-injection, slowing to $2.2 \pm 0.1 \mu\text{m}^2\text{s}^{-1}$ following 2 hours incubations, with fractional recoveries of 0.63 and 0.57 respectively. After 24 hours of incubation the fractional recovery is less than 0.25, and thus the diffusion value is meaningless (Figure S5C and D). Next, we compared FRAP of the cells treated with 2 and 6 μM quinacrine solutions after 2- and 24-hours incubations (Figure S6A-F). The concentration of quinacrine had only a marginal effect on the results. Again, following 24 hours of incubation, most of the quinacrine is in the lysosomes.

Mitoxantrone, primaquine and amidoquine all aggregate within cells

Mitoxantrone (anti-cancer) and primaquine phosphate (antimalarial) fluorescence with low quantum yields and weak photo-stability. However, due to their slow diffusion, the temporal resolution of standard XY-FRAP was sufficient for FRAP measurements (**Figure 6A-H**). **Figures 6A** and **6E** show very low fraction recovery for these two drugs, suggesting that these weakly basic drugs are also sequestered in the cellular cytoplasm, as indeed shown in **Figures 6B** and **6F**. Colocalization with LysoTracker red-633 show that Primaquine (S7A-B) and Amidoquine (S7C-D) are sequestered in the lysosomes. For amidoquine unfortunately, we could not collect any FRAP profiles as the dye did not survive the photobleached pulse. While we selected bleach spots to be small and located outside of lysosomes, this does not assure that some of the bleached area does not include smaller lysosomes. Therefore we investigated whether inhibiting lysosomal trapping will eliminate slow diffusion of cationic drugs.

Inhibition of lysosomal sequestration is only slightly increasing diffusion in cells

Lysosome internal pH of ~ 4.5 is maintained by vacuolar proton ATPase (V-H⁺-ATPase) (de Duve et al., 1974; Kaufmann and Krise, 2007). The low pKa of weakly basic small molecule drugs drive their accumulation in the lysosome. There, they become protonated (ionic form is less-permeable), which hinders their back-diffusion to the cytosol. The phenomenon is known as cationic ion trapping (Asokan and Cho, 2002; Kaufmann and Krise, 2007) and results in drug accumulation in lysosomes up to millimolar concentrations (de Duve et al., 1974). We blocked lysosomal

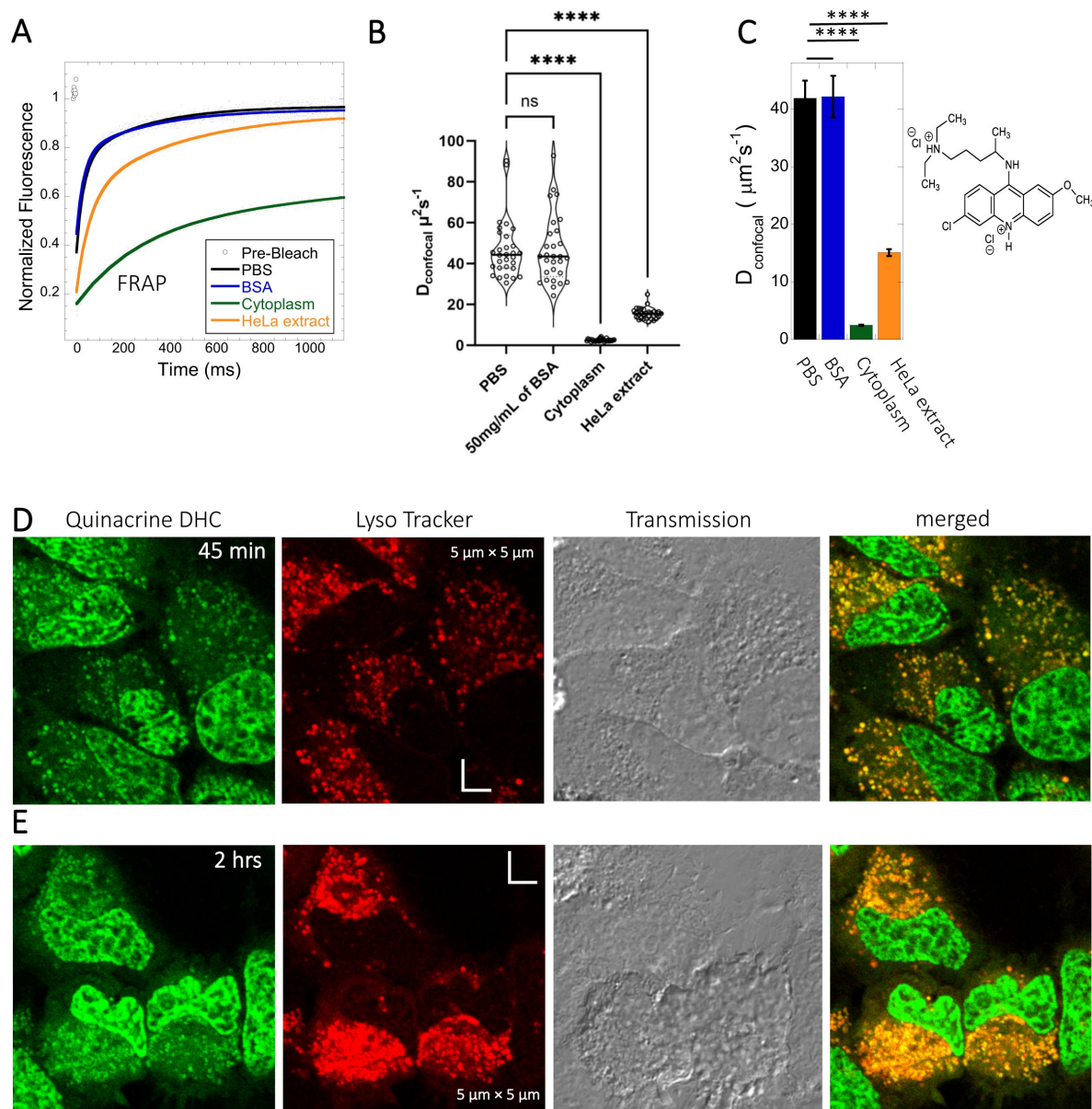


Figure 5.

Diffusion of Quinacrine in PBS and in HeLa cells.

Comparative (A) averaged FRAP recovery profiles with exponential fits for Quinacrine dihydrochloride ($N = 30$; $R = 0.99$ for each of the fits), (B) D_{confocal} values calculated from individual FRAP curves. (C) D_{confocal} values calculated from merger of 30 independent FRAP curves measured in PBS buffer, 50mg/mL of BSA, HeLa cell extract and in live HeLa cells. (D-E) Colocalization of Quinacrine DHC in lysosomal compartments are shown. HeLa cells were treated with Lyso Tracker 633 for 30 mins. Quinacrine DHC treatment was done for (D) 45 mins and (E) 2 hours respectively at 10 μM concentration. Error bars represent SE calculated from fitting the FRAP progression curves, which are averaged over at least 30 independent measurements. Statistical significance calculations are detailed in the materials and methods section.

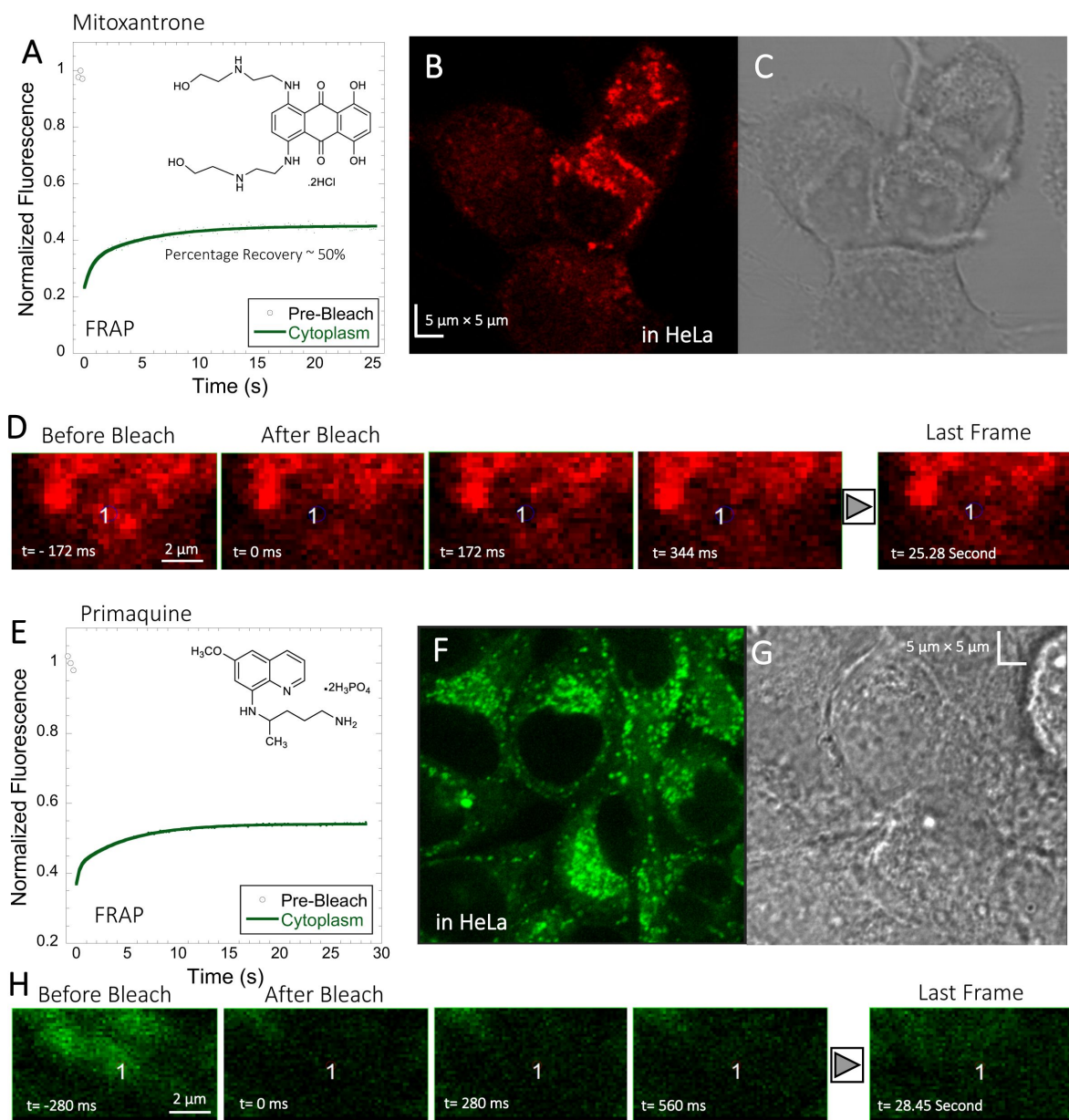


Figure 6.

Mitoxantrone and Primaquine diffusion inside HeLa cells.

Averaged XY-FRAP recovery profiles with exponential fits ($N=20$; $R=0.98$ for each of the fits) for (A) Mitoxantrone and (E) Primaquine in HeLa cells. Fluorescent and Transmission channel images of HeLa cells after 45 mins incubations with (B-C) Mitoxantrone and (F-G) Primaquine are shown. Time lapse micrographs of a portion of HeLa cell going through a classical rectangular XY-FRAP protocol for (D) Mitoxantrone and, (H) Primaquine are also shown.

acidification using either Bafilomycin A1 or sodium azide prior to drug administration. Bafilomycin A1 is a specific V-ATPase inhibitor and thus inhibits the supply of protons to the lysosomal chamber (Wang et al., 2021 [DOI](#)). Sodium azide, blocks ATP generation, resulting in the disruption of cellular activities which require ATP (Hiruma et al., 2007; Ishii et al., 2014 [DOI](#)). As a result, cationic ion trapping is inhibited. These two compounds were used to investigate whether lysosomal trapping is the cause for the 20-40-fold lower diffusion and lower fraction recovery in HeLa cells compared to buffer. Figures S8A and B, and Figure S9A and B show colocalization images for GSK3 inhibitor and quinacrine, respectively (with LysoTracker-633 used as marker) after addition of 100 nM Bafilomycin A1 (Yoshimori et al., 1991 [DOI](#)). The micrographs show that Bafilomycin A1 strongly reduces lysosomal volumes following quinacrine and GSK3 treatment (Figure S8A vs S8B and S9A vs S9B). However, D_{confocal} values were not altered much by Bafilomycin A1, as shown in Figures S8C-D and S9C-D for GSK3 inhibitor and Quinacrine, respectively. D_{confocal} for the GSK3 inhibitor is $\sim 0.60 \pm 0.02 \mu\text{m}^2\text{s}^{-1}$, and for Quinacrine $\sim 2.4 \pm 0.1 \mu\text{m}^2\text{s}^{-1}$. This suggests that Lysosomal accumulation of these two drugs is not the reason for the slow diffusion measured in the HeLa cell cytoplasm. It is notable that the D_{confocal} for Quinacrine remained consistent regardless of Bafilomycin treatment, 2 hours after incubation (Fig. S9D, $2.4 \pm 0.1 \mu\text{m}^2\text{s}^{-1}$). However, when measured immediately after injection, the diffusion coefficient was higher at $4.2 \pm 0.2 \mu\text{m}^2\text{s}^{-1}$ (Fig. S5D). This result does not support the notion that the faster diffusion measured immediately after cellular injection relates to lysosomal aggregation, and would better support self-aggregation, or aggregation with other molecules in the cell, which increases over time. This notion is further supported by the almost complete lack in FRAP observed 24 hours after injection (Fig. S5C).

To verify the Bafilomycin results, HeLa cells were pre-treated with Sodium azide at 1 and 50 mM concentrations (Figure 7A and E [DOI](#)), followed by treatment with GSK3 inhibitor or Quinacrine (Poste and Papahadjopoulos, 1976 [DOI](#)). The micrographs in Figures 7A [DOI](#) and 7E [DOI](#), show that sodium azide inhibited lysosomal accumulation of these drugs. Comparative FRAP profiles and diffusion coefficients (Figure 7B-D [DOI](#) and 7F-H) were slow, but conversely to Bafilomycin, sodium azide treatment did cause a further reduction in rates from D_{confocal} $2.4 \pm 0.1 \mu\text{m}^2\text{s}^{-1}$ to $1.8 \pm 0.1 \mu\text{m}^2\text{s}^{-1}$ for quinacrine and from 0.6 to $0.45 \mu\text{m}^2\text{s}^{-1}$ for the GSK3 inhibitor (Figure 7G and C [DOI](#)). Both Bafilomycin and sodium azide treatments resulted in elimination of drug confinement in the lysosome, and the small difference in diffusion rates may be a result of the de-acidification of the lysosomes by sodium azide, which may increase the protons in the cytosol upon treatment.

5-amino fluorescein diffuses slowly in HeLa cells and is accumulated within lysosomes

To experimentally test the relation between the pK_a of small molecules and their in-cell diffusion, we measured the diffusion of 5-amino fluorescein (AM-Fluorescein, M.W. = 347.3 Da, Figure 8A [DOI](#)), which is structurally similar to fluorescein disodium salt, but with an additional $-\text{NH}_2$ group that increases its pK_a from ~ 3 to ~ 9 (Figure 8A [DOI](#)). Figures 8B and C [DOI](#) show comparative FRAP traces for these molecules, and for CCF2. While fluorescein disodium salt has a fast FRAP rate, and is completely recovering, the cationic charged AM-fluorescein recovers much slower with a fractional recovery of 0.4. Moreover, while fluorescein disodium salt is not sequestered in lysosomes, 5-Amino fluorescein is. This sequestration is alleviated by adding sodium azide to the cells (Figure 8D [DOI](#)) but has only a small effect on the FRAP rate and fractional recovery (Figure 8B [DOI](#)). This experiment strongly suggests that the functional primary amine group is the main contributor towards the slow diffusion and low FRAP recovery of AM-fluorescein in the cell cytoplasm.

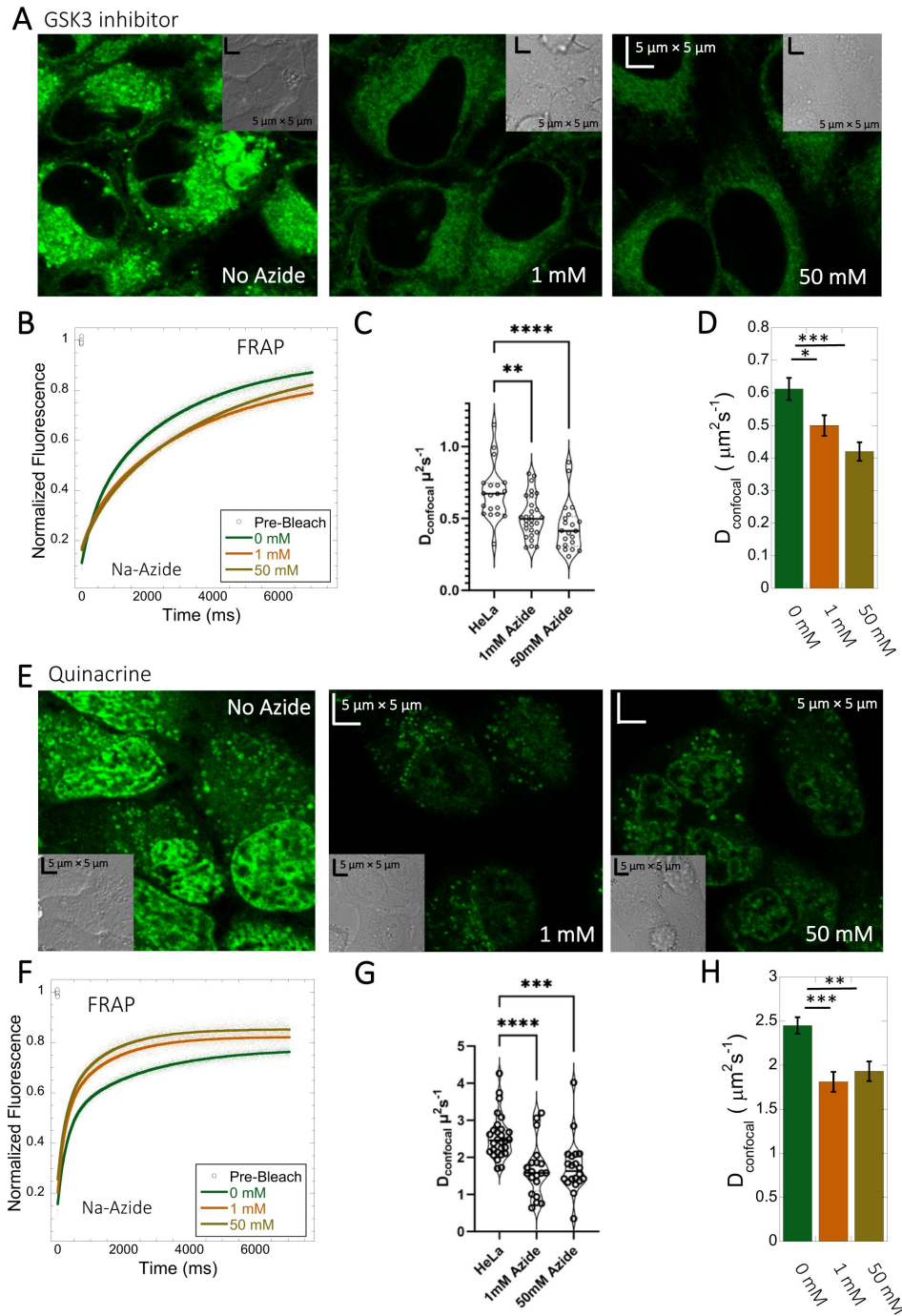


Figure 7.

Effect of Sodium Azide on the diffusion of GSK3 inhibitor and Quinacrine inside HeLa cells.

(A) GSK3 inhibitor and (E) Quinacrine dihydrochloride treated HeLa cells with or without sodium azide. Comparison of merged FRAP profiles with exponential fits ($N=30$; $R=0.99$ for each of the fits) and calculated D_{confocal} values from individual fits and combined averaged fit for (B-D) GSK3 inhibitor and (F-H) Quinacrine dihydrochloride are shown. D_{confocal} estimations from individual FRAP curves are shown in (C,G). Error bars represent SE calculated from fitting the FRAP progression curves, which are averaged over at least 30 independent measurements. Statistical significance calculations are detailed in the materials and methods section.

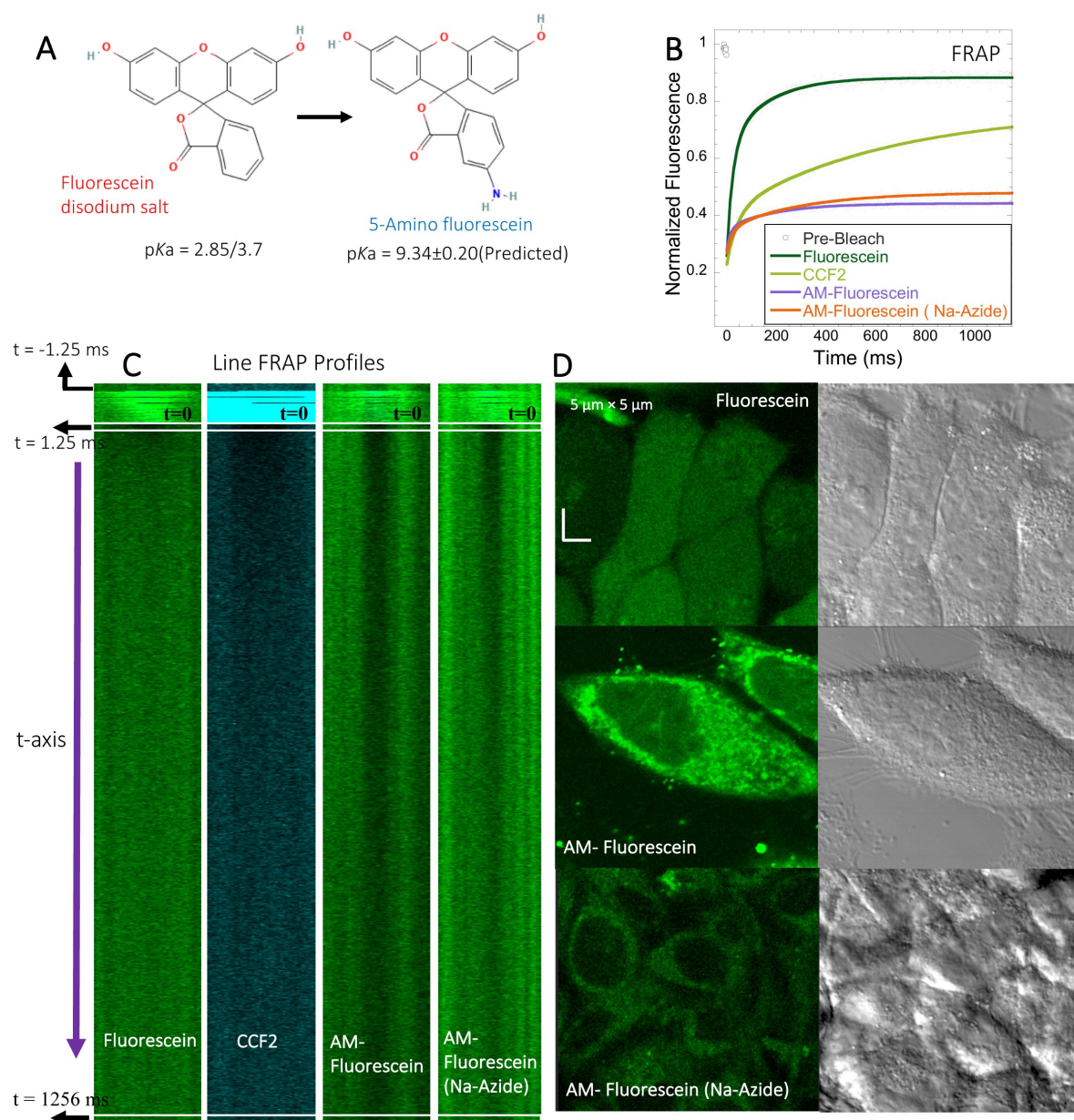


Figure 8.

FRAP of Fluorescein analogues inside HeLa cells.

(A) Structures of Fluorescein analogues with pKa values. (B) FRAP recoveries with exponential fits for Fluorescein, CCF2 and 5-amino Fluorescein, with/without Na-Azide treatments in HeLa cells (N= 30; R= 0.99 for each of the fits). (C) Line FRAP profiles with time lapses. Micrograph images of treated HeLa cells after small molecule incubations.

Acidification of primaquine and BODIPY increases their diffusion and FRAP recovery

We substituted the NH_2 of primaquine (antimalarial drug) and BODIPY compounds for the charge-neutral-NHAc functional group (Figure S10A and S11A). Due to their low quantum yield, we could not implement the faster Line-FRAP protocol, therefore, we used standard FRAP, which suits the slow diffusion of both compounds. As seen from the FRAP profiles (time-dependent monographs) in Figures S10B-D for primaquine derivatives and in Figures S11B-C for BODIPY derivatives, the recovery is much slower for free amine-containing moieties. Figure S10I-J shows that the fractional recovery for primaquine is only ~ 0.27 , whereas, for primaquine-NHAc, the recovery is ~ 0.6 . Comparing the measurements using micro-injecting versus incubation of the cells for FRAP measurements for primaquine-NHAc gave similar results (Figure S10E-H). Unfortunately, the classical FRAP is not fast enough to calculate the diffusion coefficients, as the dead time was 150 ms. For the BODIPY analogues, the observations were similar, but to a smaller extent. For BODIPY- NH_2 , the recovery percentage is $\sim 57\%$, whereas, for BODIPY-NHAc, the recovery is $\sim 67\%$ (Figure S11H). The estimated recovery half-life also supports the faster diffusion of BODIPY-NHAc, as shown in Figure S11I.

Basic small molecules do not colocalize with lipid droplets, the ER or nucleic acids

To identify the location of colocalization of basic small molecules, we tested colocalization of GSK3 inhibitor in lipid droplets using the Nile red dye (Figure S12) and in the ER using a specific BFP/mCherry tagged fluorescent antibody markers (Figure S13). Next, we looked for colocalization of Mitoxantrone with nucleic acids present in cytoplasm using SYTO blue (Figure S14). No colocalization was found between the drugs and either lipid droplets, the ER or nucleic acids.

Super resolution images of GSK3 inhibitor treated HeLa cells were taken to improve the spatial distribution (Figure S15); which however, this did not contribute new information to the puzzle. Therefore, we could not pinpoint the location of sequestration of weakly basic small molecule drugs within the cellular cytoplasm, except the lysosome.

Discussion

In this study, we selected well-known and widely used small molecule drugs to study their behavior in aqueous solutions and living cells. All drug molecules used here are fluorescent, allowing for their FRAP measurements. Previously, we found that small-molecule drugs diffuse differently in crowded media than in a simple buffer solution, even if aggregation is not an issue (Dey et al., 2022 [DOI](#)). Here, we show that many small molecules get trapped within the cellular cytoplasm, resulting in extremely slow diffusion and lower fractional recovery after FRAP. This is particularly observed for cationic-charged small molecules, with D_{confocal} values of 20-40-fold lower than observed for fluorescein or CF514 (Figure 9A [DOI](#) vs 9B), which, in many cases is accompanied by low recovery after photobleaching. The latter suggests that most molecules are occluded to components of the HeLa cell cytoplasm. Even dense cell extract from HeLa cells is not mimicking the *in vivo* results (Figure 4A-C [DOI](#) and 5A-C). This is very different from protein diffusion, where none of the 16 proteins measured here showed slow diffusion or low fractional recovery, despite their *E.coli* origin, expressed in HeLa cells. Slow diffusion and lower fractional recovery were limited to basic small molecules, showing D_{confocal} values of $0.2\text{--}2\ \mu\text{m}^2\text{s}^{-1}$. While the fractional recovery for all proteins is ~ 0.9 , for many basic small molecules it is 0.2-0.5.

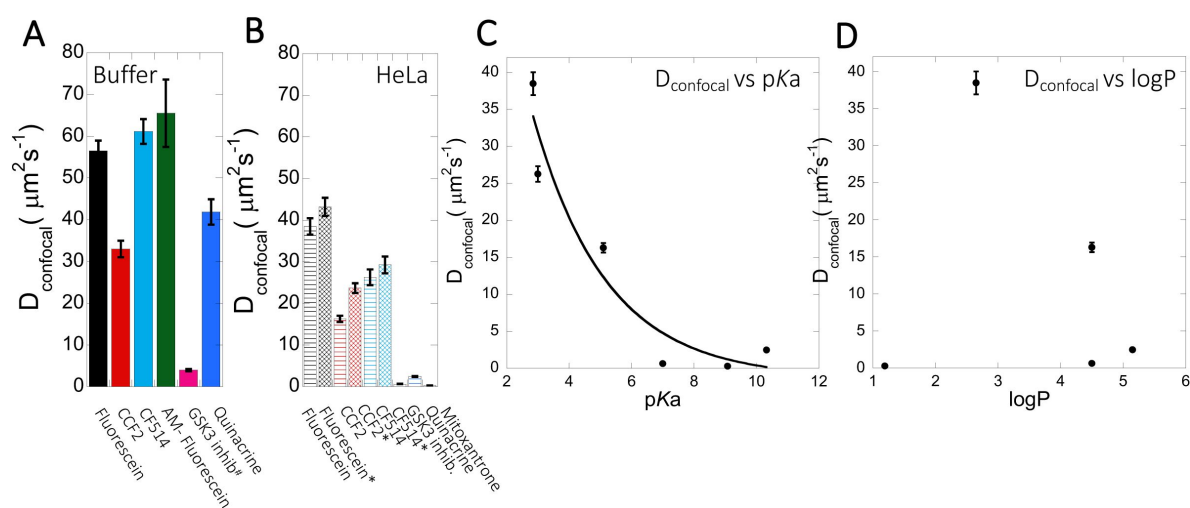


Figure 9.

D_{confocal} of small molecules in PBS buffer and inside HeLa cells.

Comparison of D_{confocal} values of small molecule drugs in (A) PBS buffer and in (B) HeLa cells. Error bars represent SE calculated from fitting the FRAP progression curves, which are averaged over at least 30 independent measurements. Dependency of D_{confocal} values on (C) pKa and (D) logP values in HeLa cells. # in (A) denotes DMEM media (instead of PBS), * in (B) denotes measurements done in the nucleus.

As primary culprit for the slow diffusion of the basic small molecule drugs we suspected sequestration in the lysosomes. Indeed, high level of accumulation in the lysosomes were found for all the basic drugs. However, pre-treatment with sodium azide or Bafilomycin A1, which we showed to inhibit lysosomal accumulation, had no or only a small effect on diffusion or fractional recovery after FRAP, which stayed low. Therefore, we evaluated colocalization of GSK3 inhibitor with lipid droplets, the ER or nucleic acids present in cytoplasm (Figure S12-14), but did not detect any.

The conclusive evidence that slow diffusion and lower FRAP recovery results from the drugs being basic came by altering three molecules from being basic to acidic and vice versa. The best example here is fluorescein, where transformation to 5-amino fluorescein drastically reduced its diffusion as well as its fractional recovery after FRAP. Two additional examples here were Primaquine that was modified to Primaquine Acetate and BODIPY-NH₂ that was modified to BODIPY-NHAc. In both cases, the diffusion became faster, and the fractional recovery increased significantly. We did not find any specific correlation between individual average diffusion rate constants with the concentration of small molecule drugs inside the cell (Figure S3A-D). For fraction recovery, we saw some decrease at high concentrations for the GSK3 inhibitor and quinacrine, but not fluorescein (Figure S3E-H), which would support our notion that they are sequestered in the cell. Correlating the calculated D_{confocal} values with pKa shows a clear trend, where diffusion is strongly slowed down at higher pKa values (Figure 9C [↗](#)). Conversely, logP, molecular weights or number of aromatic rings present in the drug molecules did not relate to D_{confocal} values (Figure 9D [↗](#) and S16A-B). This finding is important, as the activity of a drug within the cell is dictated by its active concentration. When the drug is sequestered, its active concentration is reduced. Even, if it is not fully sequestered, but only diffuses much slower, it will result in lower association rate constant, and thus lower affinity towards the drug target. As a result, to keep the drug active, its dose has to be increased, which can have negative implications on side effects, due to off-target binding.

Conclusions

Our study shows that *in vitro* biophysical crowding studies for small molecule drugs are of limited value if one wants to understand the biophysical behavior of the same drug within the cell. While it is true that availability of a drug within a cell is only one factor dictating its biological activity, it is a crucial one. Our findings here raise an important limitation on the standard rules for drug design, as these do not consider the stickiness of basic small molecules within the cell. We were able to directly address this question by using fluorescent molecule drugs, and measuring their location and diffusion within cells. Most importantly, we also show that by blocking protonation of a number of basic compounds we were able to increase substantially their diffusion and recovery after photobleaching, and thus potentially their activity. These findings may be consequential in future drug development.

Materials and methods

All the reagents used are described in a table format (Table S2, Key resources). The supplementary method sections describe the synthetic procedures, characterizations, and purity of the synthesized compounds.

Selection of drugs used in this study

To identify small molecule drugs that have fluorescence in the visible light, we screened in high-throughput 384 well plates 1600 commonly used, small molecule drug compounds for their penetration into HeLa cells and fluorescence at the DAPI (377/447), FITC (475/520), TexasRed

(560/624) and Cy5 (631/692) channels. As positive control, we used Pyrvinium (red), Doxorubicin (green) and CSB (blue). Out of those, 16 compounds gave positive results. These compounds were then verified for their photostability, availability and solubility. This resulted in the list of compounds shown in **Table 1** [↗](#), which includes also compounds used by us in previous studies.

Mammalian Cell Culture. HeLa cells were grown in 35-mm glass-bottomed dishes (MatTek Corporation) in DMEM (1X) Gibco (Life Technologies Limited) supplemented with 1X pyruvate, penicillin/streptomycin (BioIndustries), and 10% fetal bovine serum (Life Technologies Limited). The cells were subcultured when 80% confluence was reached using trypsin-EDTA for cell detachment. 2×10^5 HeLa cells in 2.5 ml of DMEM were pipetted into glass-bottomed dishes and incubated overnight. The cells were cultured in a humid atmosphere at 37°C and 5% CO₂. The cells were imaged 24–30 h after seeding. Before the microinjection, the medium was aspirated, and fresh medium was supplemented with 25 mM HEPES, pH 7.4. For drug treatment, HeLa cells were incubated with 10 µM of drug (diluted 1:1000 from stock) for indicated times at 37 °C, followed by three times PBS 1X wash prior to imaging. For dose-dependent studies, the drug concentrations varied typically between 2–12 µM. Sodium azide concentration was 1 and 50 µM. Bafilomycin A1 was used at 100 nM concentration. Similarly, during the colocalization study, HeLa cells were treated with LysoTracker dye (1X) with 1:1000 dilution for 30 mins to 1 h at 37 °C, followed by 3x washing with PBS. Quinacrine, GSK3 inhibitor, Mitoxantrone, 5-amino Fluorescein, Amidoquine, Primaquine and BODIPY analogues were incubated with cells, while Fluorescein, Quinacrine, Primaquine-Ac, and CF514 dye were also micro-injected inside the HeLa cells.

Microinjection into HeLa cells

Microinjections were performed using the Eppendorf FemtoJet microinjector attached to the Eppendorf InjectMan NI2 micromanipulator. The fluorescein sodium salt, CCF2, CF514, quinacrine, primaquine analogues, BODIPY analogues were dissolved in DMSO and highly concentrated stock aliquots were made. Diluted PBS solutions of small molecules were injected into cells using glass capillaries from Warner instruments and pulled by a vertical puller (Narishige). For every measurement, a single pressure pulse was applied to deliver the sample into the cell. Air was administrated at 15– 25 hPa for 0.1–0.3 s. For injections, single cells containing morphologically healthy and well-connected HeLa cells were selected. Before and after the microinjection, cell morphology and membrane integrity were confirmed by visually inspecting the injected cells.

HeLa cell extract preparation

Cytoplasmic HeLa cell extracts were prepared as described previously with slight modifications (Zotter et al., 2017). HeLa cells from 4×10 cm plates at 80% confluency were washed, treated with trypsin and collected. Pellets were mixed with 500µl of RIPA buffer, IMP40, 5 µL of protease inhibitor and stored in ice. After 20 min, the mixture was centrifuged at 13k×g. This process was repeated twice to obtain concentrated extract. A BSA calibration curve determined the total protein concentration; the final concentration was up to 100 mg/mL. After liquid nitrogen freezing, the final cell extract solution was stored at -80° C.

Protein purification and dye labeling

Bacterial proteins used in this study were purified as described by us previously (Marciano et al., 2022 [↗](#)). The dye labelling procedure with CF514 for purified proteins and removal of excess dye was described in detail (Dey et al., 2021 [↗](#)).

Confocal microscopy and FRAP Analysis

Confocal microscopy. Images were collected with an Olympus IX81 FluoView FV1000 Spectral/SIM Scanner confocal laser-scanning microscope, using 60X DIC oil-immersion objective, N.A. 1.35. For fluorescein sodium salt fluorescence measurements, excitation was done at 440 nm, using a diode

laser at an output power of 1-4 % of maximal intensity for high to low concentrations. In contrast, emission was recorded from 520 to 550 nm using the spectral detection system. For CCF2-FA, excitation was done with the laser at 440 nm using 1-2 % of the maximal intensity, while emission was collected from 470-560 nm with SDM560 emission dichromator as cut-off filter. For quinacrine DHC and GSK3 inhibitor, excitation was done at 488 nm laser using 1-2% of the maximal intensity, while emission was collected from 502-560 nm with SDM560 emission dichromator as cut-off filter. During colocalization experiments of quinacrine DHC and GSK3 inhibitor with LysoTracker-633 dye, the same setting was used for detecting the small molecule drugs in Channel 1 (green). LysoTracker-633 dye was detected in Channel 2 (red) by exciting at 635 nm laser using 1-2 % of the maximal intensity, while emission was collected from 655-755 nm. Colocalization for Mitoxantrone (red channel) was done by using quinacrine to track the Lysosomes (green channel). Percentage of volume and drug intensity (material) above threshold is defined as percent colocalized (Figure S16C-D). For mitoxantrone and BODIPY analogues, excitation was done at 635 nm laser using 1-2 % of the maximal power, and emission was collected from 655-755 nm in Channel 2 (red). In the blue channel, excitation was done by 400 nm laser using 2-4% of maximal power, whereas the emission was collected from 420-480 nm with SDM480 emission dichromator as cut-off filter. The same setting is used in the blue channel for the SYTO blue maker (to visualize free nucleic acids) and blue fluorescent protein marker (BFP to visualize the ER). For the mcherry protein marker (to visualize the ER), the same setting for the red channel (Channel 2) is used. For every colocalization experiments, proper blank experiments were performed separately in all the respective channels with the same microscope settings to avoid any bleeding/leakage of emission signals from one Channel to another Channel, as shown in Figure S17. Image analyses were performed using FluoView/Imaris software, and data analyses were performed using Kaleidagraph software version 4.1 (Synergy). Colocalization coefficients as a percentage of material/volume above threshold colocalized are calculated using Imaris software version 7.

Line-FRAP and Classical XY-FRAP

Line-FRAP was carried out in liquid drops and HeLa cells. For photobleaching, “Tornado” of 4 pixels (2x2, 1-pixel = 0.207 micrometer) as default laser settings by the Olympus Fluoview confocal system (Dey et al., 2021 [DOI](#)). It is the smallest area achievable using Tornado. The bleach circle was kept precisely in the middle of the scanning line. The lines we scanned were unidirectional with time intervals of 1.256 ms 1000 times (equivalent to 1.256 s) in the majority of the measurements. The number of scans prior to, during, and after the photobleaching was 10, 42, and 948, respectively. Only in some specific cellular measurements (for GSK3 inhibitor and Quinacrine DHC, where recovery is very slow) the unidirectional lines were scanned 5000 times. Photobleaching was achieved by using the laser at 405-nm or with 635-nm excitations for the 63-millisecond duration at full intensity (100%). The simultaneous scanner moved at a 100 μ s/pixel speed to perform an efficient photo-bleach. We have used two simultaneous scanners during the Line-FRAP experiments: one scanner (at 405 nm with the full intensity of 100%) for photobleaching and another scanner (at 440/515/635 nm with weak intensity) for data acquisition. For all the drugs except Mitoxantrone and BODIPY analogues (emission in red wavelengths), photo-bleach was performed by 405 nm laser. For fluorescence signal detection: fluorescein 440 nm laser (1-4%); CCF2 440 nm laser (1-2%); GSK3 inhibitor 440 nm laser (0-2%); quinacrine (1-2%) of maximal intensity were used. Emission collections were done from 520-550 nm for quinacrine DHC and GSK3 inhibitor.

Line FRAP was not possible for primaquine analogues, as it suffers from strong photo-bleach during recovery. Here classical XY FRAP was performed, with photo-bleach done by 405 nm laser with full intensity. The frame rate is maintained at 172 ms with a bleach pulse duration was 150 ms. Due to microscope setting limitations, Line FRAP is also not feasible for Mitoxantrone and BODIPY analogues. A classical XY-FRAP using two simultaneous scanners is done for these molecules. Here, bleach was performed by simultaneous 635 nm laser with the full intensity of 100%, and the main excitation was done by 635 nm laser (with the power of 4%). Emission collections were done from 655-755 nm. For BODIPY analogues FRAP recoveries, the bleach pulse

time was 150 ms. The bleach pulse time was also 150 ms for primaquine analogues for comparative FRAP profile collections. The frame rate is maintained at 172 ms. Using the Olympus IX81 FluoView FV1000 Spectral/SIM Scanner confocal laser-scanning microscope, using Tornado (which requires SIM scanner to be loaded) greatly enhances bleaching efficiency. In addition, it shortens the time to obtain the first measurement after bleach (which is immediate in this mode). This property is highly beneficial for Line-FRAP measurements, where the time scale of data acquisition plays an important role. The Fluoview SIM scanner unit synchronizes laser light simulation with confocal and multiphoton imaging to avoid interruption to image observation during laser stimulation or manipulation. We have varied the intensity of the lasers to achieve an optimal signal/noise ratio. Fluorescence recovery plots were fitted to a double exponent growth curve. FRAP experiments were also performed inside the PBS buffer drops and crowding conditions. Line FRAP was done on the cells taken from different plates, treated independently. A coverslip was introduced at the top of the drop to stop evaporation during measurements. Our previously developed method on Line FRAP protocol has been employed to calculate diffusion coefficients from the FRAP rates and averaged bleach sizes. Diffusion rates cannot be calculated using XY-FRAP. However, FRAP parameters were kept constant so that qualitatively we could compare the average half-life and percentage of recoveries. When the areas of bleach were selected in the drug-treated cell cytoplasm, we avoided the lysosomes as much as possible, within the resolution limits of the confocal microscope. Lysosomes themselves were measured to move within the cytoplasm with an diffusion coefficient of $0.03\text{--}0.071\ \mu\text{m}^2\ \text{s}^{-1}$ (Bandyopadhyay et al., 2014 [DOI](#)), which is much slower than the diffusion measured for even the slowest compounds using fast Line FRAP, further validating that we did not measure lysosome diffusion.

Determination of D_{confocal} from FRAP results

We previously developed (Dey et al., 2021 [DOI](#)) the FRAP in Line mode (Line FRAP) method to monitor the diffusion rates of proteins in various environments. Line-FRAP allows a much faster data acquisition rate compared to classical XY FRAP. It is more suitable for measuring diffusion rates for fast diffusing molecules. To calculate D_{confocal} , 20–30 independent measurements on different cells (n) were binned, and the curve fit of the progression curve was used to obtain $t_{1/2}$, and r_e values and their associated errors as determined from the fit of the curve. r_n was obtained from bleaching a fixed sample. The D_{confocal} values were then calculated using equation 1 [DOI](#) (see introduction and (Dey et al., 2021 [DOI](#))). The standard errors (SE) of the individual parameters were combined to obtain $D_{\text{confocal}} \pm \text{SE}$. The errors and means were used to calculate the significance level of the difference between two treatments, with ns for non-significant, * being for 0.05–0.01, ** being for 0.01–0.005, *** for 0.005–0.001 and **** <0.0001. D_{confocal} values from individual Line FRAP profiles were calculated the same way, and one-way ANOVA statistical tests were performed to calculate the significance between treatments using the standard function available in GraphPad Prism v10.1 using Dunnett for multiple comparisons. To verify SE values, we repeated the ~20–30 measurements independently multiple times, which gave the same SE as obtained from individual curve fits of $n = 30$ cells.

Additional information

Author Contributions

D.D. performed all the experiments. S.M. purified all the bacterial proteins. A.P, O.G., L.W., and T.S. designed the synthetic targets and synthesized primaquine acetate and BODIPY derivatives. D.D. and G.S. conceptualized the idea, analyzed the data and wrote the original draft of the manuscript. D.D., G.S., and T.S. reviewed and edited the paper. G.S. supervised the project. The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Funding Sources

The Israel Science Foundation grant number 1268/18 (GS). The Czech Science Foundation (project No. 22-20319S) for funding this project (T.S.)

Acknowledgements

We would like to thank Dr. Ori Avinoam from Department of Biomolecular Sciences, and Prof. Hagen Hofmann from Department of chemical & structural Biology, Weizmann Institute of Science for their valuable discussions and advices. We want to acknowledge Dr. Reinat Nevo for her help with Imaris software to analyze the microscopy images.

Abbreviations

- FRAP: fluorescence recovery after photo-bleaching
- CCF2: coumarin cephalosporin fluorescein
- BODIPY: boron dipyrromethene analogues
- NMR: Nuclear magnetic resonance.

Supplementary Information

Protein entry	Isoelectric point: pI	Net charge at pH 7.4: Z	D _{confocal}	Oligomeric state	Uniprot (kDa) Monomer
NfuA	4.534	-15.767	24.8±1.1	M/D	20.93
NadK	6.387	-4.808	29.8±1.4	M/D/T	32.57
FabG	6.250	-1.66	30.33±1.4	D/T/H	25.56
E-fts	5.228	-9.527	22.5±1.0	M/D	30.29
SodA	6.511	-2.481	25.5±1.2	D	22.97
tpiA	5.688	-7.851	27.9±1.7		
GpmA	5.913	-4.262	21.7±1.0	D	28.43
Can	6.230	-5.908	18.7±1.4	T	25.1
Upp	5.335	-6.829	23.0±1.1	M/D	22.53
SpeB	5.160	-14.788	30.4±1.4	Tri/H	33.56
IspD	6.257	-5.445	24.1±1.2	D/T/H	25.61
BaeR	5.554	-6.032	21.9±1.1	M/D	27.66
AcuI	5.669	-5.986	28.0±1.3	M/D/T/H	34.73
PyrF	5.874	-3.9	26.0±1.20	M/D	26.35
ThiD	5.793	-6.842	28.7±1.3	M/D	28.64
NadE	5.603	-4.993	23.2±1.1	D	27.16
CRP	7.979	+1.388	30.1±1.3	D/T	23.64
DES3im-4rp4	5.041	-5.659	29.4±1.4		
DES7im-3rk6	4.599	-10.654	28.8±1.3		
6E5C_denovobeta30	6.336	-0.714	30.0±1.4		
BSA	4.5–4.8	-15	29.6±1.4	M/D	66.5

Isoelectric points and net-charges were predicted using Prot Pi. Oligomeric states were taken from (Marciano et al., 2022) native mass spectrometry measurements.

Table S1

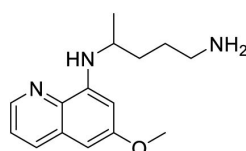
Oligomeric state, Isoelectric point and Net charge at pH 7.4

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, peptides, and recombinant proteins		
Dulbecco's phosphate-buffered saline (PBS 1X)	Biological Industries	Cat# 02-023-1A
HEPES buffer	Fisher BioReagents	LOT# 170358
DMEM(1X) Gibco	Life Technologies Limited	REF# 41965-039
Fetal Bovine Serum (Gibco)	Life Technologies Limited	REF# 12657-029
Trypsin-EDTA Solution A	Biological Industries	REF# 03-050-1B
Penicillin/ Streptomycin	Biological Industries	REF# 03-031-1B
Sodium Pyruvate Solution	Biological Industries	REF# 03-042-1B
35-mm glass-bottomed dishes	MatTek Corporation	P35G-0-14-C
Pierce™ Dye Removal Columns	Thermo-Fisher scientific	Cat# 22858
Gebaflex tubes 3.5KDa (GeBa)	TIVAN BIOTECH	Cat# MID13-100
Capillary Glass Tubing	Warner Instruments	Model No. G120TF-4 203-776-0664
96- well Black F-Bottom Plate	Greiner-bio-one	REF# 655076
Disposable cuvettes	Fisher Scientific	ZEN0040
BSA (Albumin Bovine, fraction V)	MP Biomedicals, LLC	CAS# 9048-46-8
HEWL (Lysozyme from chicken egg white)	Merck	CAS# 12650-88-3
Myoglobin (from equine heart)	Merck	CAS# 100684-32-0
Doxorubicin	AdooQ Bioscience	Cat# A14403
Fluorescein disodium salt	chemcruz	Cat# sc-206026
GSK3 inhibitor SB216763	Abcam	Cat# ab120202
Quinacrine dihydrochloride	Abcam	Cat# ab120749
Primaquine biphosphate	Merck	CAS# 63-45-6
CCF2-FA	Thermo-Fisher Scientific	Cat# K1039a
CF514 Labelling dye	Biotium	CF514 Dye Alternative green fluorescent dye
Amodiaquine	AdooQ Bioscience	Cat# A17660
Sodium Azide	Merck	CAS# 26628-22-8
Mitoxantrone	Abcam	Cat# ab141041
Bafilomycin A1	Abcam	Cat# ab120497
5-amino Fluorescein	Abcam	Cat#ab145305

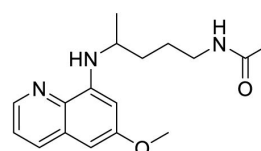
Table S2.

Key resources table

Characterization of Primaquine (Prim) and N-acetylated Primaquine (PrimAc)



Primaquine (Prim)



Primaquine acetate (PrimAc)

The absorption and emission spectra of Primaquine analogues are shown in Figure S18 A& B respectively. They were measured in the mixture of 10% DMSO in PBS or in Glycine/NaOH buffer.

Materials & Reagent

Glycine/NaOH (pH = 10) (preparation: 80 mL of distilled water, 0.6 g of glycine, 0.25 g of NaOH, pH adjusted to 10 using 35% HCl, diluted to 100 mL)

Synthesis of PrimAc

To the stirred solution of Primaquine bisphosphate (100 mg, 220 μ mol, 1 eq) and TEA (92 μ L, 660 μ mol, 3 eq) in DMF acetic anhydride (21 μ L, 220 μ mol, 1 eq) was added at RT. Reaction mixture was stirred for 5 minutes and the reaction mixture was diluted with dichloromethane (50 mL) and extracted 2x with water. Water phase was extracted with 20 mL of DCM and combined organic layers were extracted with brine. The organic phase was evaporated to the volume of approximately 1 mL and the solution was liquid loaded to the column. Gradient chromatography was done using cyclohexane, ethyl acetate and methanol (grad.: 100% cyclohexane – 30% cyclohexane, 60% ethyl acetate, 10% methanol). The chromatography was repeated to obtain the target compound as yellowish oil in 94% yield and 98.5% purity.

¹H NMR (401 MHz, CDCl₃) δ 8.53 (dd, J = 4.2, 1.6 Hz, 1H), 7.92 (dd, J = 8.3, 1.6 Hz, 1H), 7.31 (dd, J = 8.2, 4.2 Hz, 1H), 6.34 (d, J = 2.5 Hz, 1H), 6.28 (dd, J = 2.6, 0.6 Hz, 1H), 5.99 (d, J = 8.4 Hz, 1H), 5.50 (s, 1H), 3.69 – 3.58 (m, 1H), 3.36 – 3.18 (m, 2H), 1.92 (s, 3H), 1.72 – 1.63 (m, 5H), 1.30 (d, J = 6.4 Hz, 3H).

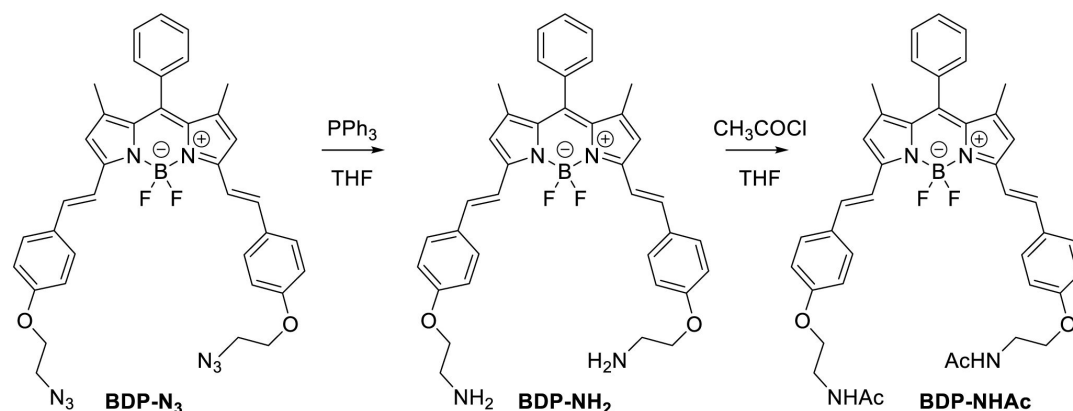
¹³C NMR (101 MHz, CDCl₃) δ 170.03, 159.45, 144.92, 144.37, 135.36, 134.84, 121.91, 96.84, 91.75, 55.24, 47.85, 39.61, 34.01, 26.30, 23.33, 20.65. (Figure S19 A&B). HRMS spectrum of Primaquine-Ac is shown in Figure S20.

Hydrolytic experiment

The compound PrimAc (1 mg) was dissolved in 20 μ L DMSO and diluted with 1.98 mL 0.1M PBS. The solution was measured by HPLC-MS (0-day, 1 day, 7 days, 1 month). The sample was stored in the dark at 25 °C between measurements. (Figure S18 C&D)

Characterization of BODIPY-NH₂ (BDP-NH₂) and BODIPY-NHAc (BDP-NHAc)

The Structures and photo-physical properties of BODIPY analogues are shown in Figure S21A. The absorption and emission spectra of BODIPY-NH₂ and BODIPY-NHAc in PBS/DMSO are shown in Figures S21 B&C, respectively.



BDP-NH₂: To a solution of a starting **BDP-N₃** (60 mg, 0.1 mmol, 1 eq.) and PPh_3 (47 mg, 0.2 mmol, 2 eq.) in THF (8 mL) water (1 mL) was added. The reaction mixture was stirred for 24 h and transferred on 20 g of silica. First fractions were eluted using a MeOH/DCM, 6/1 mixture followed by elution of the product using MeOH/DCM/NH₃ aq, 90/7/3, mixture. Organic solvents were removed under vacuum and an aqueous layer was extracted using DCM. The final crude product was reprecipitated from DCM using pentane, yielding blue crystals (34 mg, 61 % yield).

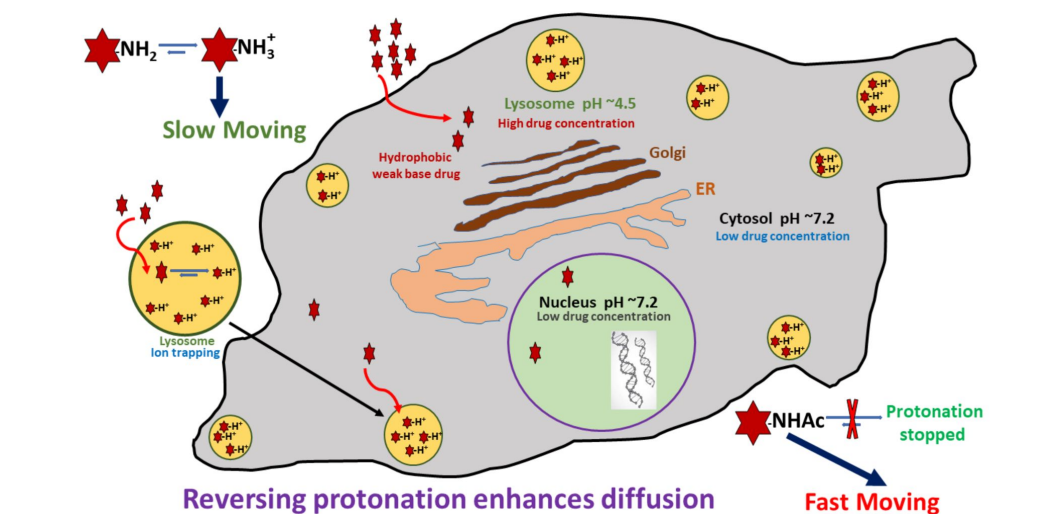
^1H NMR (401 MHz, CD_2Cl_2) δ 7.64 – 7.49 (m, 11H), 7.26 (d, J = 16.3 Hz, 2H), 7.01 – 6.92 (m, 4H), 6.66 (s, 2H), 4.03 (t, J = 5.2 Hz, 4H), 3.07 (t, J = 5.2 Hz, 4H), 1.64 (s, 4H), 1.46 (s, 6H). ^{13}C NMR (101 MHz, CD_2Cl_2) δ 160.5, 152.9, 142.6, 136.2, 135.5, 133.6, 132.2, 130.1, 129.9, 129.5, 129.3, 129.0, 117.8, 117.3, 115.3, 114.9, 71.0, 41.9, 14.8. HR-MS (ESI+), m/z : $[\text{M}+\text{H}]^+$, calcd for $[\text{C}_{37}\text{H}_{38}\text{O}_2\text{N}_4\text{BF}_2]^+$ 619.30571, found 619.30504. ^1H NMR and ^{13}C NMR of **BODIPY-NH₂** are shown Figures S22 A, B respectively. HRMS spectrum of **BODIPY-NH₂** is shown in Figure S23.

BDP-NHAc: To a solution of **BDP-NH₂** (5 mg, 0.008 mmol, 1 eq.) in THF (2.0 ml), Et₃N (15 mg, 0.14 mmol, 17 eq.) and CH_3COCl (11 mg, 0.15 mmol, 18 eq) were added. The reaction mixture was stirred at r.t. for 2 h. Water was added and a product was extracted using DCM. The solvent was removed under reduced pressure and the crude product was 3 times reprecipitated from DCM using pentane, yielding blue crystals (4.5 mg, 79 %).

^1H NMR (401 MHz, CD_2Cl_2) δ 7.78 – 7.46 (m, 10H), 7.40 – 7.31 (m, 3H), 7.26 (d, J = 16.3 Hz, 2H), 7.01 – 6.89 (m, 4H), 6.65 (s, 2H), 4.09 (t, J = 5.1 Hz, 4H), 3.65 (q, J = 5.4 Hz, 4H), 1.97 (s, 3H), 1.46 (s, 6H). ^{13}C NMR (101 MHz, CDCl_3) δ 160.0 (C), 152.9 (C), 142.7 (C), 139.1 (C), 136.0 (CH), 135.5 (C), 130.2 (C), 129.5 (CH), 129.3 (CH), 128.9 (CH), 117.8 (CH), 117.5 (CH), 115.2 (CH), 67.5 (CH₂), 39.3 (CH₂), 19.1 (CH₃), 14.8 (CH₃). HR-MS (ESI+), m/z : $[\text{M}+\text{H}]^+$, calcd for $[\text{C}_{41}\text{H}_{41}\text{O}_4\text{N}_4\text{BF}_2\text{Na}]^+$ 725.30841, found 725.30811.

^1H NMR and ^{13}C NMR of **BODIPY-NHAc** are shown Figures S24 A, B respectively. HRMS spectrum of **BODIPY-NHAc** is shown in Figure S25.

Supplementary figures



Graphical abstract

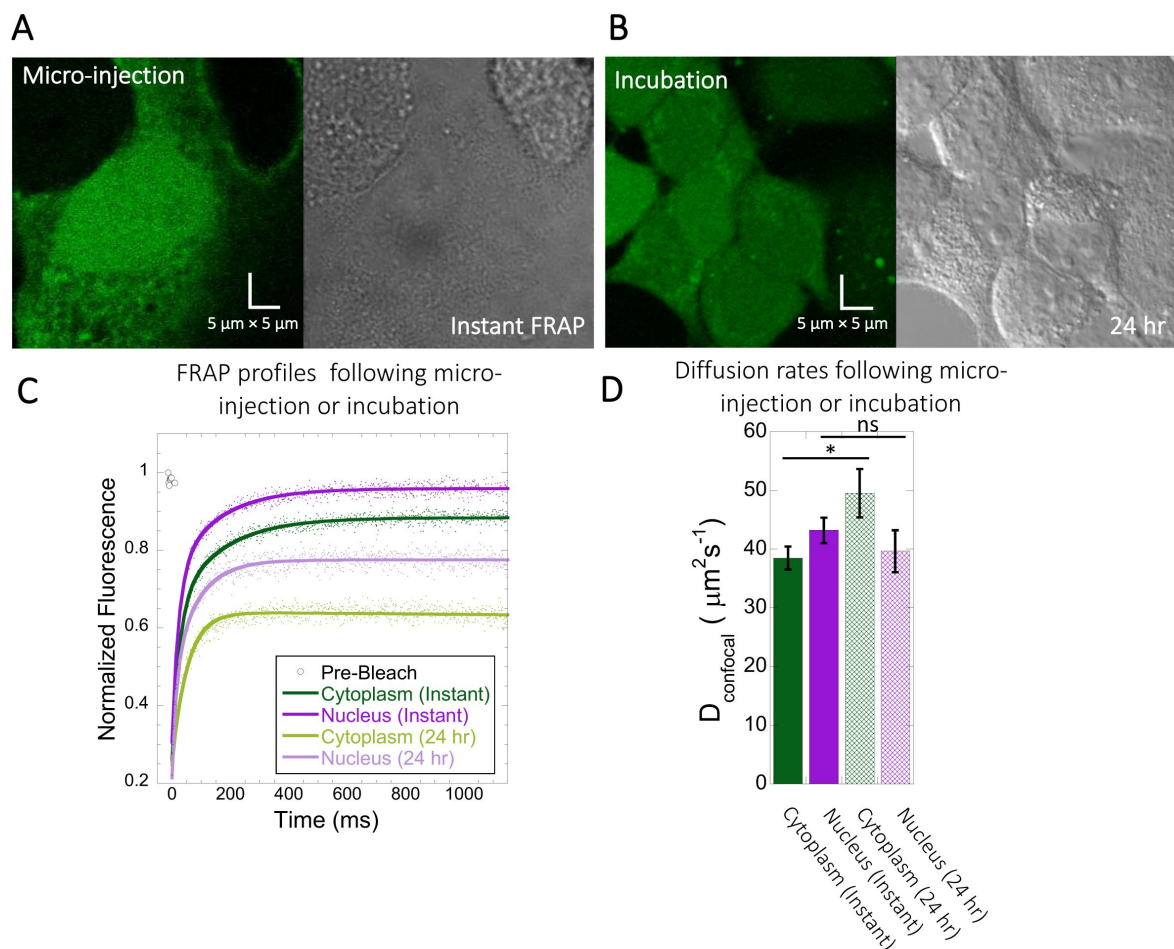


Figure S1

FRAP measurements for Fluorescein following different incubation times.

(A) Micrographs of HeLa cells instantaneously after micro-injection vs (B) FRAP after 24 hours incubation. (C) Comparative merged FRAP profiles with fits ($N=30$; $R=0.99$ for each of the fits) and (D) their D_{confocal} values. Error bars represent SE calculated from fitting the FRAP progression curves averaged over at least 30 independent measurements.

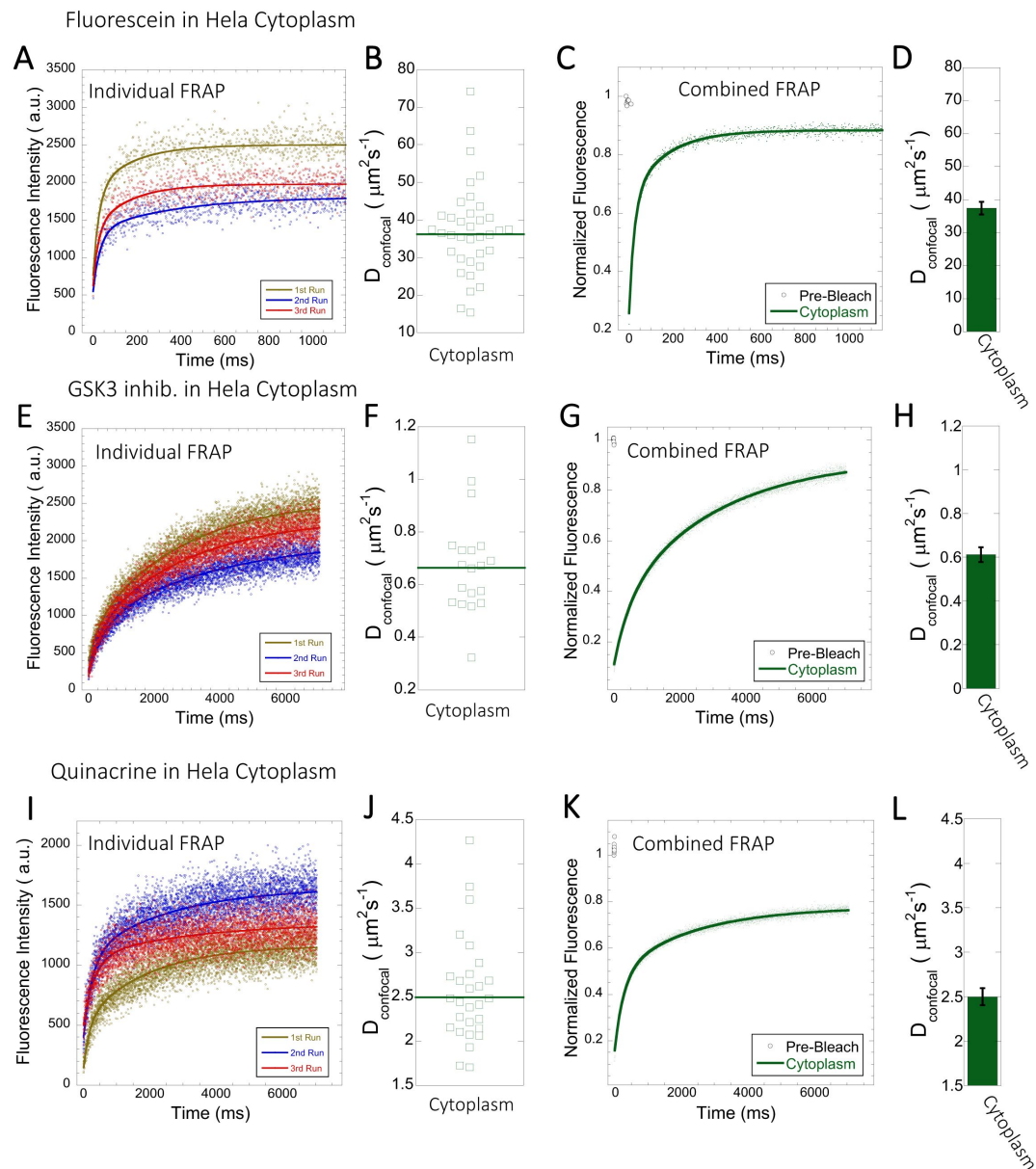


Figure S2

D_{confocal} values calculated from individual FRAP and merged FRAP curves.

FRAP curves and D_{confocal} values for Fluorescein (A-D), GSK3 inhibitor (E-H) and Quinacrine (I-L) inside HeLa cell cytoplasm are shown. Three representative individual FRAP profiles with exponential fits with $R=0.80-0.90$ for each of the compounds respectively are shown in (A), (E), and (I). Individual D_{confocal} values and their geometric means from individual FRAP curves are shown for Fluorescein (B), GSK3 inhibitor (F) and Quinacrine (J). (C), (G) and (K) show the merged FRAP curves over 30 independent measurements. This provides an improved fit ($R=0.99$). Panels (D), (H) and (L) show D_{confocal} values calculated from the merged FRAP profiles. Error bars represent SE calculated from fitting the FRAP progression curves.

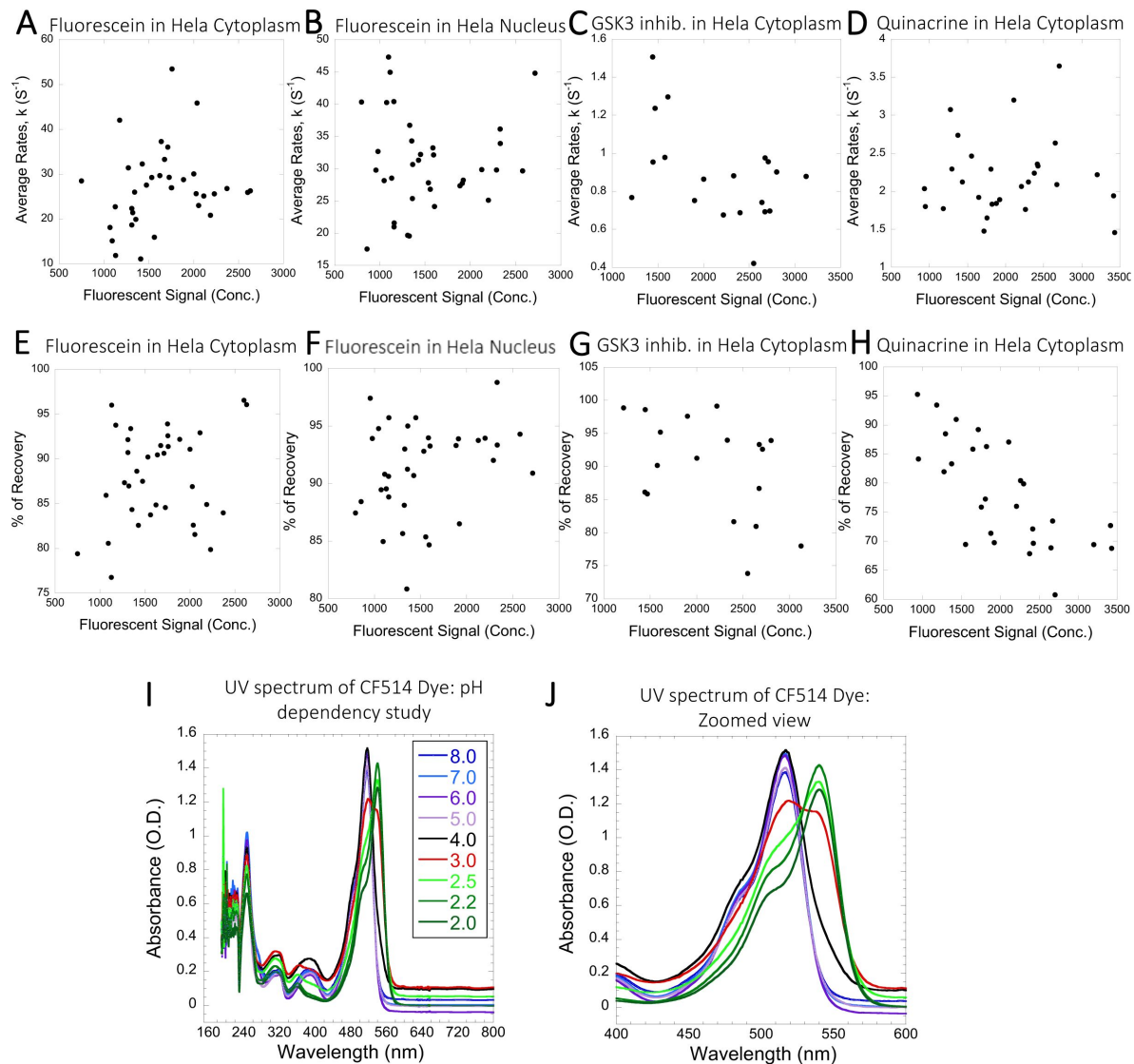


Figure S3

Diffusion rates and percent recovery as a function of drug concentration inside Hela Cells.

The diffusion rates were calculated from individual FRAP progression curves. The drug concentration is linearly related to the fluorescence intensity. (A-D) show FRAP rates as a function of the fluorescence signal for Fluorescein in the cytoplasm. (B) Fluorescein in Nucleus, (C) GSK3 inhibitor in cytoplasm, (D) Quinacrine in cytoplasm. (E-H) are as (A-D) but the x-axis denotes percent recovery after FRAP. (I-J) Comparative UV spectrum of CF514 labelling dye in 20 mM sodium phosphate buffer at different pH values, from which a pK_a value of 3 was calculated.

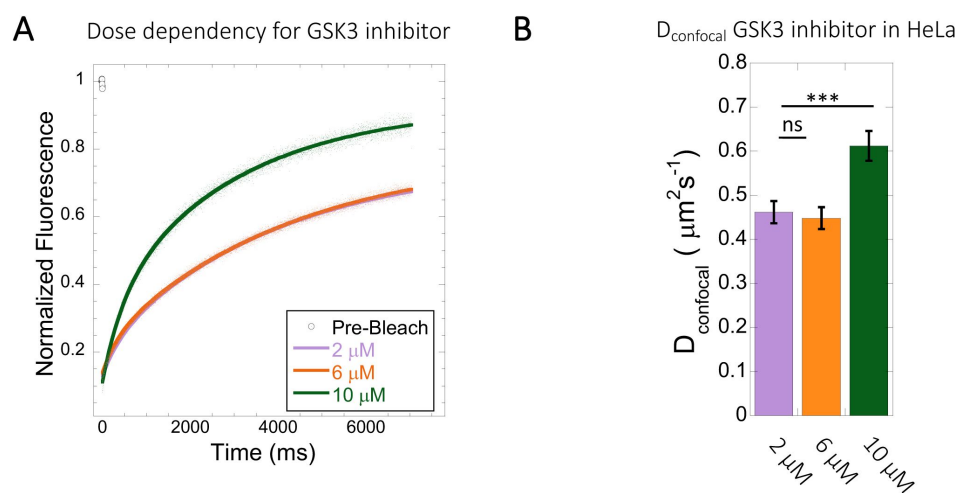


Figure S4

Diffusion rates at different doses of GSK3 inhibitor in HeLa cytoplasm.

Effect of drug dosage (2-10 μM) on (A) averaged FRAP profiles with fits ($N=30$; $R=0.99$ for each of the fits) and (B) D_{confocal} values. Error bars represent SE calculated from fitting the FRAP progression curves, which are averaged over at least 30 independent measurements.

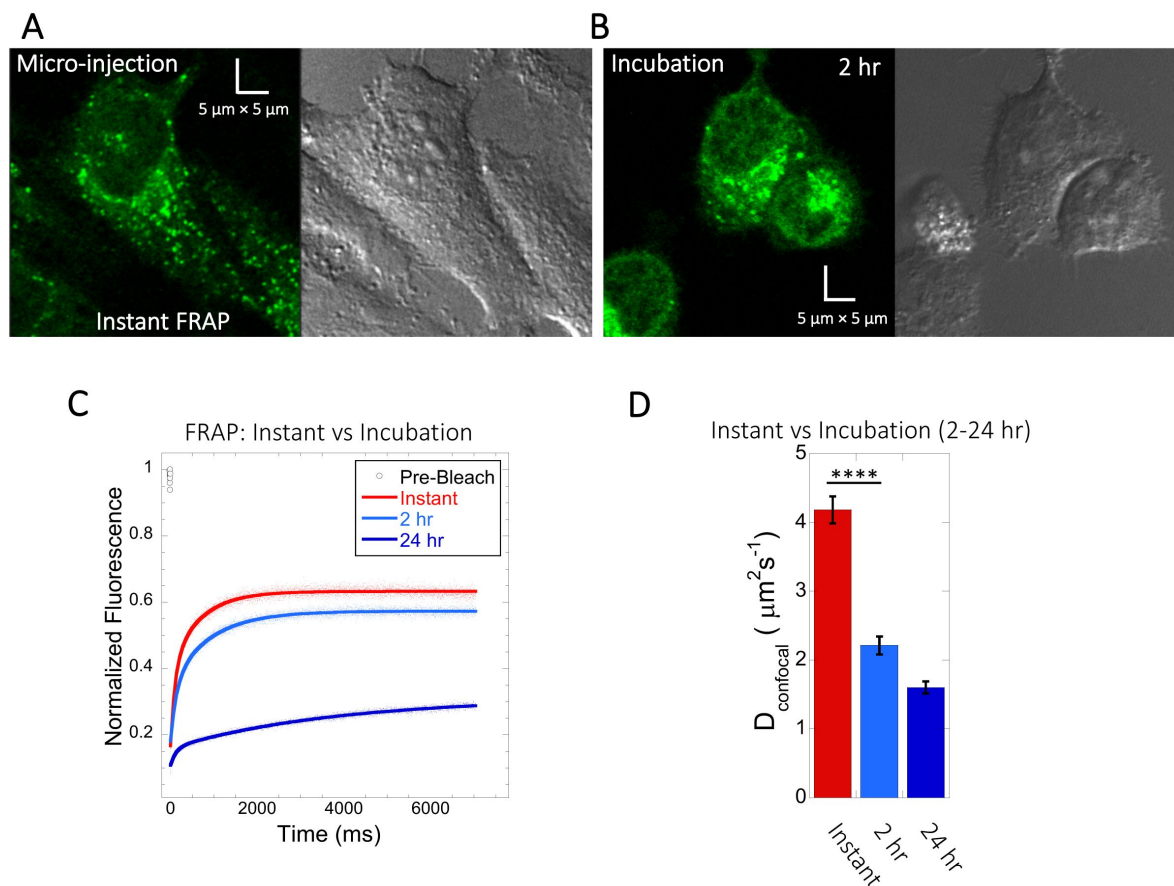


Figure S5

Quinacrine diffusion in HeLa after micro-injection or incubation.

Comparison of FRAP immediately following micro-injection (instant FRAP) and FRAP following 2-24 hr incubation. HeLa cell micrographs following (A) micro-injection and (B) 2 hours of incubation. Comparative (C) averaged FRAP profiles with exponential curve fits ($N=30$; $R=0.99$ for each of the fits) and (D) calculated D_{confocal} values. Error bars represent SE calculated from fitting the FRAP progression curves, which are averaged over at least 30 independent measurements.

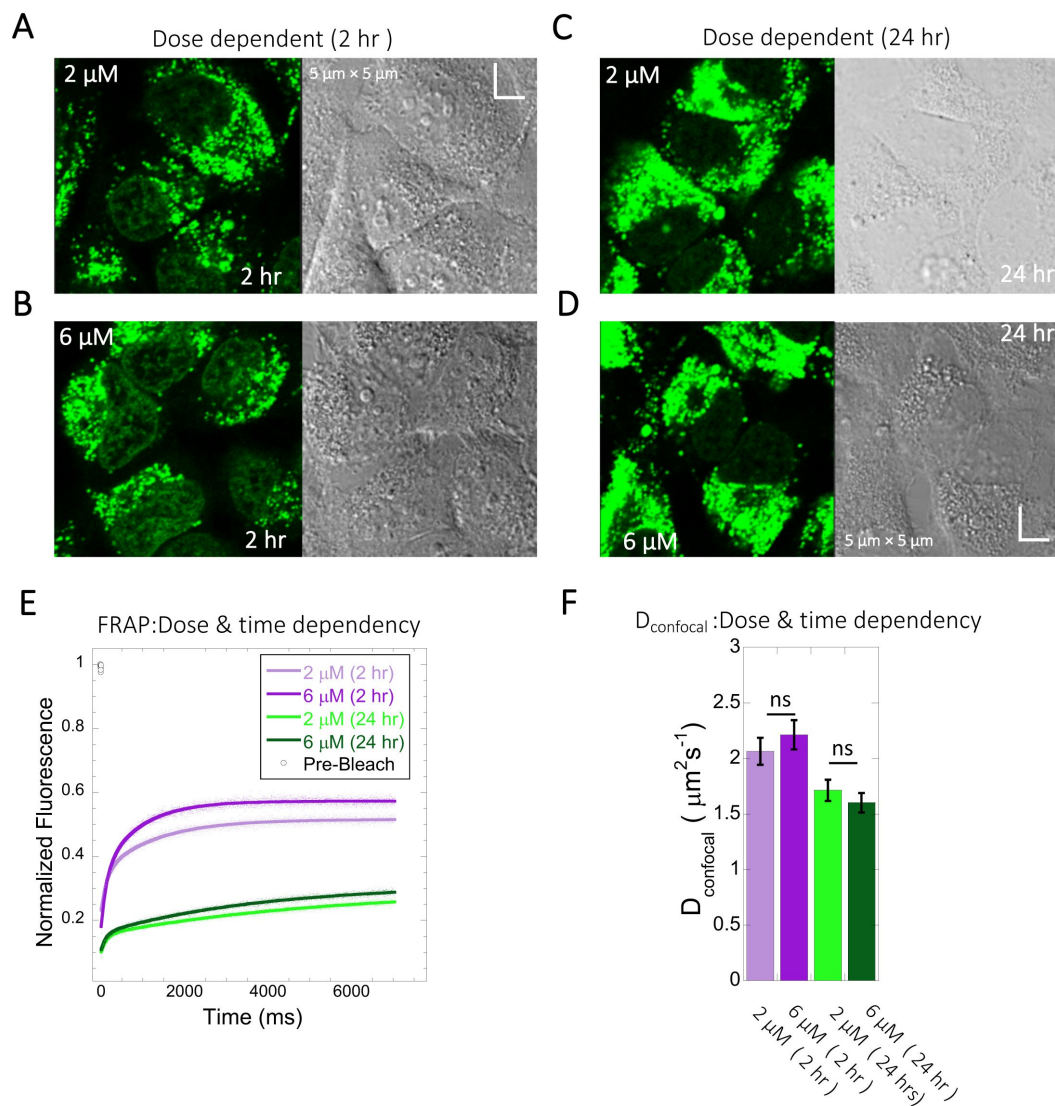


Figure S6

Dose and time dependency of Quinacrine diffusion in HeLa cells.

HeLa cell micrographs after 2 hours of incubation with (A) 2 μM and (B) 6 μM of Quinacrine. (C) and (D) are micrographs of HeLa cells after 24 hours of incubation with 2 μM and 6 μM Quinacrine respectively. (E) Effect of drug dose and time dependency on comparative FRAP profiles, with fits ($N=30$; $R=0.99$ for each of the fits) and (F) Calculated D_{confocal} values from the fits in (E). Error bars represent SE calculated from fitting the FRAP progression curves, which are averaged over at least 30 independent measurements.

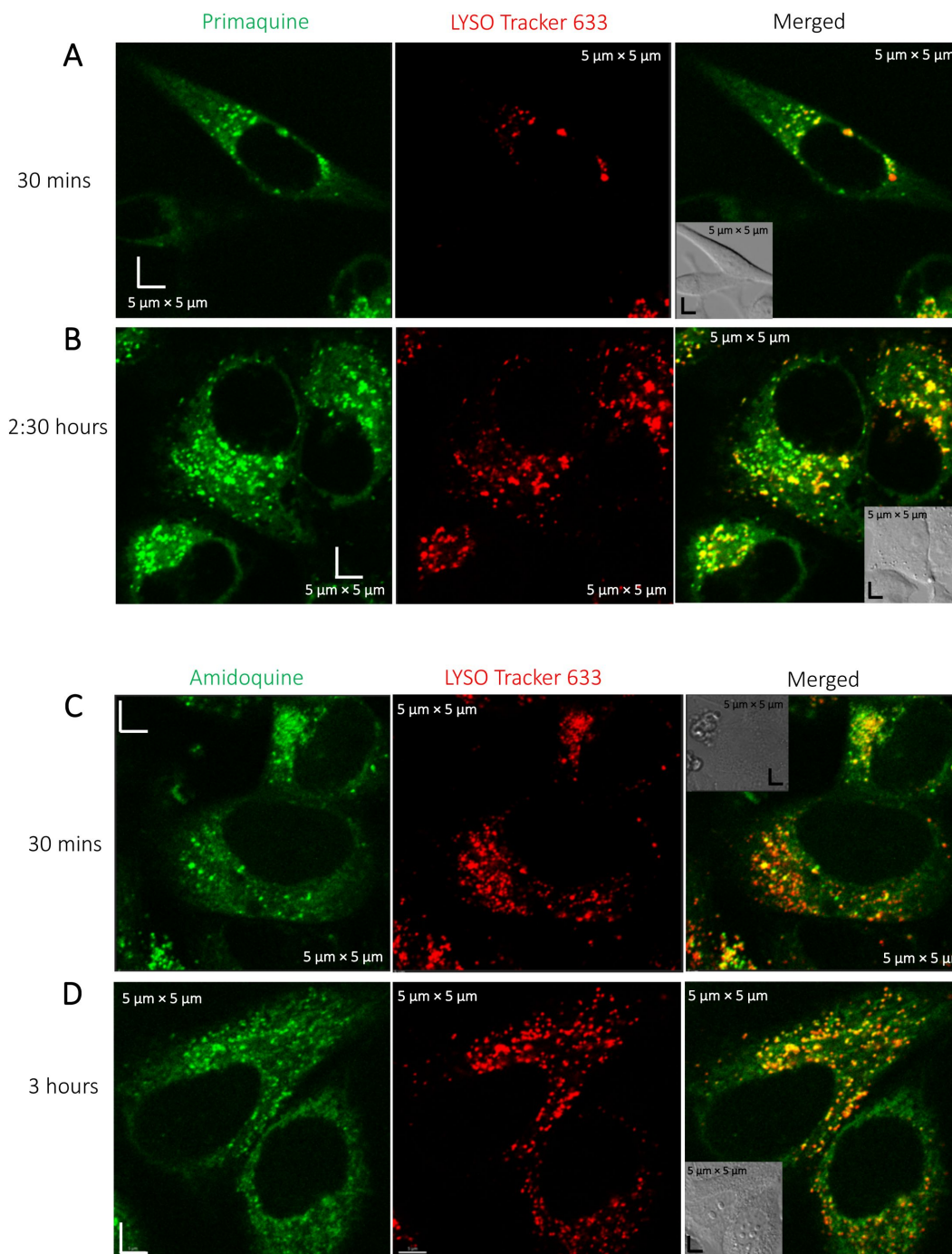


Figure S7

Colocalization of Primaquine and Amidoquine with Lyso-tracker.

Colocalization of Primaquine (A-B) and Amidoquine (C-D) with Lyso Tracker 633 in HeLa cells. Time dependent measurements show higher aggregation of both drugs in the lysosomes over time.

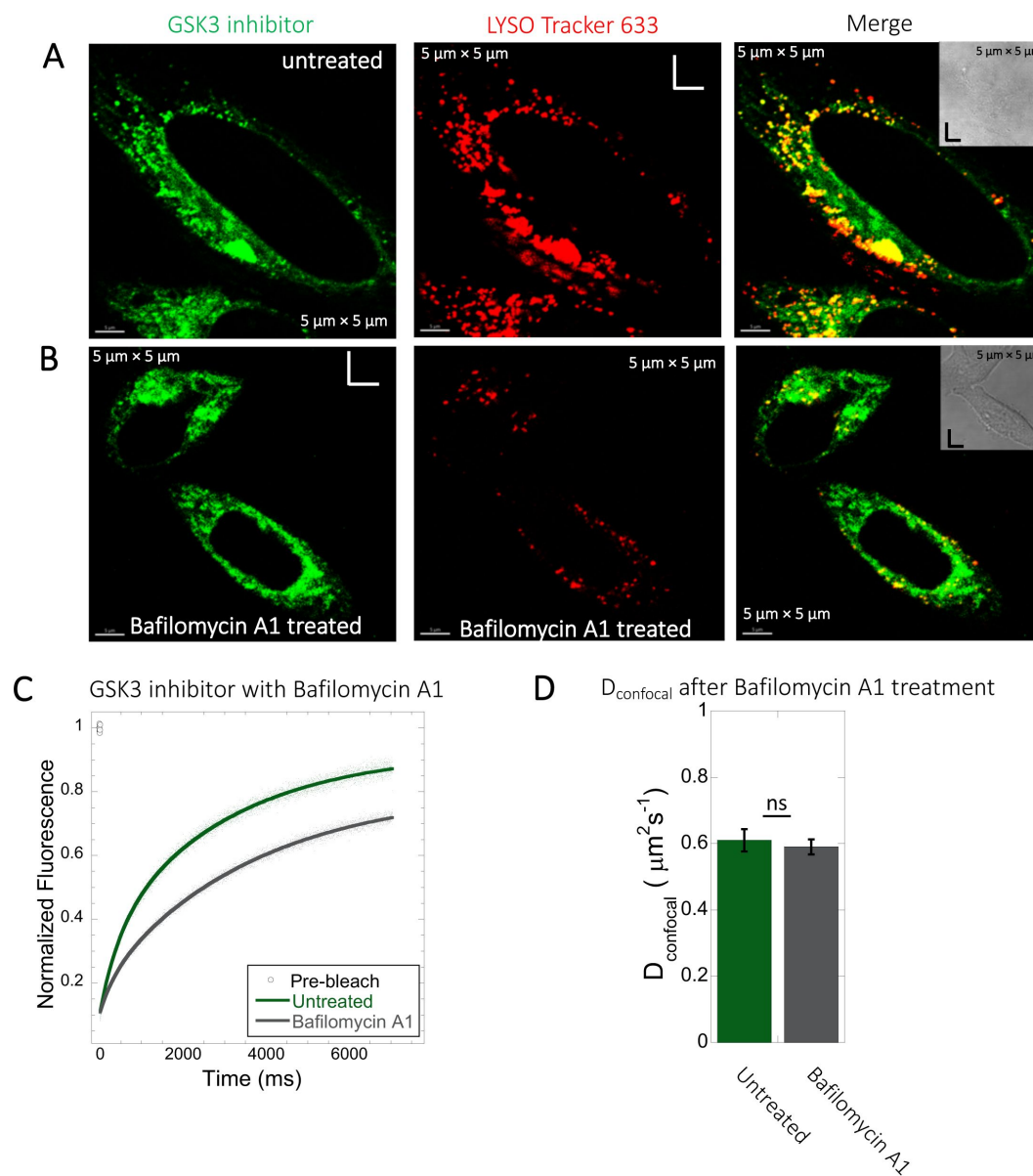


Figure S8

Inhibition of lysosomal accumulation of GSK3 inhibitor by Bafilomycin A1.

(A-B) Colocalization of GSK3 inhibitor with Lyso Tracker 633 in HeLa cells. (A) Untreated and (B) treated with 100 nM Bafilomycin A1 and corresponding colocalizations are shown. (C) Comparative averaged FRAP profiles with fits ($N = 30$; $R = 0.99$ for each of the fits) and (D) D_{confocal} diffusion values with and without the Bafilomycin A1 treatment. Error bars represent SE calculated from fitting the FRAP progression curves, which are averaged over at least 30 independent measurements.

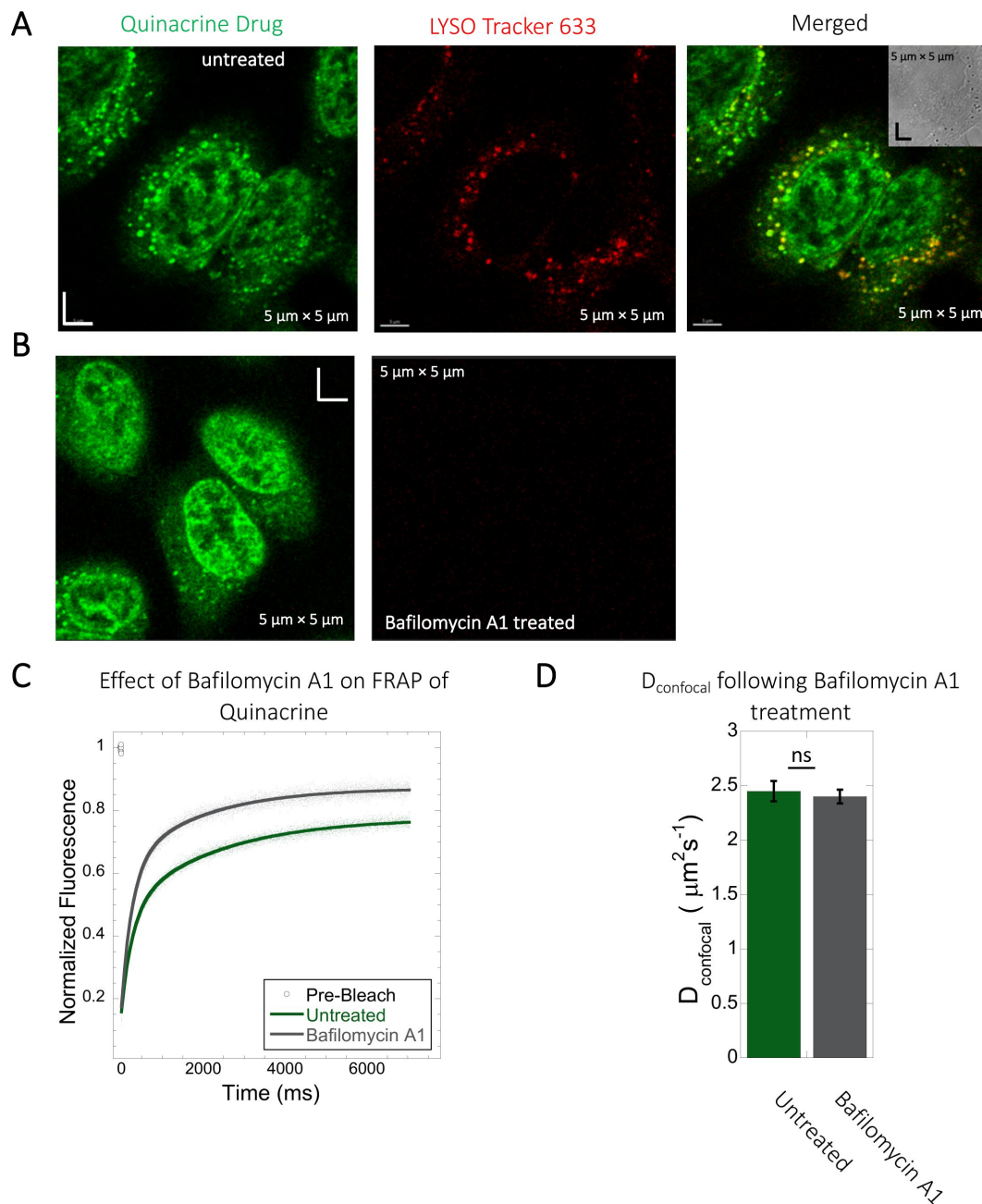


Figure S9

Inhibition of lysosomal accumulation of Quinacrine by Bafilomycin A1.

(A-B) Colocalization of Quinacrine with Lyso Tracker 633 in HeLa cells. HeLa cells (A) untreated and (B) treated with 100 nM Bafilomycin A1 and corresponding colocalizations. (C) Comparative averaged FRAP profiles with fits ($N=30$; $R=0.99$ for each of the fits) and (D) D_{confocal} values with and without the Bafilomycin A1 treatment. Error bars represent SE calculated from fitting the FRAP progression curves, which are averaged over at least 30 independent measurements.

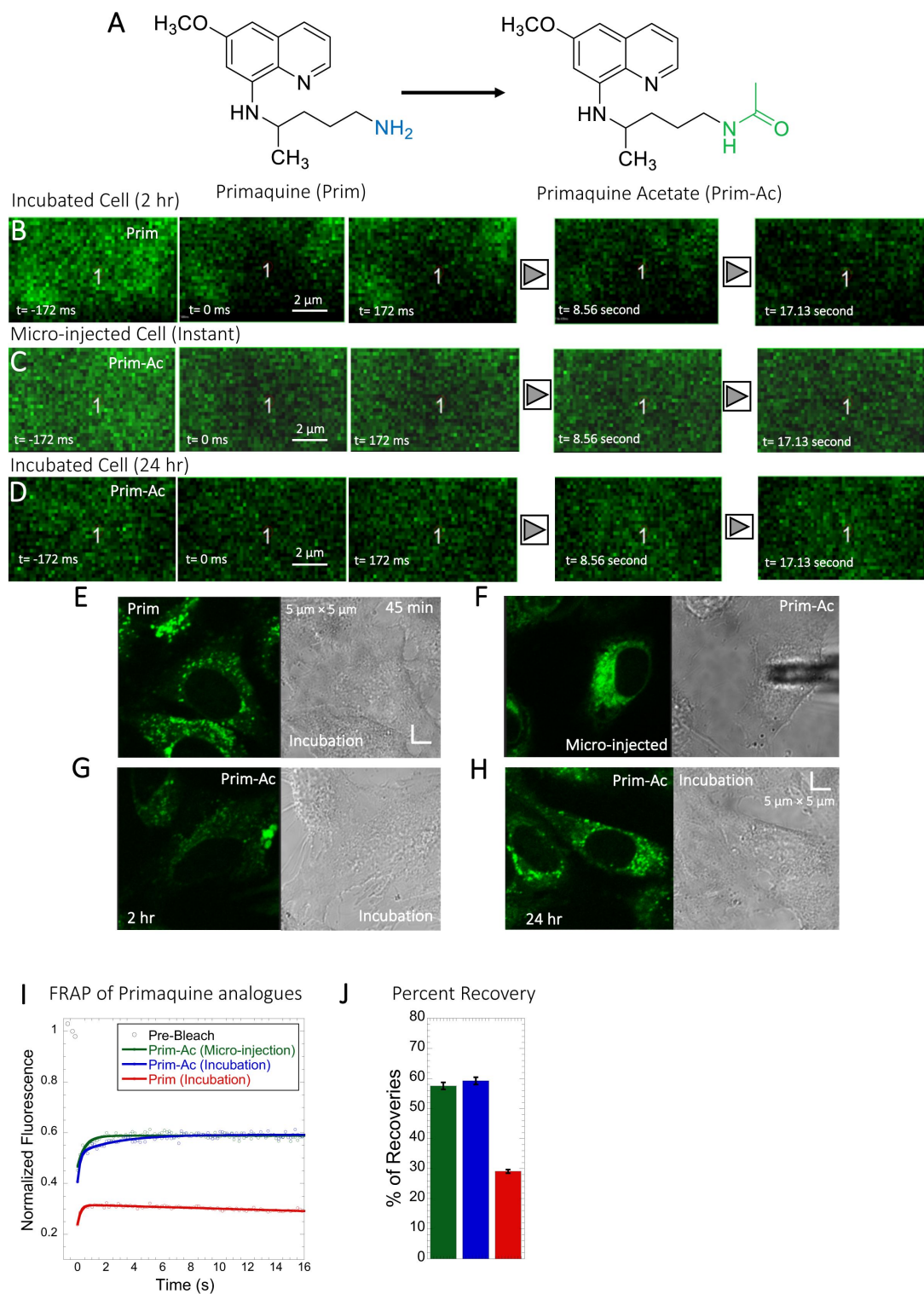


Figure S10

Comparative FRAP of Primaquine analogues in HeLa cell cytoplasm.

(A) Structures of Primaquine analogues with pKa values. Comparative (B-D) XY FRAP micrographs with time lapses in HeLa cell. (E-H) Micrographs of treated HeLa cells either with micro-injection or after drug incubations with Prim or Prim-Ac. Comparative (I) XY-FRAP recovery profiles with fits (N= 20; R= 0.98 for each of the fits) and (J) percentage of recoveries for Primaquine analogues in HeLa cytoplasm (N= 20).

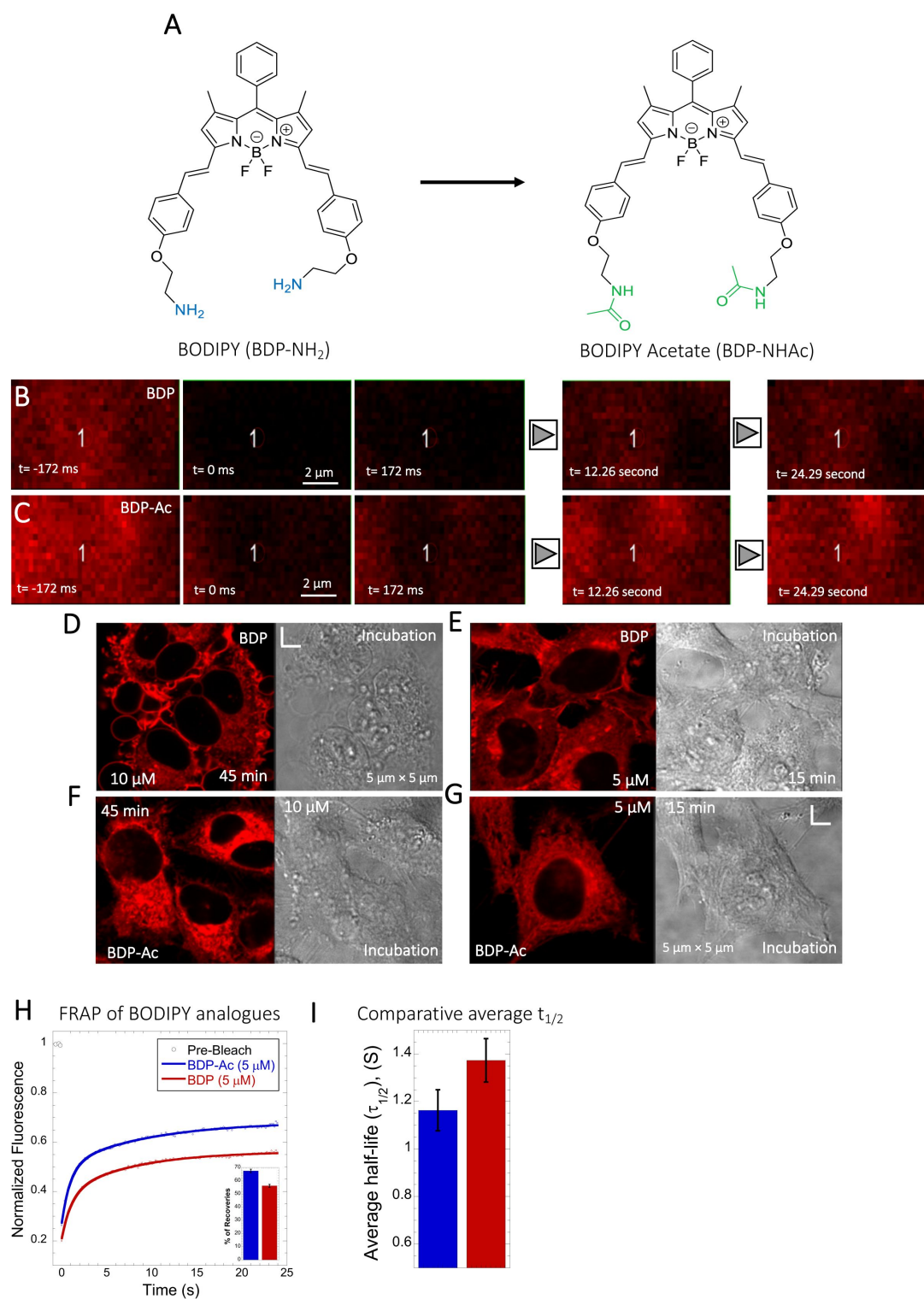


Figure S11

Comparative FRAP of BODIPY analogues in HeLa cell cytoplasm.

(A) Structures of BODIPY analogues. (B-C) Comparative XY FRAP micrographs with time lapses in HeLa cell cytoplasm. (D-G) Micrographs of treated HeLa cells after incubations (with BDP/BDP-Ac) and, Comparative (H) averaged XY-FRAP recovery profiles with fits (N= 30; R= 0.99 for each of the fits), (I) averaged half-life and percentage of recoveries for BODIPY analogues in HeLa cytoplasm (H) (N= 30).

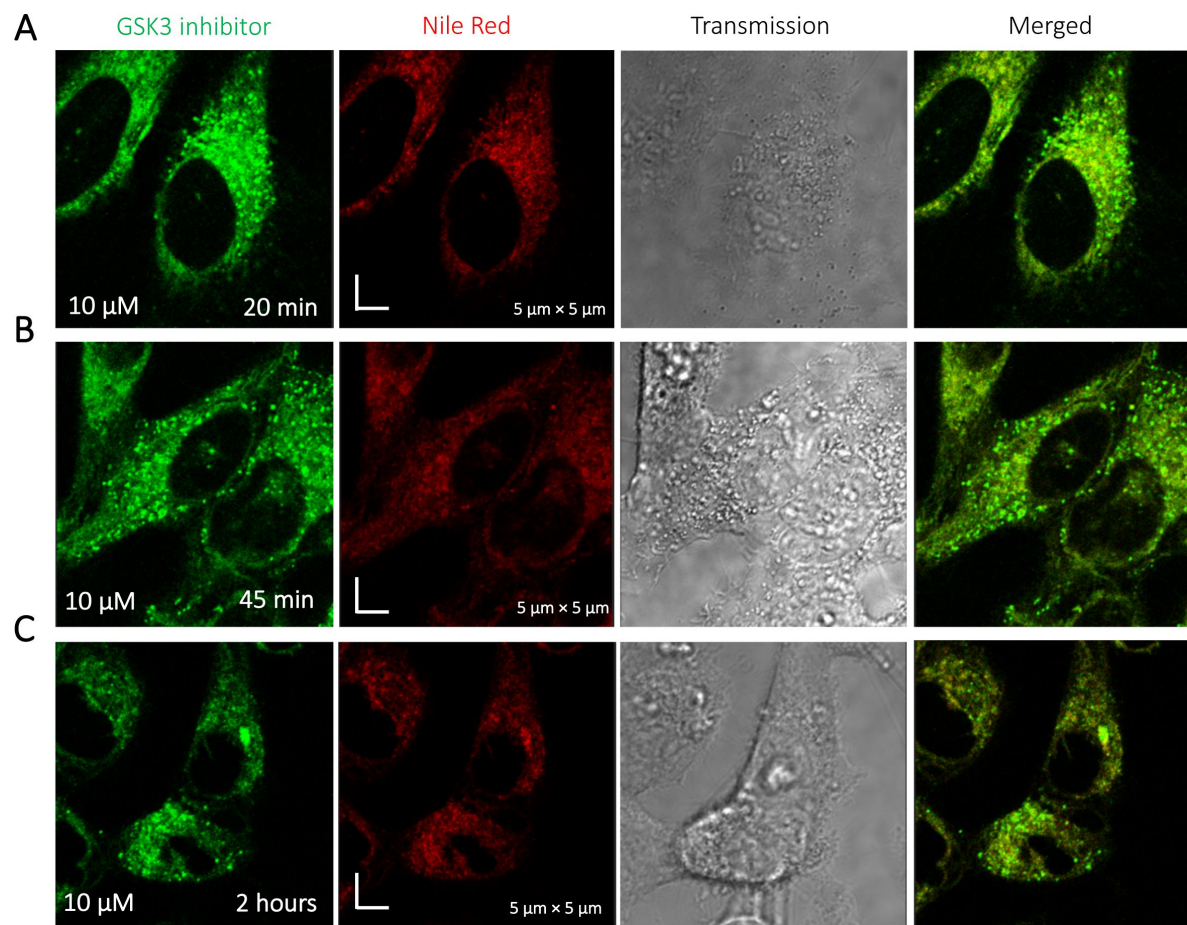


Figure S12

Time dependent colocalization of GSK3 inhibitor and Nile Red in HeLa cell cytoplasm.

Panels (A-C) show Nile red accumulating in lipid droplets, while GSK3 inhibitor does not colocalize.

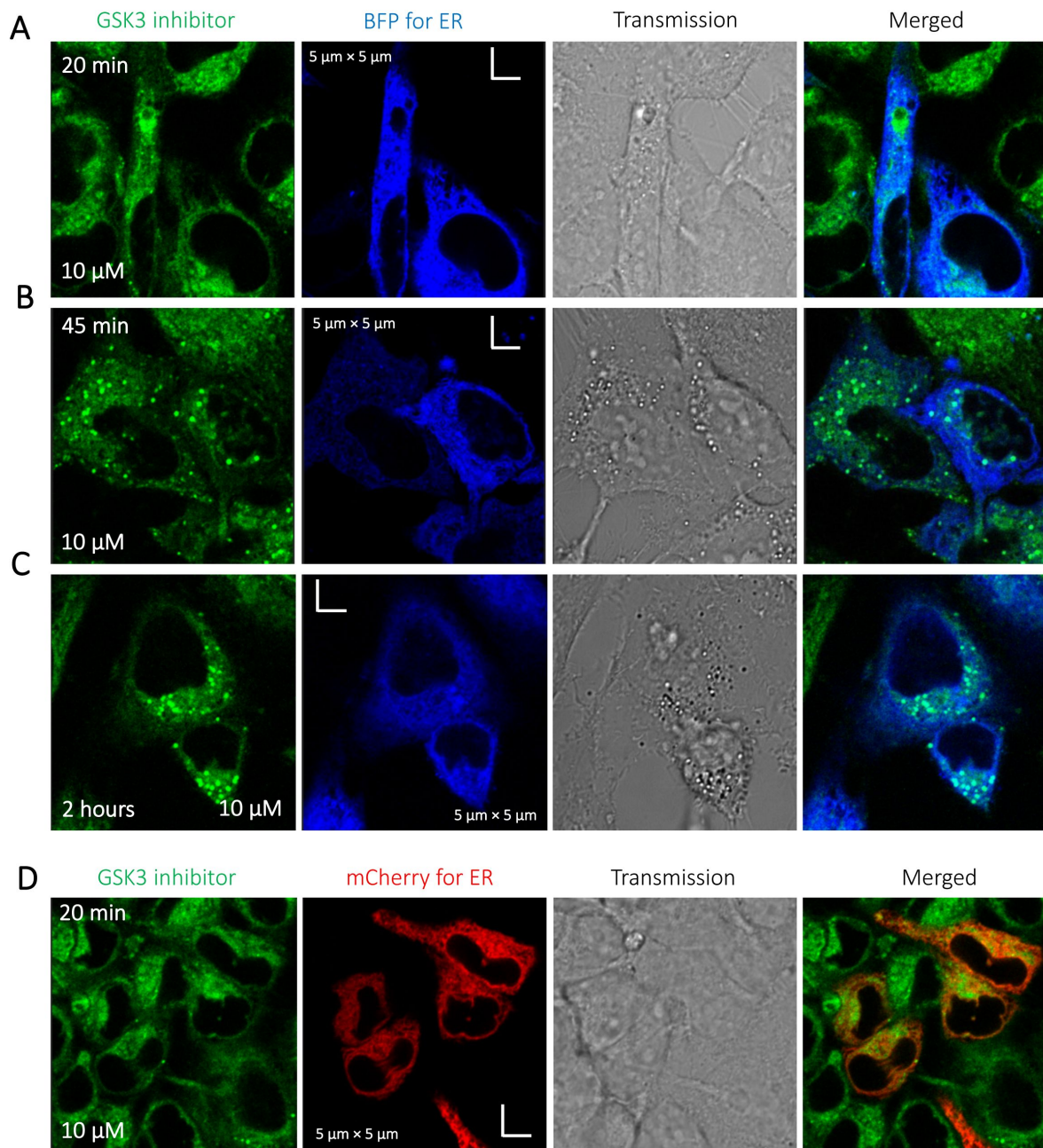


Figure S13

Time dependent colocalization of GSK3 inhibitor and ER marker in HeLa cell cytoplasm.

Panels (A-D) show BFP/mCherry fused antibody marker accumulating in ER, but GSK3 inhibitor does not colocalize. HeLa cells were first transiently transfected with BFP/mCherry fused antibody markers, then treated with GSK3 inhibitor.

Figure S14

Time dependent colocalization study of Mitoxantrone and SYTO blue marker in HeLa cells.

Panels (A-B) show SYTO blue marker accumulating in nucleic acids, but Mitoxantrone is not colocalized. HeLa cells were first treated with SYTO blue, then treated with Mitoxantrone.

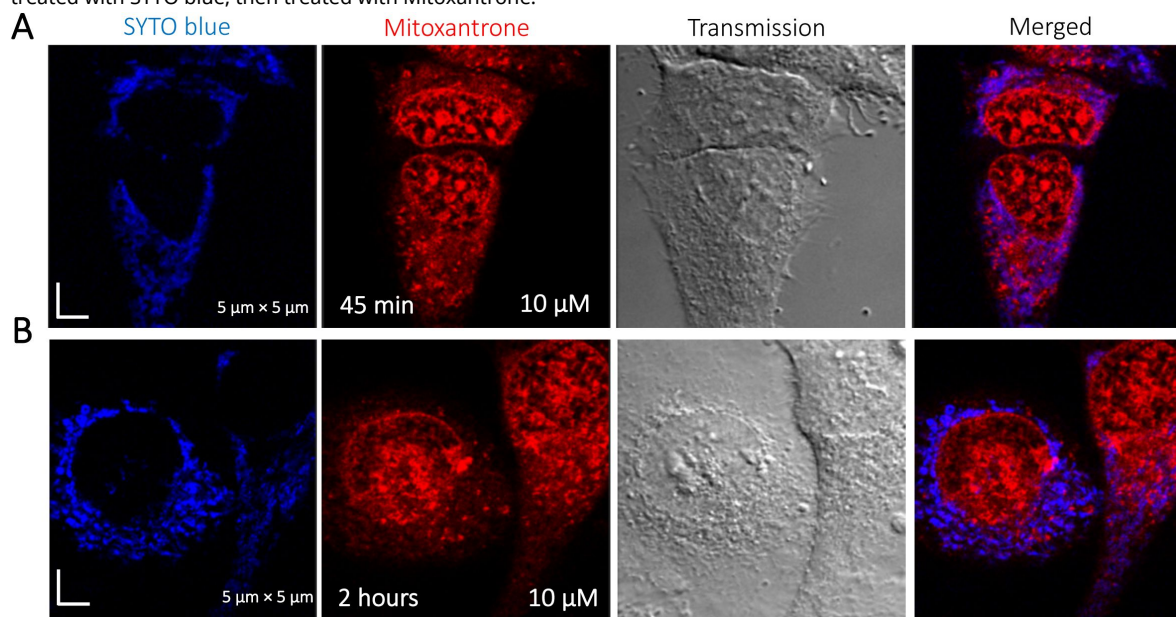
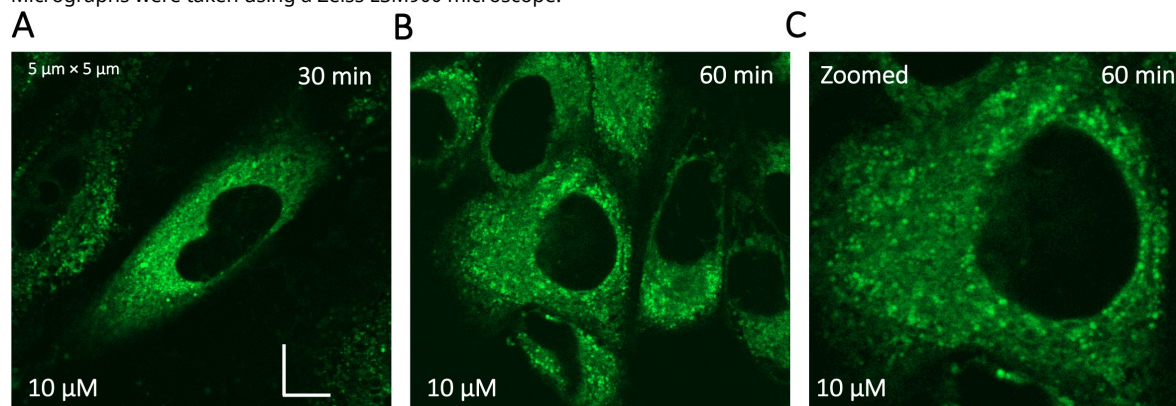


Figure S15

Super resolution images of GSK3 inhibitor treated HeLa cells.

Micrographs were taken using a Zeiss LSM900 microscope.



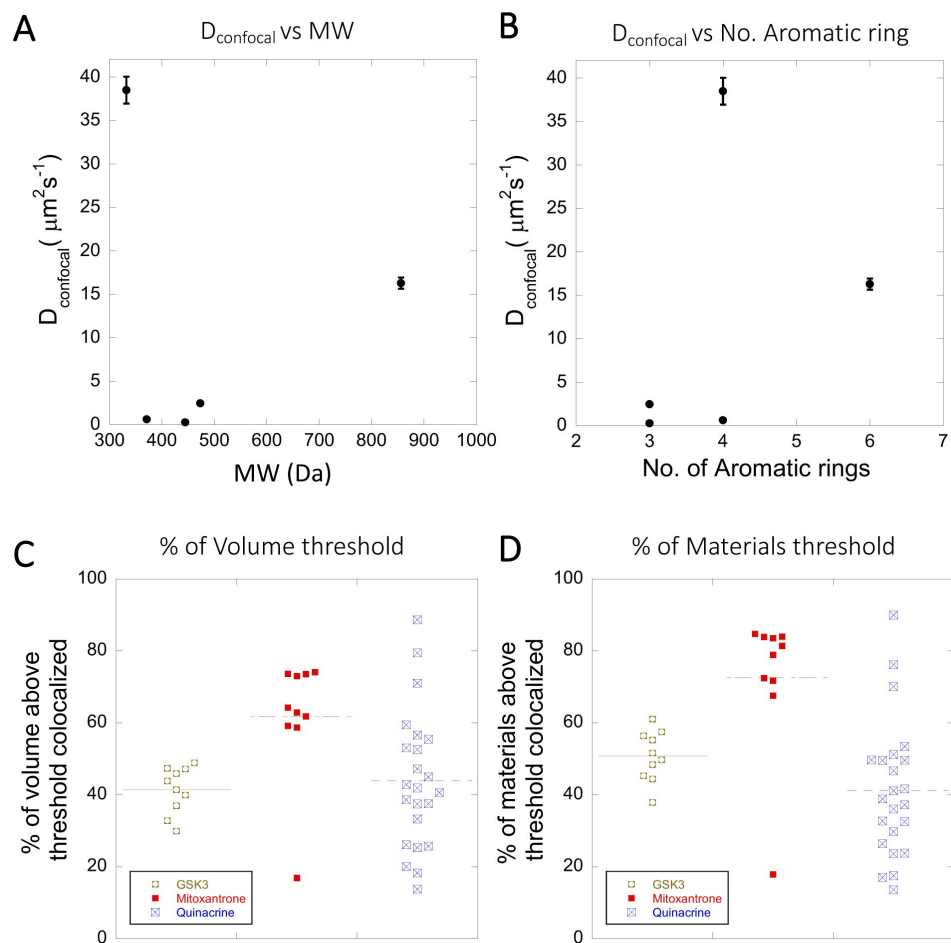


Figure S16

Dependency of D_{confocal} rates on MW, number of aromatic rings and colocalization within Lysosomes.

(A) Dependency of D_{confocal} on molecular weight (MW). (B) Dependency of D_{confocal} on the number of aromatic rings in the drugs. (C-D) Comparison of percentage of colocalization among GSK3 inhibitor vs Mitoxantrone vs Quinacrine in acidic lysosomes are shown. Each dot represents 4-5 stained cells in a single frame acquisition. Colocalization study for GSK3 inhibitor and Quinacrine was done with commercial Lyso tracker-633, and for Mitoxantrone using Quinacrine itself used as a Lyso tracker dye.

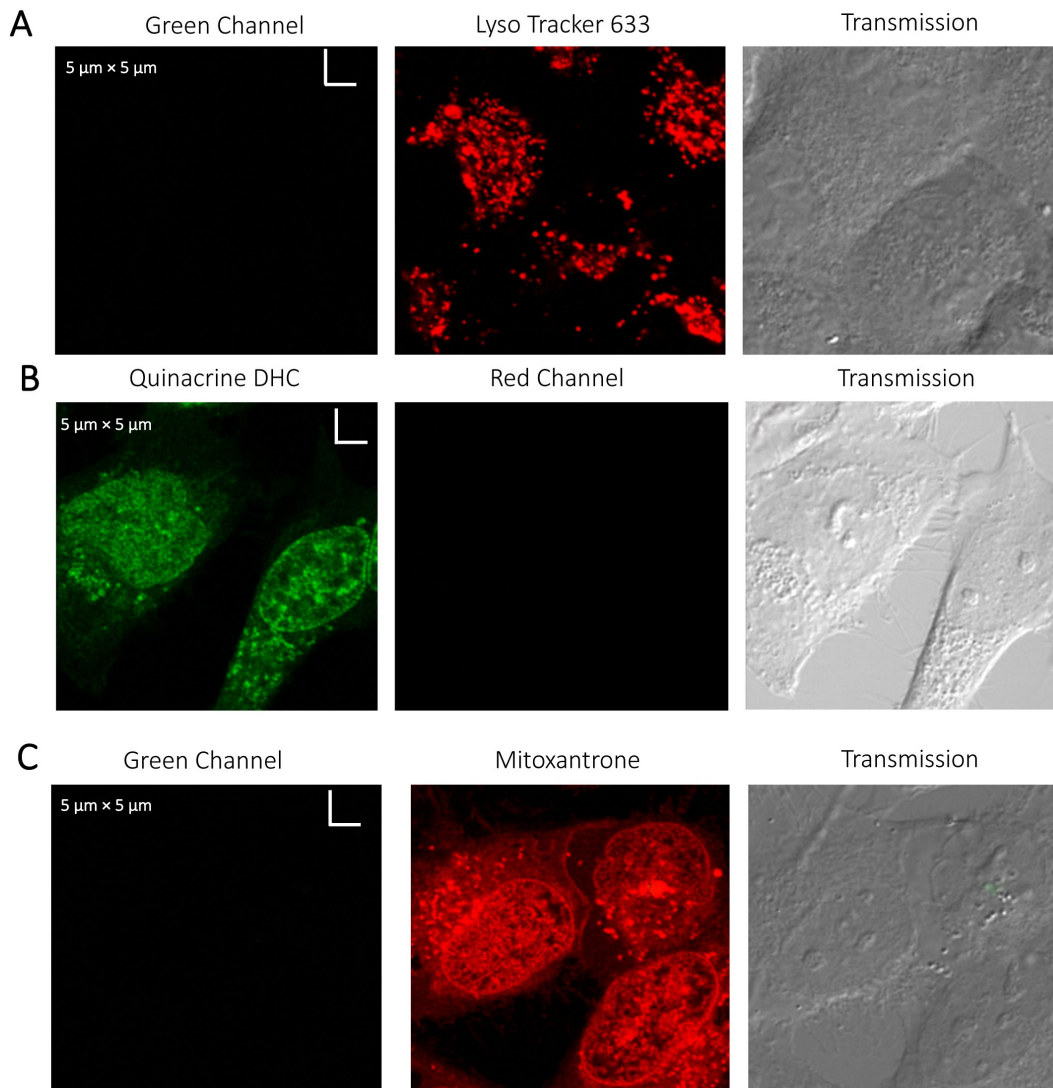


Figure S17

Evaluating channel leakage.

Control experiments showing that channel leakage did not occur during colocalization studies. (A) Lyso-Tracker only, (B) Quinacrine only and (C) Mitoxantrone only.

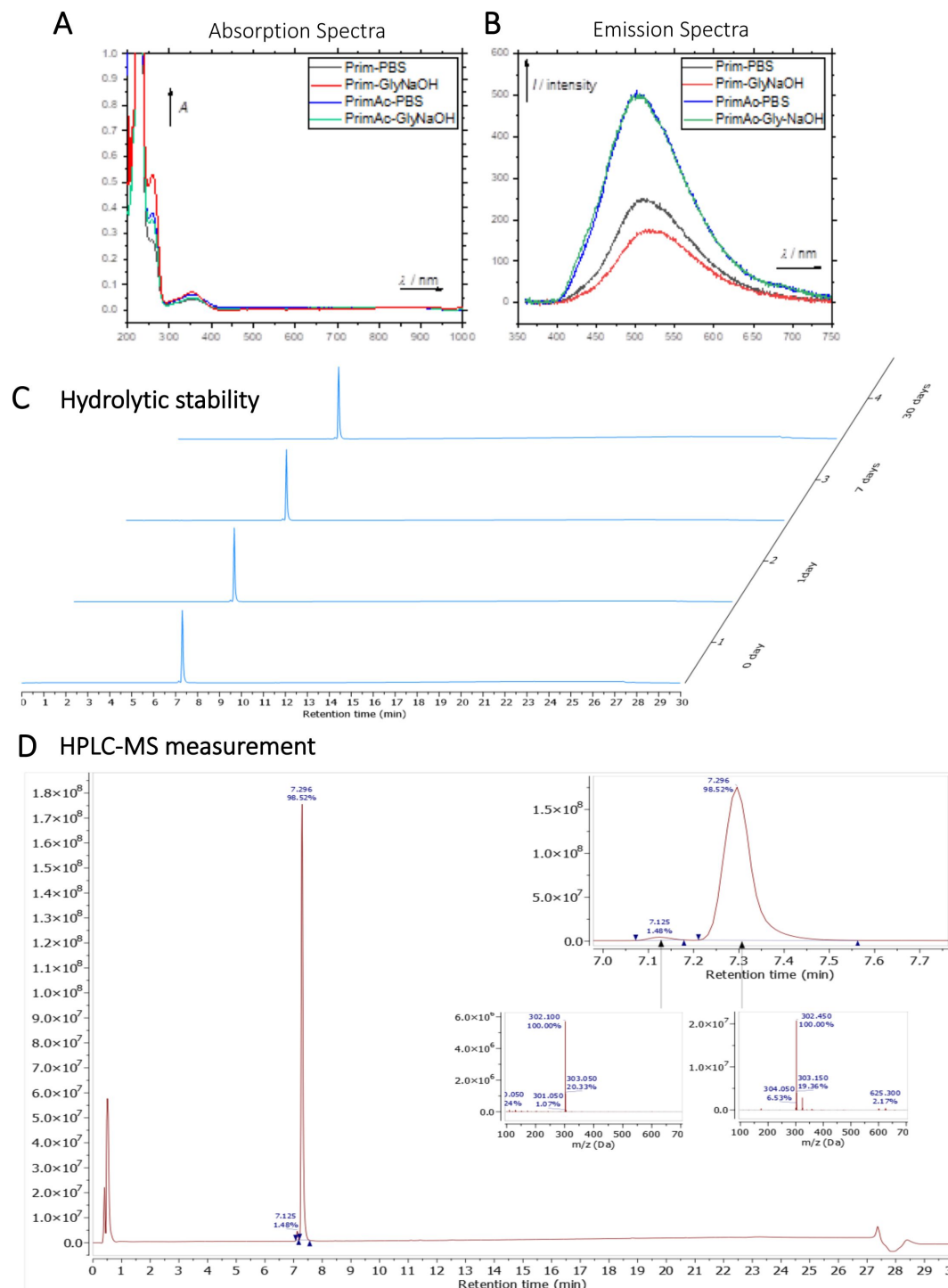
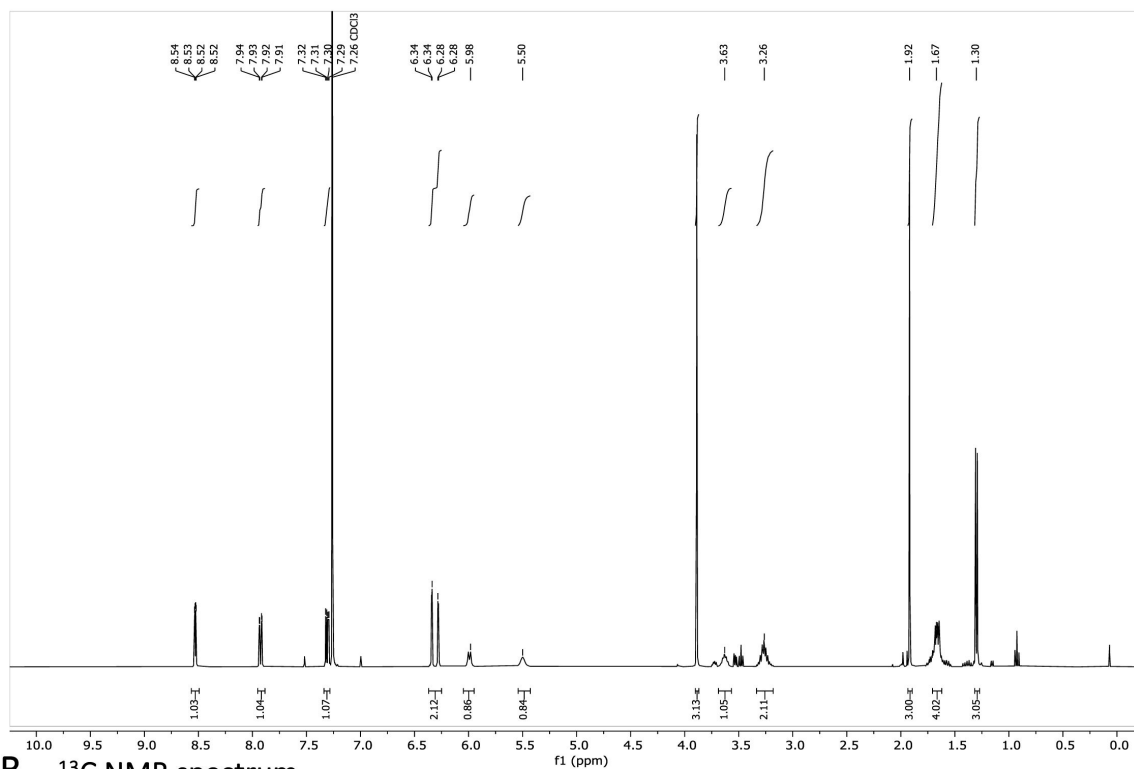


Figure S18

Characterization of Primaquine (Prim) and N-acetylated Primaquine (PrimAc).

(A) Absorption ($\epsilon \approx 1 \cdot 10^{-3}$) and (B) emission ($\epsilon \approx 3 \cdot 10^{-4}$) spectra. Both experiments were measured in the mixture of 10% DMSO in PBS or in Glycine/NaOH buffer. (C) Measurement of hydrolytic stability by HPLC. (D) HPLC-MS measurement of PrimAc are shown. According to HPLC-MS data the peak at 7.125 corresponds to 302 m/z (ESI +) so as the main peak at 7.296. The molar mass of PrimAc corresponds to the measured value.

A ^1H NMR spectrum.



B ^{13}C NMR spectrum

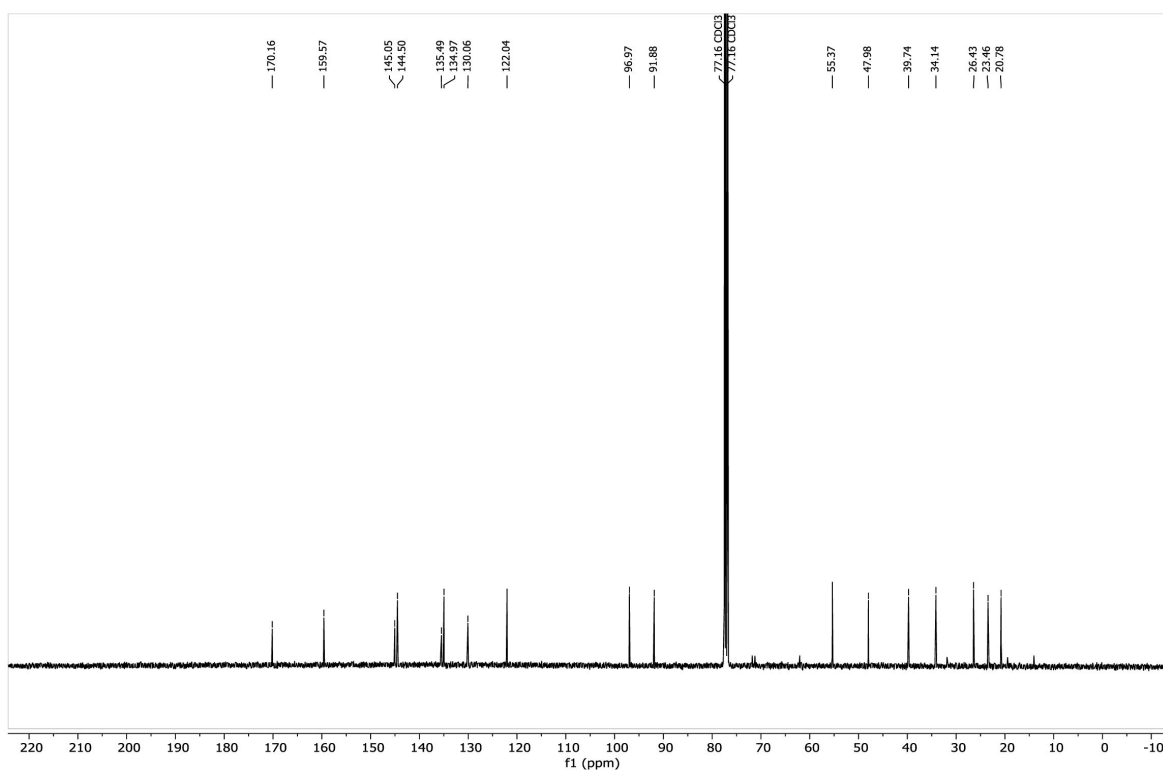


Figure S19

N-acetylated Primaquine.

(A) ^1H NMR spectrum is shown. The presence of impurities is caused by the degradation in the used solvent. (B) ^{13}C NMR spectrum is shown.

HRMS spectrum

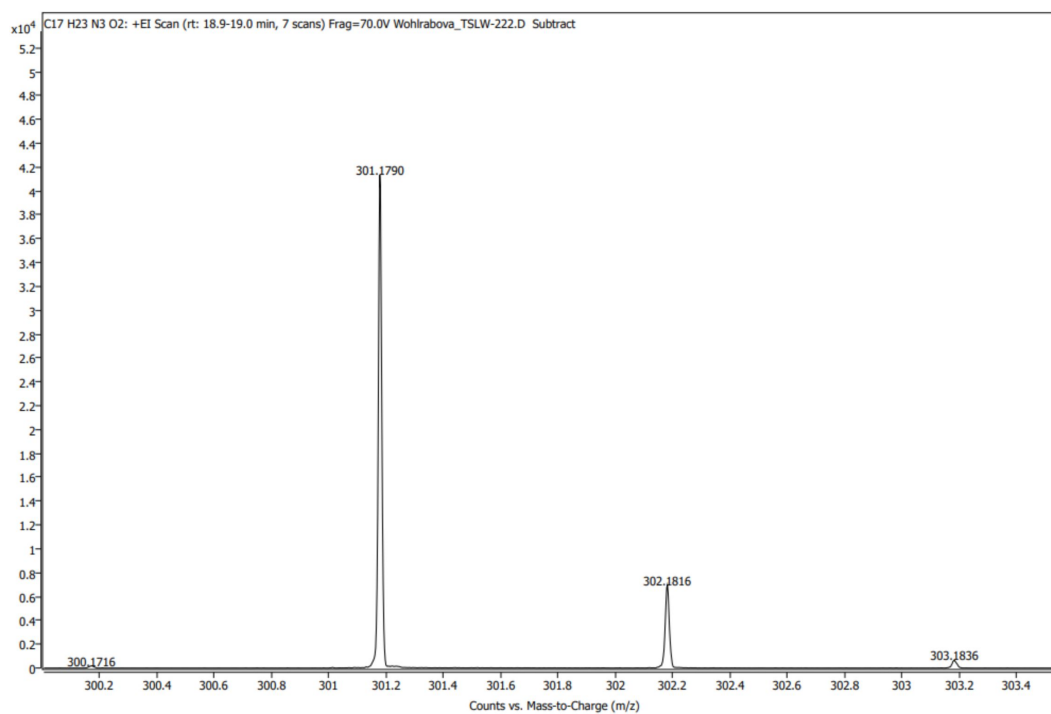


Figure S20

N-acetylated Primaquine.

HRMS spectrum (+EI) is shown

BODIPY analogues

A

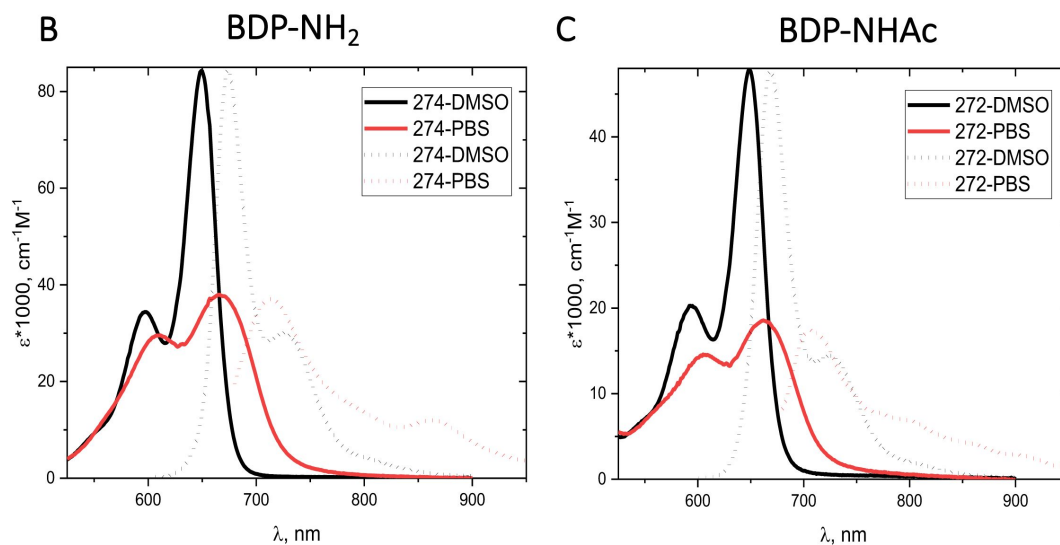
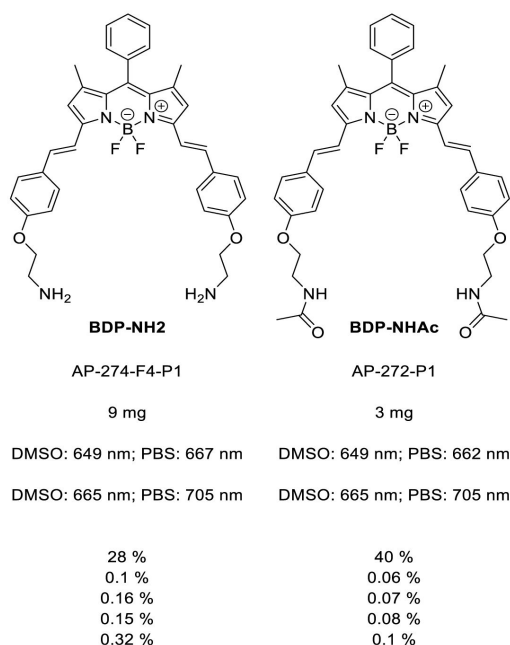
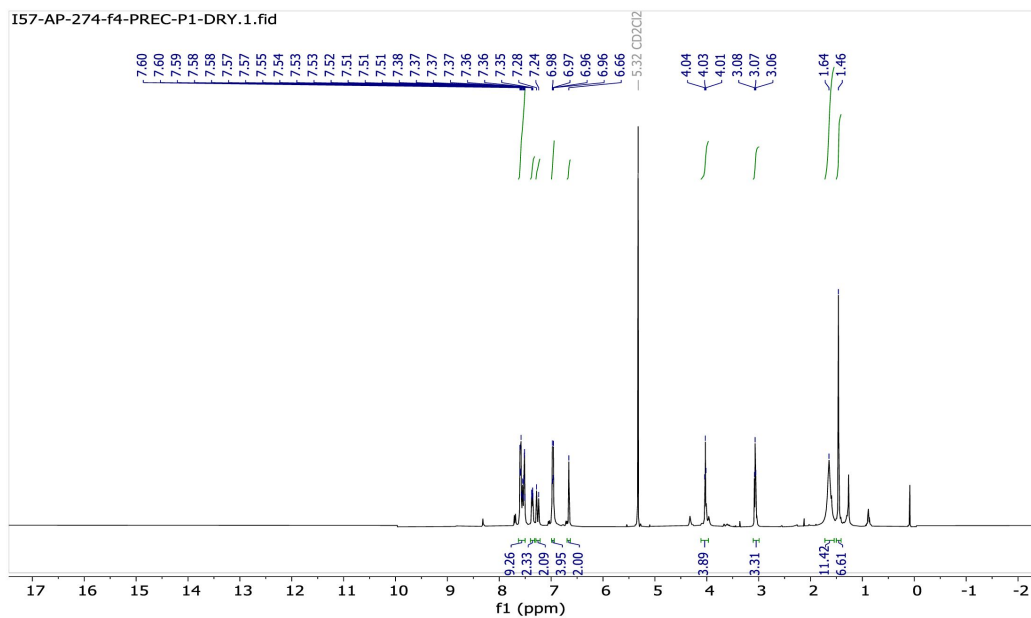


Figure S21

(A) Structure and photo-physical properties of BODIPY analogues. (B) Absorption (full line) and emission spectra (dashed line) of BDP-NH₂ in DMSO (black) and PBS (red) are shown. (C) Absorption (full line) and emission spectra (dashed line) of BDP-NHAc in DMSO (black) and PBS (red) are shown.

A ^1H NMR spectrum



B ^{13}C NMR spectrum

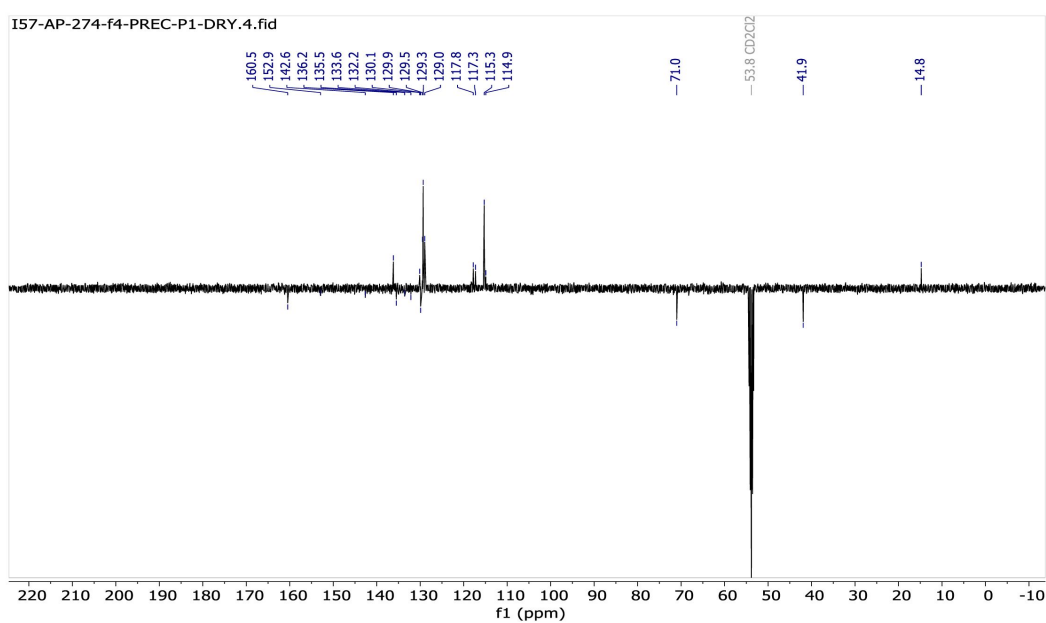


Figure S22

Characterization of BODIPY-NH₂.

(A) ^1H NMR spectrum and (B) ^{13}C NMR spectrum are shown.

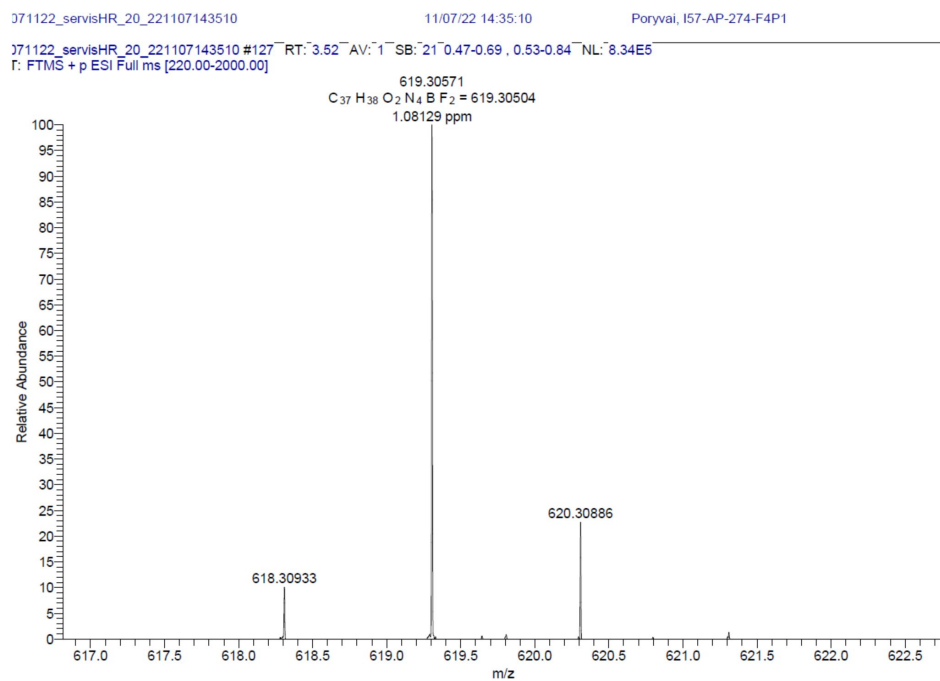
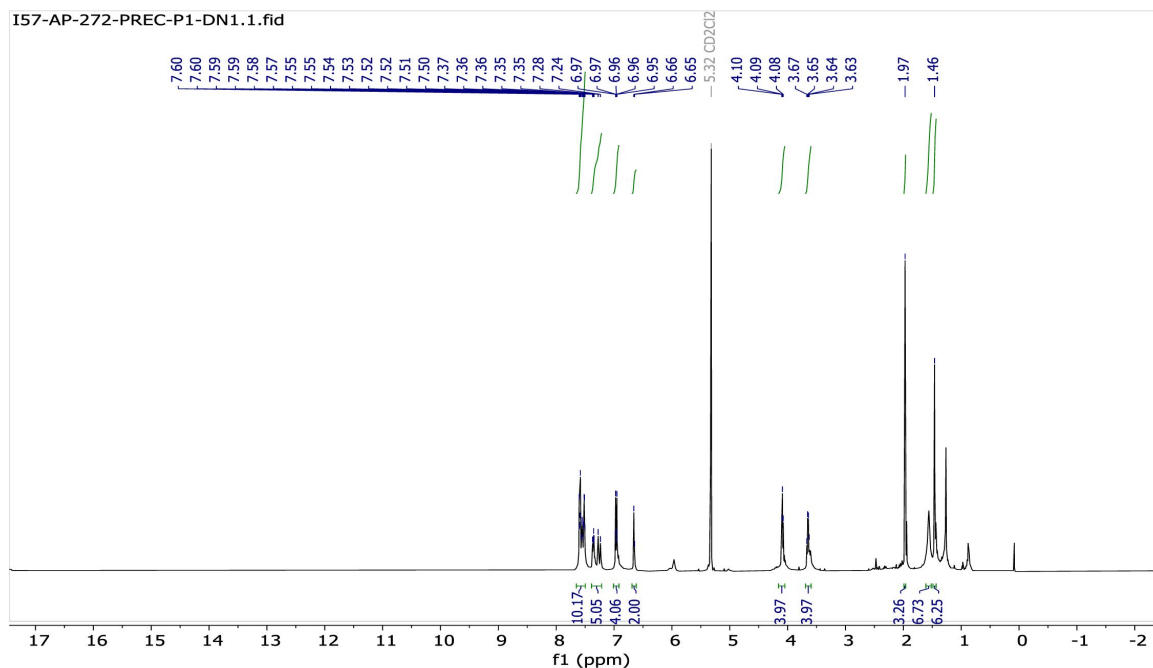


Figure S23

BODIPY-NH₂.

HRMS spectrum (+EI) is shown.

A ^1H NMR spectrum



B ^{13}C NMR spectrum

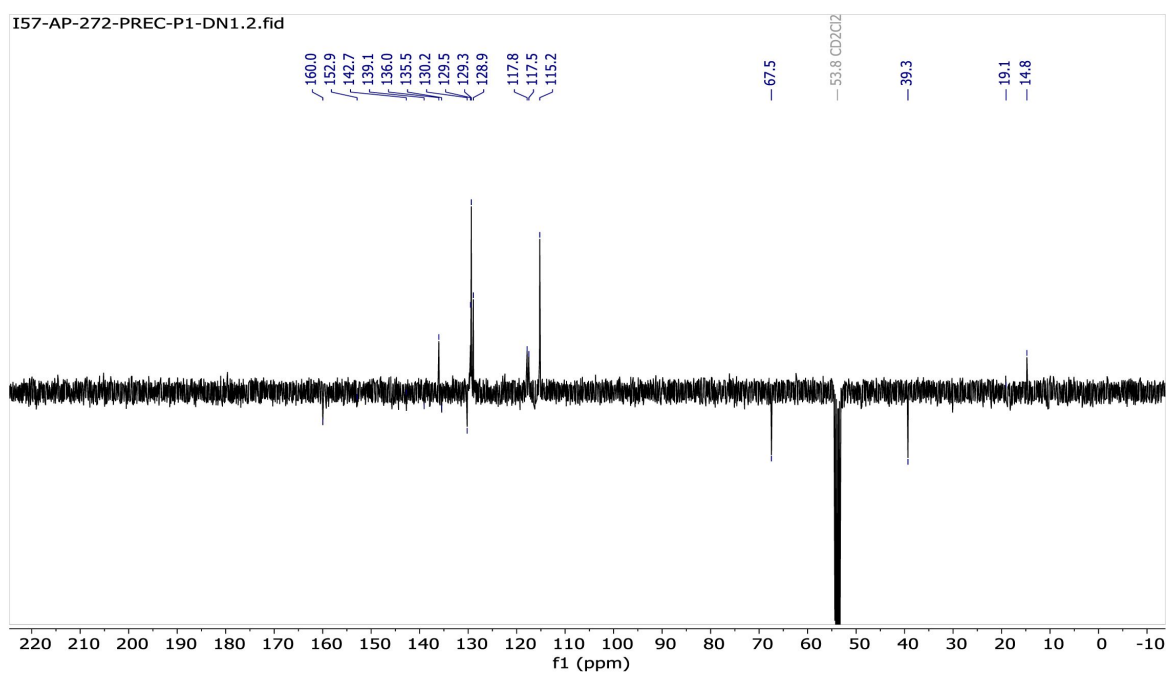


Figure S24

Characterization of BODIPY-NHAc.

(A) ^1H NMR spectrum and (B) ^{13}C NMR spectrum are shown.

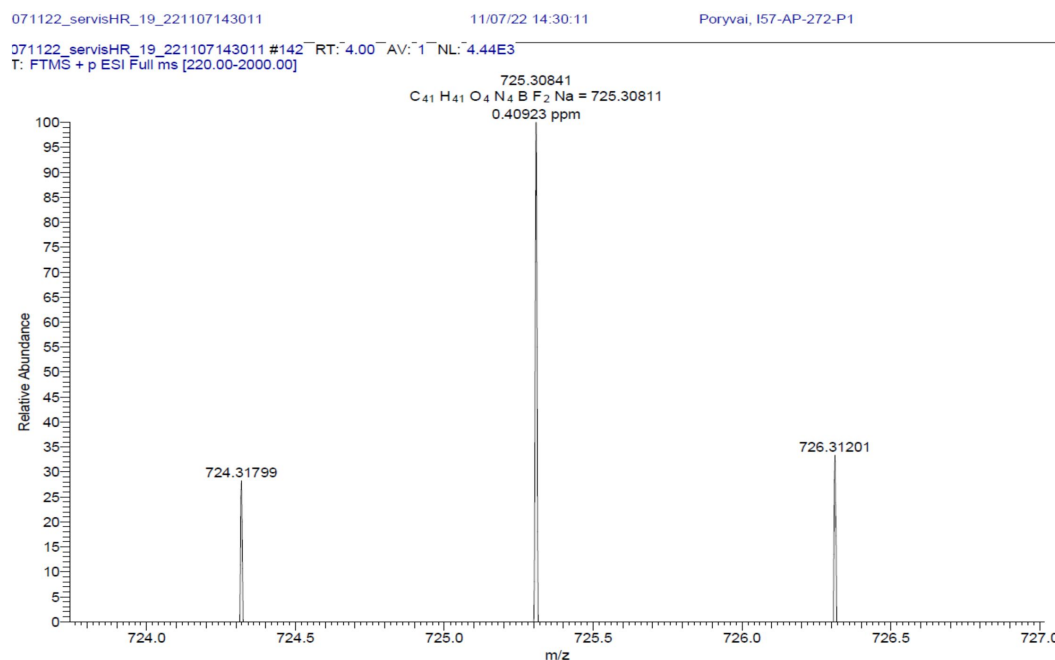


Figure S25

BODIPY-NHAc.

HRMS spectrum (+EI) is shown

References

- Schreiber SL (2005) **Small Molecules: The Missing Link in The Central Dogma** *Nat Chem. Biol* **1**:64–66
- Stockwell BR (2004) **Exploring Biology with Small Organic Molecules** *Nature* **432**:846–854
- Ong SE *et al.* (2009) **Identifying the proteins to which small-molecule probes and drugs bind in cells** *Proc Natl. Acad. Sci. U. S. A* **106**:4617–4622
- Ngo HX, Garneau-Tsodikova S (2018) **What are the drugs of the future?** *Med chem comm* **9**:757–758
- Veber DF, Johnson SR, Cheng HY, Smith BR, Ward KW, Kopple KD (2002) **Molecular Properties That Influence the Oral Bioavailability of Drug Candidates** *J. Med. Chem* **45**:2615–2623
- Charifson PS, Walters WP (2014) **Acidic and Basic Drugs in Medicinal Chemistry: A Perspective** *J Med. Chem* **57**:9701–9717
- Lipinski CA, Lombardo F, Dominy BW, Feeney PJ (1997) **Experimental and Computational Approaches to Estimate Solubility and Permeability in Drug Discovery and Development Settings** *Adv. Drug Deliv. Rev* **23**:3–25
- Manallack DT (2007) **The pK(a) Distribution of Drugs: Application to Drug Discovery** *Perspect Medicin Chem* **1**:25–38
- Meanwell NA (2011) **Improving drug candidates by design: a focus on physicochemical properties as a means of improving compound disposition and safety** *Chem Res Toxicol* **24**:1420–1456
- Di L *et al.* (2012) **Evidence-Based Approach to Assess Passive Diffusion and Carrier-Mediated Drug Transport.** *Drug Discov. Today* **17**:905–912
- Dobson PD, Kell DB (2008) **Carrier-Mediated Cellular Uptake of Pharmaceutical Drugs: An Exception or the Rule?** *Nat. Rev. Drug Discov* **7**:205–220
- Sugano K *et al.* (2010) **Coexistence of Passive and Carrier-Mediated Processes in Drug Transport** *Nat. Rev. Drug Discov* **9**:597–614
- Dey D, Nunes-Alves A, Wade RC, Schreiber G (2022) **Diffusion of small molecule drugs is affected by surface interactions and crowder proteins** *iScience* **25**
- Model MA, Hollembeak JE, Kurokawa M (2021) **Macromolecular crowding: A hidden link between cell volume and everything else** *Cell. Physiol. Biochem* **55**:25–40
- Fulton AB (1982) **How crowded is the cytoplasm?** *Cell* **30**:345–347
- Ellis RJ (2001) **Macromolecular crowding: Obvious but underappreciated Trends in Biochemical Sciences** **26**:597–604

- Milo R, Phillips R (2015) **Cell Biology by the Numbers** *Cell Biology by the Numbers*
- Neurohr GE, Amon A (2020) **Relevance and Regulation of Cell Density** *Trends Cell Biol* **30**:213–225
- Madshus IH (1988) **Regulation of intracellular pH in eukaryotic cells** *The Biochem. J* **250**:1–8
- Asokan A, Cho MJ (2002) **Exploitation of intracellular pH gradients in the cellular delivery of macromolecules** *J. Pharm. Sci* **91**:903–913
- Proksch E (2018) **pH in nature, humans and skin** *J. Dermatol* **45**:1044–1052
- Long X *et al.* (2022) **Nucleus-Targeting Phototherapy Nanodrugs for High-Effective Anti-Cancer Treatment** *Front. pharmacol* **13**:1–8
- Tiwari R, Jain P, Asati S, Haider T, Soni V, Pandey V (2018) **State-of-art based approaches for anticancer drug-targeting to nucleus** *J. Drug Deliv. Sci. Technol* **48**:383–392
- Jansen G, Barr H, Kathmann I, Bunni MA, Priest DG, Noordhuis P, Peters GJ, Assaraf YG (1999) **Multiple mechanisms of resistance to polyglutamatable and lipophilic antifolates in mammalian cells: Role of increased folypolyglutamylation, expanded folate pools, and intralysosomal drug sequestration** *Mol. Pharmacol* **55**:761–769
- Gotink KJ *et al.* (2011) **Lysosomal sequestration of sunitinib: A novel mechanism of drug resistance** *Clin. Cancer Res* **17**:7337–7346
- Zhitomirsky B, Assaraf YG (2015) **Lysosomal sequestration of hydrophobic weak base chemotherapeutics triggers lysosomal biogenesis and lysosome dependent cancer multidrug resistance** *Oncotarget* **6**:1143–1156
- Zotter A Bäuerle, Dey D F, Kiss V, Schreiber G (2017) **Quantifying enzyme activity in living cells** *J. Biol. Chem* **292**:15838–15848
- Dey D, Marciano S, Nunes-Alves A, Kiss V, Wade RC, Schreiber G (2021) **Line-FRAP, A Versatile Method to Measure Diffusion Rates In Vitro and In Vivo** *J. Mol. Biol* **433**
- Ehsanian R, van Waes C, Feller SM (2011) **Beyond DNA binding - A review of the potential mechanisms mediating quinacrine's therapeutic activities in parasitic infections, inflammation, and cancers** *Cell Commun. Signal* **9**:1–18
- Camarda G *et al.* (2019) **Antimalarial activity of primaquine operates via a two-step biochemical relay** *Nat. Commun* **10**:1–9
- Olliaro P, Mussano P (2003) **Amodiaquine for treating malaria** *Cochrane Database of Systematic Reviews* **2**:1–69
- Shenkenberg TD, von Hoff DD (1986) **Mitoxantrone: A new anticancer drug with significant clinical activity** *Ann. Intern. Med* **105**:67–81
- Wagman A, Johnson K, Bussiere D (2005) **Discovery and Development of GSK3 Inhibitors for the Treatment of Type 2 Diabetes** *Curr. Pharm. Des* **10**:1105–1137
- Jampol LM, Cunha-Vaz J (1984) **Diagnostic Agents in Ophthalmology: Sodium Fluorescein and Other Dyes** *Pharmacology of the Eye Handbook of Experimental Pharmacology* **69**:699–714

Zuverink M, Barbieri JT (2015) **From GFP to β -lactamase: advancing intact cell imaging for toxins and effectors** *Pathog. Dis.* **73**

Poryvai A, Galkin M, Shvadchak V, Slanina T (2022) **Red-Shifted Water-Soluble BODIPY Photocages for Visualisation and Controllable Cellular Delivery of Signaling Lipids** *Angew. Chemie-Int* **61**

Wang R, Wang J, Hassan A, Lee CH, Xie XS, Li X (2021) **Molecular basis of V-ATPase inhibition by bafilomycin A1** *Nat. Commun* **12**:1–8

Spugnini EP, Citro G, Fais S (2010) **Proton pump inhibitors as anti-vacuolar-ATPases drugs: A novel anticancer strategy** *J. Exp. Clin. Cancer Res* **29**

Hiruma H, Katakura, Takenami T T, Igawa S, Kanoh M, Fujimura T, Kawakami T (2007) **Vesicle disruption, plasma membrane bleb formation, and acute cell death caused by illumination with blue light in acridine orange-loaded malignant melanoma cells** *J. Photochem. Photobiol. B Biol* **86**:1–8

Marciano S *et al.* (2022) **Protein quaternary structures in solution are a mixture of multiple forms** *Chem Sci* **13**:11680–11695

Zlokarnik G, Negulescu PA, Knapp TE, Mere L, Burres N, Feng L, Whitney M, Roemer K, Tsien RY (1998) **Quantitation of transcription and clonal selection of single living cells with beta-lactamase as reporter** *Science* **279**:84–88

de Duve C, de Barsey T, Poole B, Trouet A, Tulkens P, van Hoof F (1974) **Commentary. Lysosomotropic agents** *Biochem. Pharmacol* **23**:2495–2531

Kaufmann AM, Krise JP (2007) **Lysosomal sequestration of amine-containing drugs: Analysis and therapeutic implications** *J. Pharm. Sci* **96**:729–746

Ishii H, Shirai T, Makino C, Nishikata T (2014) **Mitochondrial inhibitor sodium azide inhibits the reorganization of mitochondria-rich cytoplasm and the establishment of the anteroposterior axis in ascidian embryo** *Dev. Growth Differ* **56**:175–188

Yoshimori T, Yamamoto A, Moriyama Y, Futai M, Tashiro Y (1991) **Bafilomycin A1, a specific inhibitor of vacuolar-type H⁺-ATPase, inhibits acidification and protein degradation in lysosomes of cultured cells** *J. Biol. Chem* **266**:17707–17712

Poste G, Papahadjopoulos D (1976) **Lipid vesicles as carriers for introducing materials into cultured cells: influence of vesicle lipid composition on mechanism(s) of vesicle incorporation into cells** *Proc. Natl. Acad. Sci. U. S. A* **73**:1603–1607

Bandyopadhyay D, Cyphersmith A, Zapata JA, Kim YJ, Payne C. K (2014) **Lysosome transport as a function of lysosome diameter** *PLoS One* **9**

Author information

Debabrata Dey

Department of Biomolecular Sciences, Weizmann Institute of Science, Rehovot, Israel
ORCID iD: [0000-0003-3794-6476](https://orcid.org/0000-0003-3794-6476)

Shir Marciano

Department of Biomolecular Sciences, Weizmann Institute of Science, Rehovot, Israel

Anna Poryvai

Institute of Organic Chemistry and Biochemistry of the Czech Academy of Sciences, Prague, Czech Republic

Ondřej Groborz

Institute of Organic Chemistry and Biochemistry of the Czech Academy of Sciences, Prague, Czech Republic

Lucie Wohlrábová

Institute of Organic Chemistry and Biochemistry of the Czech Academy of Sciences, Prague, Czech Republic

Tomás Slanina

Institute of Organic Chemistry and Biochemistry of the Czech Academy of Sciences, Prague, Czech Republic

Gideon Schreiber

Department of Biomolecular Sciences, Weizmann Institute of Science, Rehovot, Israel
ORCID iD: [0000-0002-2922-5882](https://orcid.org/0000-0002-2922-5882)

For correspondence: gideon.schreiber@weizmann.ac.il

Editors

Reviewing Editor

Felix Campelo

Institute of Photonic Sciences, Barcelona, Spain

Senior Editor

Felix Campelo

Institute of Photonic Sciences, Barcelona, Spain

Reviewer #1 (Public review):

Summary:

The authors set out to measure the diffusion of small drug molecules inside live cells. To do this, they selected a range of fluorescent drugs, as well as some commonly used dyes, and used FRAP to quantify their diffusion. The authors find that drugs diffuse and localize within the cell in a way that is weakly correlated with their charge, with positively charged molecules displaying dramatically slower diffusion and a high degree of subcellular localization.

The study is important because it points to an important issue related to the way drugs behave inside cells beyond the simple "IC50" metric (a decidedly mesoscopic/systemic value). The authors conclude, and I agree, that their results point to nuanced effects that are governed by drug chemistry that could be optimized to make them more effective.

Strengths:

- (1) The work examines an understudied aspect of drug delivery.
- (2) The work uses well-established methodologies to measure diffusion in cells
- (3) The work provides an extensive dataset, covering a range of chemistries that are common in small molecule drug design
- (4) The authors consider several explanations as to the origin of changes in cellular diffusion

Comments on revised version:

In general, my comments were addressed, new discussions were added, and the paper has been improved significantly, which is great.

However, despite providing very clear instructions, a lot of my comments re statistical treatment were disregarded. Bar charts still do not show the repeats as individual points. Errors bars still represent SEM, which gives a wrong idea about the spread of the data. FRAP lines are still averages, and still do not show the spread of the data.

Significance assignments are done based on average and SEMs, as opposed to the full dataset. There is nothing technically wrong with this, but it generally creates an impression that things are more reproducible/rigorous/significant than they would be if the data was shown completely.

<https://doi.org/10.7554/eLife.97255.2.sa2>

Reviewer #2 (Public Review):

Summary:

Blocking a weak base compound's protonation increased intracellular diffusion and fractional recovery in the cytoplasm, which may improve the intracellular availability and distribution of weakly basic, small molecule drugs and be impactful in future drug development.

Strengths:

- (1) The intracellular distribution of drugs and the chemical properties that drive their distribution are much needed in the literature. Thus, the idea behind this paper is of relevance.
- (2) The study used common compounds that were relevant to others.
- (3) Altering a compound's pKa value and measuring cytosolic diffusion rates certainly is inciteful on how weak base drugs and their relatively high pKa values affect distribution and pharmacokinetics. This particular experiment demonstrated relevance to drug targeting and drug development.
- (4) The manuscript was fairly well written.

Comments on revised version:

After reviewing the authors' responses to my questions and concerns, they have adequately corrected the errors, added new information and data based off the reviewers suggestions that improved the manuscript. The manuscript in its current form would add quality information to a part of the literature that is lacking much needed information.

<https://doi.org/10.7554/eLife.97255.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:**Reviewer #1 (Public Review):***Summary:*

The authors set out to measure the diffusion of small drug molecules inside live cells. To do this, they selected a range of fluorescent drugs, as well as some commonly used dyes, and used FRAP to quantify their diffusion. The authors find that drugs diffuse and localize within the cell in a way that is weakly correlated with their charge, with positively charged molecules displaying dramatically slower diffusion and a high degree of subcellular localization.

The study is important because it points at an important issue related to the way drugs behave inside cells beyond the simple "IC50" metric (a decidedly mesoscopic/systemic value). The authors conclude, and I agree, that their results point to nuanced effects that are governed by drug chemistry that could be optimized to make them more effective.

We are grateful to the reviewer for summarizing the work and appreciate him/her pointing out that it is high time to consider the drug aggregation and high degree of subcellular localization while optimizing to make them more effective beyond the mesoscopic value like "IC50".

Strengths:

The work examines an understudied aspect of drug delivery.

The work uses well-established methodologies to measure diffusion in cells

The work provides an extensive dataset, covering a range of chemistries that are common in small molecule drug design

The authors consider several explanations as to the origin of changes in cellular diffusion

We are grateful to the reviewer for pointing out the strengths of the manuscript.

Weaknesses:

The results are described qualitatively, despite quantitative data that can be used to infer the strength of the proposed correlations.

The statistical treatment of the data is not rigorous and not visualized according to best practices, making it difficult for readers to assess the significance of the findings.

Some important aspects of drug behavior are not discussed quantitatively, such as the cell-to-cell or subcellular variability in concentration.

It is unclear if the observed behavior of each drug in the cell actually relates to its efficacy - though this is clearly beyond the scope of this specific work.

We have addressed the weaknesses found by the reviewer (see below in Reviewer #1 Recommendations For The Authors). Concerning the last point, It would have been indeed very valuable to find a relation between drug's observable behavior and their efficacy, but as the reviewer indicates, it is beyond the scope of this work.

Reviewer #2 (Public Review):

Summary:

Blocking a weak base compound's protonation increased intracellular diffusion and fractional recovery in the cytoplasm, which may improve the intracellular availability and distribution of weakly basic, small molecule drugs and be impactful in future drug development.

We are thankful to the reviewer for summarizing our work and acknowledging that the points raised above can be impactful in future drug development.

Strengths:

- (1) The intracellular distribution of drugs and the chemical properties that drive their distribution are much needed in the literature. Thus, the idea behind this paper is of relevance.*
- (2) The study used common compounds that were relevant to others.*
- (3) Altering a compound's pKa value and measuring cytosolic diffusion rates certainly is inciteful on how weak base drugs and their relatively high pKa values affect distribution and pharmacokinetics. This particular experiment demonstrated relevance to drug targeting and drug development.*
- (4) The manuscript was fairly well written.*

We are thankful to the reviewer for pointing out the strengths of the manuscript like the intracellular distribution of drugs and properties that drive it, which are missing in the literature.

Weaknesses:

- (1) Small sample sizes. 2 acids and 1 neutral compound vs 6 weak bases (Figure 1).*

We fully agree with the reviewer on this point. However, the major limitation we have faced here is the small number of drug/drug-like molecules that fluorescent with sufficient high quantum yields. For this study, we initially screened 1600 drugs for their fluorescence in the visible spectrum, and penetration into cells, resulting in 16 drugs. Of those, a small number was suitable for FRAP due to low quantum yield. For some of the molecules (Mitoxantrone, Priaquine), recovery was minimal, making them challenging to study. We added this information in the materials and method section under “Selection of drugs used in this study” (p.10).

- (2) A comparison between the percentage of neutral and weak base drug accumulation in lysosomes would have helped indicate weak base ion trapping. Such a comparison would have strengthened this study.*

For weakly basic compounds, the ionic form and the non-ionic form of the molecules always remain in equilibrium. The direction of the equilibrium depends on the pH of the medium, which determines the major form of the drug molecules in the solution. Our examples of GSK3 inhibitor (neutral compound, $pK_a \sim 7.0$, as predicted by Chemaxon), shows behaviour very similar to the other basic drugs ($pK_a > 8$) inside the cells. As lysosome pH is about 5.0, the neutral drug also gets protonated inside the lysosomes, as the colocalization study reveals (Figure 4). We added Fig S16 C-D, where we show co-localization of three drugs within the

lysosomes showing that all the three weak base drugs colocalize to acidic lysosomes from moderately to extensively. See also in p. 11 under “Confocal microscopy and FRAP Analysis section”.

(3) When cytosolic diffusion rates of compounds were measured, were the lysosomes extracted from the image using Imaris to determine a realistic cytosolic value? In real-time, lysosomes move through the cytosol at different rates. Because weak base drugs get trapped, it is likely the movement of a weak base in the lysosome being measured rather than the movement of a weak base itself throughout the cytosol. This was unclear in the methods. Please explain.

We want to thank the reviewer for pointing this out. To clarify the point, we added to the material and method section in p. 13 the following text: “When the areas of bleach were selected in the drug-treated cell cytoplasm, we avoided the lysosomes as much as possible, within the resolution limits of the confocal microscope. Lysosomes themselves were measured to move within the cytoplasm with an diffusion coefficient of $0.03\text{--}0.071\text{ }\mu\text{m}^2\text{ s}^{-1}$ (Bandyopadhyay et al., 2014), which is much slower than the diffusion measured for even the slowest compounds using fast Line FRAP, further validating that we did not measure lysosome diffusion.” In addition, we show that in cells after Bafilomycin A1 or Na-Azide treatments the number of lysosomes was reduced drastically (Figures S8& S9, and Figure 7), while the rates of diffusion remain very slow, similar to those measured without lysosomal inhibitors.

(4) Because weak base drugs can be protonated in the cytoplasm, the authors need to elaborate on why they thought that inhibiting lysosome accumulation of weak bases would increase cytosolic diffusion rates. Ion trapping is different than “micrometers per second” in the cytosol. Moreover, treating cells with sodium azide de-acidifies lysosomes and acidifies the cytosol; thus, more protons in the cytosol means more protonation of weak base drugs. The diffusion rates were slowed down in the presence of lysosome inhibition (Figure 7), which is more fitting of the story about blocking protonation increases diffusion rates, but in this case, increasing cytosolic protonation via lysosome de-acidification agents decreases diffusion rates. Please elaborate.

We thank the reviewer for the comment. We added to the results in p. 7 (top) the following “While we selected bleach spots to be small and located outside of lysosomes, this does not assure that some of the bleached area does not include smaller lysosomes. Therefore we investigated whether inhibiting lysosomal trapping will eliminate slow diffusion of cationic drugs.” In addition, we added to the results in p. 7-8 the following: “Comparative FRAP profiles and diffusion coefficients (Figure 7B-D and 7F-H) were slow, but conversely to Bafilomycin, sodium azide treatment did cause a further reduction in rates from Dconfocal $2.4\pm 0.1\text{ }\mu\text{m}^2\text{s}^{-1}$ to $1.8\pm 0.1\text{ }\mu\text{m}^2\text{s}^{-1}$ for quinacrine and from 0.6 to $0.45\text{ }\mu\text{m}^2\text{s}^{-1}$ for the GSK3 inhibitor (Figure 7C and G). Both Bafilomycin and sodium azide treatments resulted in elimination of drug confinement in the lysosome, and the small difference in diffusion rates may be a result of the de-acidification of the lysosomes by sodium azide, which may increase the protons in the cytosol upon treatment.”

Reviewer : A discussion of the likely impact:

The manuscript certainly adds another dimension to the field of intracellular drug distribution, but the manuscript needs to be strengthened in its current form. Additional experiments need to be included, and there are clarifications in the manuscript that need to be addressed. Once these issues are resolved, then the manuscript, if the conclusions are further strengthened, is much needed and would be inciteful to drug development.

Reviewer #1 (Recommendations For The Authors):

Major issues:

The paper suffers from poor statistical treatment of the data. FRAP recovery curves should be shown for each repeat, overlaid by an average with SDs as errorbars or shaded regions shown. In bar plots, SEMs should be eliminated in favor of StdDevs. All datapoints should be shown for each bar in Figs. 3-8. To show differences in D_confocal appropriate statistical tests should be conducted. In addition it is unclear what an "independent repeat" is. Does this mean 30 separate imaging sessions/drug treatments/etc? Is it 30 cells on the same coverslip? Is it a combination of both? All reported errors, SD or SEM, should have a single significant digit. Guidelines and best practices for representing quantitative imaging data are all described and visualized in detail in Lord et al. JBS 2020.

We improved the statistics and added the individual progression curves and did the statistics on them as requested. See Figure S2 for individual FRAP curves of fluorescein, GSK3 inhibitor and quinacrine. Statistical analysis of the individual FRAP curves is in Figure 3B, 4B, 5B, 7C and G. For details see figures legends and material and methods p. 13 in "Determination of Dconfocal from FRAP results". Line FRAP was done from the cells taken from different plates, treated independently (see text p. 13).

The extensive (and commendable!) dataset the authors have collected can be put to better use than what is currently done. The main text figures in the current form of the preprint are mostly descriptive and their discussion is qualitative, to the point where the author's conclusions are supported only anecdotally. Instead, I would much rather see panels that collate the entire dataset (both protein and drugs) numerically, comparing diffusion values in buffer/cytoplasm/nucleus for all drugs (Like Fig. S6, which is in my opinion the most important in the paper but for some reason relegated to the SI). In addition I would like to see correlations within the dataset, such as D_confocal vs. pKa, vs. concentration (as measured by overall fluorescence signal, see my comment below), vs. mw, or vs. specific chemical moieties (number of charges, aromatic rings, etc). Such correlations should be discussed in terms of a correlation coefficient if conclusions were to be drawn from them, and include errors if available.

We want to thank the reviewer for these suggestions. We now made new Figures 9, and S16 to compare multiple parameters. Figure 9C shows a clear relation between pKa and Dconfocal, but no relation was found between logP, MW or number of aromatic rings and Dconfocal. Fig. S3 also shows the relation between drug concentration and Dconfocal values. These data are now discussed in the discussion section in p. 9 (bottom).

The drug sequestration hypothesis and other conclusions brought forth by the authors could be further tested by looking at the concentration dependence of the drugs inside each cell and/or its partitioning between different subcellular compartments. The concentration dependence of these drugs is discussed in a very anecdotal fashion using two concentrations - and despite some cases showing an effect no further studies were done. Drug concentrations in this experiment can vary between cells between repeats or even within a single repeat as a result of drug chemistry and delivery methods (microinjection/passive permeability). This is especially important since it is unclear what clinically-relevant concentrations are for each drug (or at least an IC50 for the cell types tested here). I would like to see a quantitative measure of concentrations as another metric to compare diffusion behavior (see my comment above as well).

And maybe one thing to consider in addition would be some discussion in the paper about what sub-cellular distributions might actually mean in the context of drug efficacy (asking for myself as well!) - a paragraph describing recent works on the topic with some references could be instructive.

We want to thank the reviewer for the suggestion. We added now Figure S3, showing the relation between fluorescence intensity in each cell (which is directly related to the concentration of the compound) and FRAP rates and percent recovery for fluorescein, GSK inhibitor and Quinacrine. The results show now relation between drug concentration and FRAP rates, and some relation towards percent recovery. These data are now discussed in the main text (p. 4 bottom and p.6) and in the discussion (p. 9, bottom).

Minor issues:

Readers could benefit from a schematic showing the line FRAP method. It is difficult to understand from the text.

We show now in Figure 2 the line-FRAP method, and discuss it in the introduction (p. 3 top).

Have the authors considered enrichment in the cell membrane? Summed intensity projections or co-labeling with membrane dyes could prove useful to identify if the membrane is enriched in fluorescence.

The microscopy slides, including the super-resolution image in Figure S15 do not show enrichment of membranes.

Cell extracts obtained by chemical lysis are problematic because they contain surfactants. This comparison might not be meaningful.

The reviewer is correct about surfactants; However, this is only for illustration to show the crowd density of the cell extracts compared to live cells.

Unclear why "Bleach size" plots are shown. They are not discussed in the main text.

We show now a bleach size plot in Figure 2, where we explain the method. We removed them from the other figures.

Some figure panels have a strange aspect ratio, causing text to look distorted.

We corrected the figure distortion in the revised manuscript.

How are the values of D_{confocal} in buffer compared with past literature? Should these not all be diffusion limited? BCECF - larger than many of the drugs used here - shows $\sim 100 \mu\text{m}^2/\text{s}$ in buffer (Verkman TiBS 2002).

We discussed this in our previous work (Ref. 13, iScience 2022, Dey et al.) D_{confocal} is a relative diffusion rate and should not be confused with single-molecule diffusion coefficients. FRAP cannot measure the diffusion of more than $100 \mu\text{m}^2/\text{s}$ in the buffer. However, when comparing apparent FRAP rates between different fluorophores, it is not quantitative due to the major implication of the bleach radius towards diffusion rates. The rate constant normalized by bleach radius² is the proper way to compare i.e., our D_{confocal} . (Ref. JMB 2021, iScience 2022 by Dey et al.).

Reviewer #2 (Recommendations For The Authors):

Recommendations:

(1) Page 3 at the bottom of the Introduction states, "...sodium azide (Hiruma et al., 2007) inhibited accumulation in lysosomes, cellular diffusion...increased only slightly." However, Figure 7C, F shows a sodium azide-induced decrease in the Dconfocal cellular diffusion. Please clarify.

Thank you for pointing this out; we corrected it in the revised version, including adding statistics.

(2) Page 6 states, "Quinacrine accumulation in the lysosome was observed also immediately after micro-injection, with aggregation increasing over time. Dconfocal of $4.2 \pm 0.2 \mu\text{m}^2 \text{s}^{-1}$ was calculated from line-FRAP immediately after micro-injection, slowing to $2.2 \pm 0.1 \mu\text{m}^2 \text{s}^{-1}$ following 2 hours incubations, with fractional recoveries of 0.63 and 0.57 respectively." If lysosome sequestration does not have an effect on cytosolic diffusion rates as the manuscript concludes, why do the authors think the diffusion rate decreased here within 2 hours? A solid conclusion would strengthen the conclusions of this manuscript rather than passing over it.

Thank you for pointing this out. We added the following text to page 7: "It is notable that the Dconfocal for Quinacrine remained consistent regardless of Bafilomycin treatment, 2 hours after incubation (Fig. S9D, $2.4 \pm 0.1 \mu\text{m}^2 \text{s}^{-1}$). However, when measured immediately after injection, the diffusion coefficient was higher at $4.2 \mu\text{m}^2 \text{s}^{-1}$ (Fig. S5D). This result does not support the notion that the faster diffusion measured immediately after cellular injection relates to lysosomal aggregation, and would better support self-aggregation, or aggregation with other molecules in the cell, which increases over time. This notion is further supported by the almost complete lack in FRAP observed 24 hours after injection (Fig. S5C)."

(3) In the Results section, the subheading states, "Inhibition of lysosomal sequestration is only slightly increasing diffusion in cells", but the conclusion for bafilomycin was...Dconfocal values were not altered by Bafilomycin A1", and the conclusion for sodium azide was diffusion coefficients (Figure 7B-C and 7E-F) were not much changed for the two drugs and stayed low... similarly to what was observed with Bafilomycin." The clear question is what is the result, "slightly increased diffusion, decreased diffusion, or had no significant effect at all"? Please clarify the wording in the manuscript to accurately describe the results.

Indeed, a small difference is observed between the two treatments. We added now statistical significance to Fig. 7D and H and to Fig. S8 and S9. In addition, we clarified this point in the text in p.7-8: "Comparative FRAP profiles and diffusion coefficients (Figure 7B-D and 7F-H) were slow, but conversely to Bafilomycin, sodium azide treatment did cause a further reduction in rates from Dconfocal $2.4 \pm 0.1 \mu\text{m}^2 \text{s}^{-1}$ to $1.8 \pm 0.1 \mu\text{m}^2 \text{s}^{-1}$ for quinacrine and from 0.6 to $0.45 \mu\text{m}^2 \text{s}^{-1}$ for the GSK3 inhibitor (Figure 7C and G). Both Bafilomycin and sodium azide treatments resulted in elimination of drug confinement in the lysosome, and the small difference in diffusion rates may be a result of the de-acidification of the lysosomes by sodium azide, which may increase the protons in the cytosol upon treatment."

(4) In Figure 8B, why was the Dconfocal for AM-fluorescein with or without sodium azide not included here? Besides consistency, the results might demonstrate significance. Please elaborate on the occlusion of this data.

Fraction recovery after FRAP of AM-fluorescein was very low. Calculating Dconfocal rates with such low fraction recovery is meaningless, as in the time of measurement only a small fraction recovered. Therefore, we calculated Dconfocal only when fraction recovery was at least 0.5.

(5) Throughout the Results section, the ideas and experiments are of relevance, but the suggestions/conclusions at the end of each paragraph of this section seem lightly thought out. For example, as stated on Page 8, "...however, this did not contribute new information to the puzzle." For a chemistry paper, a chemical suggestion strengthens the manuscript.

We want to thank the reviewer for these suggestions. We now made new Figures 9, and S16 to compare multiple parameters. Figure 9C shows a clear relation between pKa and Dconfocal, but no relation was found between logP, MW or number of aromatic rings and Dconfocal. Fig. S16 also shows the relation between drug concentration and Dconfocal values. We revised the discussion section to give more weight to these quantitative assessments. These data are now discussed in p. 9.

In conclusion, the manuscript's ideas are needed, but the conclusions drawn from the experiments need to be strengthened, more explanatory, and consistent with the main conclusion of the manuscript.

See answer to point 5.

<https://doi.org/10.7554/eLife.97255.2.sa0>